

Appendices of *On the Geometric Emergence of the Electromagnetic Field in Teleparallel Spacetime*

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1 Appendix 1: On Notation

Throughout this appendix, the coordinates will be taken as

$$x^\mu = (ct, x, y, z)$$

Greek indices

$$\mu, \nu, \rho, \dots = 0, 1, 2, 3$$

denote spacetime coordinate indices, while Latin indices

$$a, b, c, \dots = 0, 1, 2, 3$$

denote internal Lorentz indices. For spatial indices,

$$i, j, k, \dots = 1, 2, 3$$

will be used. The internal Minkowski metric is chosen as

$$\eta_{AB} = \text{diag}(-1, 1, 1, 1)$$

Moreover, the SI unit system will be used.

2 Appendix 2: Connection Calculation in Teleparallel Geometry, Derivation of the Chosen Tetrad, Inverse Tetrad, Metric, and Torsion

2.1 Notation and Dimensional Analysis of the Maxwell-Adapted Temporal Tetrad

In this appendix, the temporal tetrad ansatz used in the work will be introduced explicitly. First, the general tetrad definition will be given; then, using the chosen notation for the electromagnetic potential one-form, the explicit expression of the Maxwell-adapted temporal tetrad will be obtained. Afterwards, the dimensional analysis of this expression will be performed and the tetrad matrix will be written explicitly.

2.1.1 General Tetrad Definition

In teleparallel geometry, the fundamental geometric variable is the tetrad field. The tetrad field is defined as a coframe one-form that establishes a relation between the internal Lorentz space and spacetime coordinates:

$$e^a = e^a{}_\mu dx^\mu. \quad (1)$$

Here, $e^a{}_\mu$ denotes the components of the tetrad in the coordinate basis. Through the tetrad, the spacetime metric is obtained as

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu \quad (2)$$

The inverse tetrad is denoted by $E_a{}^\mu$ and satisfies the conditions

$$E_a{}^\mu e^a{}_\nu = \delta^\mu{}_\nu, \quad E_a{}^\mu e^b{}_\mu = \delta_a{}^b \quad (3)$$

When the electromagnetic contribution is switched off, the reference limit of the temporal tetrad is taken as

$$e_{\text{ref}}^0 = dx^0 = c dt \quad (4)$$

This expression represents only the background temporal flow.

2.1.2 Electromagnetic Potential One-Form

In this work, the electromagnetic potential one-form is defined by

$$A = A_\mu dx^\mu \quad (5)$$

Since our fixed coordinate notation is Eq. 1,

$$dx^0 = c dt \quad (6)$$

is written. The electromagnetic potential one-form is taken in terms of the physical scalar and vector potentials as

$$A = -\phi dt + A_i dx^i \quad (7)$$

Using Eq. 6, it can also be written as

$$A = -\frac{\phi}{c} dx^0 + A_i dx^i \quad (8)$$

Thus, in the coordinate notation of Eq. 1, the covariant potential components are

$$A_\mu = \left(-\frac{\phi}{c}, A_1, A_2, A_3 \right) \quad (9)$$

Explicitly,

$$A_0 = -\frac{\phi}{c}, \quad A_1 = A_1, \quad A_2 = A_2, \quad A_3 = A_3. \quad (10)$$

2.1.3 Derivation of the Maxwell-Adapted Temporal Tetrad

In order to encode the electromagnetic potential geometrically within the temporal tetrad, the temporal tetrad is taken as

$$e^0 = dx^0 + \kappa_{\text{EM}} A \quad (11)$$

Here, κ_{EM} is the geometric normalization coefficient that makes the electromagnetic potential one-form compatible with the dimension of the tetrad. Since the potential one-form is

$$A = -\frac{\phi}{c} dx^0 + A_i dx^i \quad (12)$$

one obtains

$$e^0 = dx^0 + \kappa_{\text{EM}} A = dx^0 + \kappa_{\text{EM}} \left(-\frac{\phi}{c} dx^0 + A_i dx^i \right) = dx^0 - \kappa_{\text{EM}} \frac{\phi}{c} dx^0 + \kappa_{\text{EM}} A_i dx^i = \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) dx^0 + \kappa_{\text{EM}} A_i dx^i$$

If Eq. 1 is used, the same expression is written as

$$e^0 = \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) c dt + \kappa_{\text{EM}} A_i dx^i = c \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) dt + \kappa_{\text{EM}} A_i dx^i = (c - \kappa_{\text{EM}} \phi) dt + \kappa_{\text{EM}} A_i dx^i$$

Thus, the compact and explicit representations of the temporal tetrad are

$$e^0 = dx^0 + \kappa_{\text{EM}} A \quad (13)$$

and

$$e^0 = \left(1 - \kappa_{\text{EM}} \frac{\phi}{c}\right) dx^0 + \kappa_{\text{EM}} A_i dx^i \quad (14)$$

When written using dt in SI units, one obtains

$$e^0 = (c - \kappa_{\text{EM}} \phi) dt + \kappa_{\text{EM}} A_i dx^i \quad (15)$$

This result shows that, in the chosen notation, the scalar potential contribution enters the temporal tetrad with a minus sign, while the spatial vector potential contribution enters with a plus sign.

2.1.4 Dimensional Analysis

In this subsection, it will be shown that Eq. 13 is consistent in SI units. The dimension of the tetrad one-form is taken to be length:

$$[e^0] = L. \quad (16)$$

For the reference temporal tetrad term,

$$[c dt] = \frac{L}{T} T = L \quad (17)$$

Thus, the electromagnetic contribution must also have the dimension of length:

$$[\kappa_{\text{EM}} A] = L. \quad (18)$$

The electromagnetic potential one-form is

$$A = -\phi dt + A_i dx^i \quad (19)$$

For the scalar potential in SI units,

$$[\phi] = \text{V} \quad (20)$$

is taken. Since the volt unit is

$$1 \text{ V} = \frac{\text{J}}{\text{C}} = \frac{\text{kg m}^2}{\text{C s}^2} \quad (21)$$

in terms of fundamental dimensions,

$$[\text{V}] = \frac{ML^2}{QT^2} \quad (22)$$

is obtained.

Therefore, the dimension of the scalar potential term is

$$[\phi dt] = [V] [s] = \frac{ML^2}{QT^2} T = \frac{ML^2}{QT}$$

For the spatial vector potential in SI units,

$$[A_i] = \frac{Vs}{m} \quad (23)$$

and hence

$$[A_i dx^i] = [A_i] L = \frac{Vs}{m} m = Vs$$

is obtained.

Thus, the temporal and spatial parts of the electromagnetic potential one-form have the same dimension:

$$[A] = [\phi dt] = [A_i dx^i] = Vs. \quad (24)$$

In terms of fundamental dimensions,

$$[A] = \frac{ML^2}{QT} \quad (25)$$

is obtained.

Now, using the condition

$$[\kappa_{EM} A] = L \quad (26)$$

one finds

$$[\kappa_{EM}] = \frac{L}{[A]} = \frac{L}{ML^2/(QT)} = \frac{QT}{ML}$$

Thus,

$$[\kappa_{EM}] = \frac{C s}{kg m} \quad (27)$$

and equivalently

$$[\kappa_{EM}] = \frac{m}{V s} \quad (28)$$

can be written.

Indeed,

$$\frac{m}{V s} = \frac{m}{\left(\frac{kg m^2}{C s^2}\right) s} = \frac{C s}{kg m}$$

is obtained.

Now dimensional analysis is performed for the normalization coefficient used in this work and to be shown in later sections,

$$\kappa_{EM} = \frac{\sqrt{4\pi G \epsilon_0}}{c} \quad (29)$$

Here 4π is dimensionless. The dimension of Newton's constant is

$$[G] = \frac{L^3}{MT^2} \quad (30)$$

and the dimension of the vacuum permittivity is

$$[\epsilon_0] = \frac{Q^2 T^2}{ML^3} \quad (31)$$

Thus,

$$[G\epsilon_0] = \frac{L^3}{MT^2} \frac{Q^2 T^2}{ML^3} = \frac{Q^2}{M^2}.$$

From this,

$$[\sqrt{G\epsilon_0}] = \frac{Q}{M} \quad (32)$$

is obtained. Since the dimension of the speed of light is

$$[c] = \frac{L}{T} \quad (33)$$

one has

$$\left[\frac{\sqrt{G\epsilon_0}}{c} \right] = \frac{Q}{M} \frac{T}{L} = \frac{QT}{ML}.$$

This result is the same as the dimension of $[\kappa_{\text{EM}}]$ obtained above:

$$\left[\frac{\sqrt{4\pi G\epsilon_0}}{c} \right] = [\kappa_{\text{EM}}]. \quad (34)$$

Consequently, Eq. 13 is consistent in SI units.

2.2 Tetrad Matrix

In this work, the temporal tetrad is taken as

$$e^0 = \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) dx^0 + \kappa_{\text{EM}} A_i dx^i \quad (35)$$

The spatial tetrads are chosen in the reference configuration containing the cosmological scale factor as

$$e^i = a(t) dx^i \quad (36)$$

Explicitly, the tetrad one-forms are

$$e^0 = \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) dx^0 + \kappa_{\text{EM}} A_1 dx^1 + \kappa_{\text{EM}} A_2 dx^2 + \kappa_{\text{EM}} A_3 dx^3, e^1 = a(t) dx^1, e^2 = a(t) dx^2, e^3 = a(t) dx^3. \quad (37)$$

From the definition $e^a = e^a{}_\mu dx^\mu$ and for the coordinates Eq. 1, the tetrad matrix is

$$e^a{}_\mu = \begin{pmatrix} 1 - \kappa_{\text{EM}} \frac{\phi}{c} & \kappa_{\text{EM}} A_1 & \kappa_{\text{EM}} A_2 & \kappa_{\text{EM}} A_3 \\ 0 & a(t) & 0 & 0 \\ 0 & 0 & a(t) & 0 \\ 0 & 0 & 0 & a(t) \end{pmatrix}. \quad (38)$$

2.3 Inverse Tetrad

The inverse tetrad

$$E_a{}^\mu$$

satisfies the following conditions:

$$E_a{}^\mu e^b{}_\mu = \delta_a{}^b, \quad E_a{}^\mu e^a{}_\nu = \delta^\mu{}_\nu.$$

Since the chosen tetrad has an upper triangular form, the inverse tetrad can be computed directly. The temporal inverse tetrad components are

$$E_0{}^0 = \frac{1}{1 - \kappa_{\text{EM}}\phi/c}, \quad E_0{}^1 = 0, \quad E_0{}^2 = 0, \quad E_0{}^3 = 0 \quad (39)$$

In matrix form,

$$E_a{}^\mu = \begin{pmatrix} \frac{1}{1 - \kappa_{\text{EM}}\phi/c} & 0 & 0 & 0 \\ -\frac{\kappa_{\text{EM}}A_1}{a(t)(1 - \kappa_{\text{EM}}\phi/c)} & \frac{1}{a(t)} & 0 & 0 \\ -\frac{\kappa_{\text{EM}}A_2}{a(t)(1 - \kappa_{\text{EM}}\phi/c)} & 0 & \frac{1}{a(t)} & 0 \\ -\frac{\kappa_{\text{EM}}A_3}{a(t)(1 - \kappa_{\text{EM}}\phi/c)} & 0 & 0 & \frac{1}{a(t)} \end{pmatrix} \quad (40)$$

is obtained. If the inverse tetrad definition were in the form $E^\mu{}_a$, then it would be taken as

$$E^\mu{}_a = \begin{pmatrix} \frac{1}{1 - \kappa_{\text{EM}}\phi/c} & -\frac{\kappa_{\text{EM}}A_1}{a(t)(1 - \kappa_{\text{EM}}\phi/c)} & -\frac{\kappa_{\text{EM}}A_2}{a(t)(1 - \kappa_{\text{EM}}\phi/c)} & -\frac{\kappa_{\text{EM}}A_3}{a(t)(1 - \kappa_{\text{EM}}\phi/c)} \\ 0 & \frac{1}{a(t)} & 0 & 0 \\ 0 & 0 & \frac{1}{a(t)} & 0 \\ 0 & 0 & 0 & \frac{1}{a(t)} \end{pmatrix}. \quad (41)$$

Moreover, for the inverse tetrad to be well-defined, the condition

$$1 - \kappa_{\text{EM}}\frac{\phi}{c} \neq 0 \quad (42)$$

must be satisfied.

2.4 Calculation of the Metric Tensor

The metric tensor is obtained from the tetrad as

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu \quad (43)$$

Since the line element is

$$ds^2 = \eta_{ab} e^a e^b$$

one writes

$$ds^2 = -(e^0)^2 + \delta_{ij} e^i e^j \quad (44)$$

Substituting the chosen tetrad gives

$$ds^2 = - \left[\left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) dx^0 + \kappa_{\text{EM}} A_i dx^i \right]^2 + a^2(t) \delta_{ij} dx^i dx^j \quad (45)$$

Explicitly, this expression becomes

$$ds^2 = - \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right)^2 (dx^0)^2 - 2\kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) A_i dx^0 dx^i + [a^2(t) \delta_{ij} - \kappa_{\text{EM}}^2 A_i A_j] dx^i dx^j.$$

Thus, the metric components are found to be

$$g_{00} = - \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right)^2, \quad (46)$$

$$g_{0i} = g_{i0} = -\kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) A_i, \quad (47)$$

and

$$g_{ij} = a^2(t) \delta_{ij} - \kappa_{\text{EM}}^2 A_i A_j \quad (48)$$

In matrix form,

$$g_{\mu\nu} = \begin{pmatrix} - \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right)^2 - \kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) A_1 - \kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) A_2 - \kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) A_3 \\ -\kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) A_1 a^2(t) - \kappa_{\text{EM}}^2 A_1^2 - \kappa_{\text{EM}}^2 A_1 A_2 - \kappa_{\text{EM}}^2 A_1 A_3 \\ -\kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) A_2 - \kappa_{\text{EM}}^2 A_2 A_1 a^2(t) - \kappa_{\text{EM}}^2 A_2^2 - \kappa_{\text{EM}}^2 A_2 A_3 \\ -\kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) A_3 - \kappa_{\text{EM}}^2 A_3 A_1 - \kappa_{\text{EM}}^2 A_3 A_2 a^2(t) - \kappa_{\text{EM}}^2 A_3^2 \end{pmatrix}. \quad (49)$$

The same line element can also be written using $dx^0 = c dt$ as

$$ds^2 = - [(c - \kappa_{\text{EM}} \phi) dt + \kappa_{\text{EM}} A_i dx^i]^2 + a^2(t) \delta_{ij} dx^i dx^j \quad (50)$$

2.5 Inverse Metric

The inverse metric is obtained from

$$g^{\mu\nu} = \eta^{ab} E_a{}^\mu E_b{}^\nu$$

The components are explicitly

$$g^{00} = -\frac{1}{(1 - \kappa_{\text{EM}}\phi/c)^2} + \frac{\kappa_{\text{EM}}^2}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)^2} (A_1^2 + A_2^2 + A_3^2),$$

$$g^{01} = g^{10} = -\frac{\kappa_{\text{EM}}A_1}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)}, \quad (51)$$

$$g^{02} = g^{20} = -\frac{\kappa_{\text{EM}}A_2}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)}, \quad (52)$$

$$g^{03} = g^{30} = -\frac{\kappa_{\text{EM}}A_3}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)}, \quad (53)$$

and

$$g^{ij} = \frac{1}{a^2(t)}\delta^{ij} \quad (54)$$

In matrix form, the inverse metric is

$$g^{\mu\nu} = \begin{pmatrix} -\frac{1}{(1 - \kappa_{\text{EM}}\phi/c)^2} + \frac{\kappa_{\text{EM}}^2 (A_1^2 + A_2^2 + A_3^2)}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)^2} & -\frac{\kappa_{\text{EM}}A_1}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)} & -\frac{\kappa_{\text{EM}}A_2}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)} & -\frac{\kappa_{\text{EM}}A_3}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)} \\ -\frac{\kappa_{\text{EM}}A_1}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)} & \frac{1}{a^2(t)} & 0 & 0 \\ -\frac{\kappa_{\text{EM}}A_2}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)} & 0 & \frac{1}{a^2(t)} & 0 \\ -\frac{\kappa_{\text{EM}}A_3}{a^2(t)(1 - \kappa_{\text{EM}}\phi/c)} & 0 & 0 & \frac{1}{a^2(t)} \end{pmatrix} \quad (55)$$

2.6 Torsion Two-Forms

In teleparallel geometry, within the Weitzenböck gauge, the torsion two-form is defined as

$$T^a = de^a \quad (56)$$

In terms of components,

$$T^a_{\mu\nu} = \partial_\mu e^a_\nu - \partial_\nu e^a_\mu \quad (57)$$

Using this definition, the torsion two-forms can be classified separately for the temporal and spatial tetrad components.

2.6.1 Temporal Torsion Components

First, for the temporal tetrad Eq. 13,

$$T^0 = de^0 = d\left[\left(1 - \kappa_{\text{EM}}\frac{\phi}{c}\right)dx^0\right] + d(\kappa_{\text{EM}}A_i dx^i).$$

The first term is

$$d \left[\left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) dx^0 \right] = d \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) \wedge dx^0 = -\frac{\kappa_{\text{EM}}}{c} \partial_i \phi dx^i \wedge dx^0$$

Here, since $\partial_0 \phi dx^0 \wedge dx^0 = 0$, the temporal derivative contribution does not appear in this term.

The second term is

$$d (\kappa_{\text{EM}} A_i dx^i) = \kappa_{\text{EM}} \partial_0 A_i dx^0 \wedge dx^i + \frac{\kappa_{\text{EM}}}{2} (\partial_i A_j - \partial_j A_i) dx^i \wedge dx^j.$$

Thus, the temporal torsion two-form is

$$T^0 = \kappa_{\text{EM}} \left(\partial_0 A_i + \frac{1}{c} \partial_i \phi \right) dx^0 \wedge dx^i + \frac{\kappa_{\text{EM}}}{2} (\partial_i A_j - \partial_j A_i) dx^i \wedge dx^j.$$

Accordingly, the time-space components of temporal torsion are

$$T^0_{0i} = \kappa_{\text{EM}} \left(\partial_0 A_i + \frac{1}{c} \partial_i \phi \right), \quad (58)$$

and the space-space components are

$$T^0_{ij} = \kappa_{\text{EM}} (\partial_i A_j - \partial_j A_i) \quad (59)$$

Due to antisymmetry,

$$T^0_{i0} = -T^0_{0i} \quad (60)$$

Using $\partial_0 = (1/c)\partial_t$, the temporal time-space component can also be written as

$$T^0_{0i} = \frac{\kappa_{\text{EM}}}{c} (\partial_t A_i + \partial_i \phi) \quad (61)$$

2.6.2 Spatial Torsion Components

For the spatial tetrads,

$$e^i = a(t) dx^i \quad (62)$$

and therefore

$$T^i = de^i = d(a(t) dx^i) = da(t) \wedge dx^i.$$

Since $dx^0 = c dt$,

$$da(t) = \dot{a}(t) dt = \frac{\dot{a}(t)}{c} dx^0$$

is written. Thus, the spatial torsion two-form is obtained as

$$T^i = \frac{\dot{a}(t)}{c} dx^0 \wedge dx^i \quad (63)$$

Consequently, the time-space components of the spatial torsion are

$$T^i_{0j} = \frac{\dot{a}(t)}{c} \delta^i_j, \quad (64)$$

$$T^i_{j0} = -\frac{\dot{a}(t)}{c}\delta^i_j, \quad (65)$$

and the space-space components are

$$T^i_{jk} = 0 \quad (66)$$

3 Appendix-3: Geometric Formulation of the Electromagnetic Force, Representation of the Electric and Magnetic Fields, and Derivation of Maxwell's Equations

3.1 Temporal Torsion and the Electromagnetic Field Tensor

The electromagnetic field tensor is defined by the exterior derivative of the potential one-form:

$$F = dA. \quad (67)$$

In terms of components, this becomes

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (68)$$

In this work, the Maxwell-adapted temporal tetrad is taken as in Eq. 13. In the Weitzenböck gauge, the temporal torsion two-form is

$$T^0 = de^0 \quad (69)$$

Substituting the temporal tetrad gives

$$T^0 = d(dx^0 + \kappa_{\text{EM}}A) = d(dx^0) + \kappa_{\text{EM}}dA.$$

Since $d(dx^0) = 0$, one obtains

$$T^0 = \kappa_{\text{EM}}dA. \quad (70)$$

Since the electromagnetic field two-form is $F = dA$, this relation can be written in terms of components as

$$T^0_{\mu\nu} = \kappa_{\text{EM}}F_{\mu\nu} \quad (71)$$

or equivalently as

$$F_{\mu\nu} = \frac{1}{\kappa_{\text{EM}}}T^0_{\mu\nu} \quad (72)$$

This result shows that the electromagnetic field tensor can be read geometrically through the internal Lorentz component of temporal torsion.

3.2 Explicit Calculation of the Electromagnetic Field Two-Form

The electromagnetic potential one-form is given by Eq. 8. Since the electromagnetic field two-form is Eq. 67, one writes

$$dA = d\left(-\frac{\phi}{c}dx^0\right) + d(A_i dx^i) \quad (73)$$

- For the first term

$$d\left(-\frac{\phi}{c}dx^0\right) = -\frac{1}{c}d\phi \wedge dx^0. \quad (74)$$

Here

$$d\phi = \partial_0\phi dx^0 + \partial_i\phi dx^i \quad (75)$$

and therefore

$$-\frac{1}{c}d\phi \wedge dx^0 = -\frac{1}{c}(\partial_0\phi dx^0 + \partial_i\phi dx^i) \wedge dx^0 = -\frac{1}{c}\partial_i\phi dx^i \wedge dx^0 = \frac{1}{c}\partial_i\phi dx^0 \wedge dx^i.$$

- The second term is

$$d(A_i dx^i) = dA_i \wedge dx^i. \quad (76)$$

Here

$$dA_i = \partial_0 A_i dx^0 + \partial_j A_i dx^j \quad (77)$$

and therefore

$$d(A_i dx^i) = \partial_0 A_i dx^0 \wedge dx^i + \partial_j A_i dx^j \wedge dx^i. \quad (78)$$

The spatial term is written in antisymmetric form as

$$\partial_j A_i dx^j \wedge dx^i = \frac{1}{2}(\partial_i A_j - \partial_j A_i) dx^i \wedge dx^j \quad (79)$$

Thus,

$$F = dA = \left(\partial_0 A_i + \frac{1}{c}\partial_i\phi\right) dx^0 \wedge dx^i + \frac{1}{2}(\partial_i A_j - \partial_j A_i) dx^i \wedge dx^j.$$

Accordingly, the components of the electromagnetic field tensor are obtained as

$$F_{0i} = \partial_0 A_i + \frac{1}{c}\partial_i\phi \quad (80)$$

and

$$F_{ij} = \partial_i A_j - \partial_j A_i \quad (81)$$

Furthermore, since

$$\partial_0 = \frac{1}{c}\partial_t \quad (82)$$

one can also write

$$F_{0i} = \frac{1}{c}(\partial_t A_i + \partial_i\phi) \quad (83)$$

3.3 Obtaining the Electric and Magnetic Fields from Temporal Torsion

3.3.1 Expressing the Electric Field from Temporal Torsion

In SI units, the standard sign convention for the physical electric field is

$$E_i = -\partial_i \phi - \partial_t A_i \quad (84)$$

Therefore, the relation between the component F_{0i} above and the physical electric field, using Eq. 83, is

$$F_{0i} = \frac{1}{c} (\partial_t A_i + \partial_i \phi) = -\frac{1}{c} E_i \quad (85)$$

Moreover,

$$E_i = -c F_{0i} \quad (86)$$

Using the relation Eq. 72 between temporal torsion and the electromagnetic field tensor, one obtains

$$E_i = -\frac{c}{\kappa_{\text{EM}}} T^0_{0i} \quad (87)$$

3.3.2 Expressing the Magnetic Field from Temporal Torsion

The magnetic field is defined through the spatial electromagnetic field components as

$$B^k = \frac{1}{2} \epsilon^{kij} F_{ij} = \frac{1}{2} \epsilon^{kij} (\partial_i A_j - \partial_j A_i) \quad (88)$$

Using the temporal torsion relation, one obtains

$$B^k = \frac{1}{2\kappa_{\text{EM}}} \epsilon^{kij} T^0_{ij} \quad (89)$$

Thus, the electric field is encoded in the time-space component of temporal torsion, while the magnetic field is encoded in the space-space component of temporal torsion. These two fields are read as different components of the same temporal torsion two-form.

3.4 Homogeneous Maxwell Equations

When the electromagnetic field two-form is defined by Eq. 67, the homogeneous Maxwell equations arise not as an additional dynamical assumption, but as a geometric identity following from the nilpotency of the exterior derivative. Indeed,

$$dF = d(dA) = d^2 A = 0 \quad (90)$$

Thus,

$$dF = 0 \quad (91)$$

is the Bianchi identity of the electromagnetic field two-form.

In terms of components, this identity is written as

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0 \quad (92)$$

This expression gives the homogeneous Maxwell equations for appropriate choices of indices.

First, let only spatial indices be chosen:

$$\lambda = i, \quad \mu = j, \quad \nu = k. \quad (93)$$

This choice gives, inside Eq. (92),

$$\partial_i F_{jk} + \partial_j F_{ki} + \partial_k F_{ij} = 0 \quad (94)$$

Since the magnetic field is defined by Eq. 88, this identity expresses the absence of sources for the magnetic field:

$$\nabla \cdot \mathbf{B} = 0. \quad (95)$$

Second, let one temporal and two spatial indices be chosen:

$$\lambda = 0, \quad \mu = i, \quad \nu = j. \quad (96)$$

In this case, the Bianchi identity becomes

$$\partial_0 F_{ij} + \partial_i F_{j0} + \partial_j F_{0i} = 0 \quad (97)$$

Due to antisymmetry,

$$F_{j0} = -F_{0j} \quad (98)$$

and hence

$$\partial_0 F_{ij} - \partial_i F_{0j} + \partial_j F_{0i} = 0 \quad (99)$$

When the physical electric field and the magnetic field are defined as in Eq. 86 and Eq. 88, Eq. (99) corresponds to Faraday's law:

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0. \quad (100)$$

Here, the relation Eq. 82 has been used. As a result, the homogeneous Maxwell equations are obtained in the form

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0 \quad (101)$$

Thus, it is seen that, in this formalism as well, the homogeneous Maxwell equations follow directly from the definition $F = dA$ and the geometric identity $d^2 = 0$.

4 Appendix-4: Total Electromagnetic Action and Derivation of the Field Equations

If the electromagnetic action is written, it will be taken as

$$S_{\text{EM}} = \frac{1}{c} \int d^4x e \left[-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - J^\mu A_\mu \right] \quad (102)$$

Here $e = \det(e^a_\mu)$ is the tetrad determinant.

4.1 The Relation Between A_μ and the Temporal Tetrad

The electromagnetic potential A_μ is contained within the temporal tetrad, and it must be discussed whether a direct relation can be established between the variation of the tetrad and the variation of A_μ . The temporal tetrad component is given by Eq. (13) as

$$e^0_\mu = u_\mu + \kappa_{\text{EM}} A_\mu \quad (103)$$

Taking the variation of this expression gives

$$\delta e^0_\mu = \delta u_\mu + \kappa_{\text{EM}} \delta A_\mu = 0 + \kappa_{\text{EM}} \delta A_\mu.$$

Thus,

$$\delta e^0_\mu = \kappa_{\text{EM}} \delta A_\mu \quad (104)$$

and equivalently

$$\delta A_\mu = \frac{1}{\kappa_{\text{EM}}} \delta e^0_\mu \quad (105)$$

are obtained.

This relation is written in functional-derivative form as

$$\frac{\delta A_\mu}{\delta e^a_\nu} = \frac{1}{\kappa_{\text{EM}}} \delta^0_a \delta^\nu_\mu \quad (106)$$

This result is of critical importance. This is because the variation with respect to A_μ is the projection of the tetrad variation only onto the temporal internal Lorentz component. Therefore, the Maxwell operator appears directly in the tetrad variation not in all internal Lorentz components, but only in the temporal projection selected by δ^0_a .

4.2 Basic Variations to Be Used

4.2.1 Variation of the Tetrad Determinant

The variation of the tetrad determinant is standardly

$$\delta e = e E_a^\mu \delta e^a_\mu \quad (107)$$

Here E_a^μ is the inverse tetrad. This expression follows directly from the determinant identity:

$$\delta \det M = \det M \operatorname{Tr}(M^{-1} \delta M). \quad (108)$$

Taking $M^a_\mu = e^a_\mu$ for the tetrad matrix gives

$$\delta e = e E_a^\mu \delta e^a_\mu \quad (109)$$

4.2.2 Metric Variation

The metric is defined through the tetrads as

$$g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu \quad (110)$$

Its variation is

$$\delta g_{\mu\nu} = \eta_{ab} (\delta e^a_\mu e^b_\nu + e^a_\mu \delta e^b_\nu) \quad (111)$$

For the inverse metric, the identity

$$g^{\mu\alpha} g_{\alpha\nu} = \delta^\mu_\nu \quad (112)$$

is used. Since its variation gives

$$\delta g^{\mu\alpha} g_{\alpha\nu} + g^{\mu\alpha} \delta g_{\alpha\nu} = 0 \quad (113)$$

one obtains

$$\delta g^{\mu\nu} = -g^{\mu\rho} g^{\nu\sigma} \delta g_{\rho\sigma} \quad (114)$$

Substituting Eq. (111) into this expression gives

$$\delta g^{\mu\nu} = -g^{\mu\rho} g^{\nu\sigma} \eta_{ab} (\delta e^a_\rho e^b_\sigma + e^a_\rho \delta e^b_\sigma) = -(E_a^\mu g^{\nu\rho} + E_a^\nu g^{\mu\rho}) \delta e^a_\rho.$$

4.2.3 Variation of the Field Tensor

One of the basic structures used in this work is the field tensor, which enters the Lagrangian in the form $F_{\mu\nu} F^{\mu\nu}$. In this case, the variation of the field tensor also becomes important. This will be shown in this part. Since the electromagnetic field tensor is Eq. 68, its variation is

$$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu \quad (115)$$

In covariant form, since the symmetric part of the connection cancels in the antisymmetric difference, it can be written as

$$\delta F_{\mu\nu} = \nabla_\mu \delta A_\nu - \nabla_\nu \delta A_\mu \quad (116)$$

4.3 Variation of the Kinetic Term

In this section, the variation of the kinetic term will be performed, and the point to be noted is the following. At the beginning of the section, it was explained that the variation would be performed with respect to the structure e_μ ; however, through Eq. 105, there is a direct connection between the variation of A_μ and the variation of e_μ . Since the Lagrangian is taken directly as the classical electromagnetic Lagrangian, the variation with respect to A_μ will be continued first, and then the variation with respect to the tetrad will be obtained through the structure Eq. 105. For the classical Maxwell Lagrangian, when the SI unit system is used, the kinetic action is

$$S_{\text{kin}} = -\frac{1}{4\mu_0 c} \int d^4x e F_{\mu\nu} F^{\mu\nu} \quad (117)$$

First, the part coming directly from the variation of $F_{\mu\nu}$ is calculated. In this step, determinant and metric variations are temporarily kept separate:

$$\delta S_{\text{kin}}|_{\delta A} = -\frac{1}{4\mu_0 c} \int d^4x e \delta (F_{\mu\nu} F^{\mu\nu}) = -\frac{1}{4\mu_0 c} \int d^4x e 2F^{\mu\nu} \delta F_{\mu\nu} = -\frac{1}{2\mu_0 c} \int d^4x e F^{\mu\nu} (\nabla_\mu \delta A_\nu - \nabla_\nu \delta A_\mu).$$

In the second term, the index exchange $\mu \leftrightarrow \nu$ is performed:

$$F^{\mu\nu} \nabla_\nu \delta A_\mu \rightarrow F^{\nu\mu} \nabla_\mu \delta A_\nu = -F^{\mu\nu} \nabla_\mu \delta A_\nu.$$

Therefore,

$$F^{\mu\nu} (\nabla_\mu \delta A_\nu - \nabla_\nu \delta A_\mu) = F^{\mu\nu} \nabla_\mu \delta A_\nu + F^{\mu\nu} \nabla_\mu \delta A_\nu = 2F^{\mu\nu} \nabla_\mu \delta A_\nu.$$

For this reason,

$$\delta S_{\text{kin}}|_{\delta A} = -\frac{1}{\mu_0 c} \int d^4x e F^{\mu\nu} \nabla_\mu \delta A_\nu. \quad (118)$$

Partial integration is applied:

$$\int d^4x e F^{\mu\nu} \nabla_\mu \delta A_\nu = \int d^4x \nabla_\mu (e F^{\mu\nu} \delta A_\nu) - \int d^4x e (\nabla_\mu F^{\mu\nu}) \delta A_\nu. \quad (119)$$

If the boundary term is neglected, then

$$\delta S_{\text{kin}}|_{\delta A} = \frac{1}{\mu_0 c} \int d^4x e (\nabla_\mu F^{\mu\nu}) \delta A_\nu. \quad (120)$$

Using Eq. (105), one obtains

$$\delta S_{\text{kin}}|_{\delta A} = \frac{1}{\mu_0 c} \int d^4x e (\nabla_\mu F^{\mu\nu}) \frac{1}{\kappa_{\text{EM}}} \delta^0_a \delta e^a_\nu = \frac{1}{c} \int d^4x e \frac{1}{\kappa_{\text{EM}}} \delta^0_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} \right) \delta e^a_\nu.$$

4.3.1 Kinetic Contribution from Determinant and Metric Dependence

For classical electromagnetic theory, e denotes the tetrad determinant. In standard Maxwell theory, since A_μ and the tetrad are treated as independent variables, e and $g^{\mu\nu}$ are taken to be fixed in the variation with respect to A_μ . Therefore, in the classical case, determinant and metric dependence do not directly contribute to the A_μ variation.

However, in this work A_μ is treated as a structure embedded in the temporal tetrad; therefore, the variation with respect to A_μ also induces a variation on the tetrad, determinant, and metric. For this reason, the additional contribution of the kinetic term arising from determinant and metric dependence must also be taken into account separately.

Since the kinetic action is

$$S_{\text{kin}} = \frac{1}{c} \int d^4x e \left(-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \right) \quad (121)$$

the variation coming only from e and $g^{\mu\nu}$ dependence is written as

$$\delta S_{\text{kin}}|_{e,g} = \frac{1}{c} \int d^4x \delta \left[e \left(-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \right) \right]_{e,g} \quad (122)$$

Here $F_{\mu\nu}$ is held fixed, and only the determinant and the metric used in raising indices are varied.

First,

$$F_{\mu\nu} F^{\mu\nu} = F_{\mu\nu} F_{\alpha\beta} g^{\mu\alpha} g^{\nu\beta} \quad (123)$$

is written. When $F_{\mu\nu}$ is held fixed,

$$\begin{aligned} \delta (F_{\mu\nu} F^{\mu\nu})_g &= F_{\mu\nu} F_{\alpha\beta} \delta (g^{\mu\alpha} g^{\nu\beta}) \\ &= F_{\mu\nu} F_{\alpha\beta} (\delta g^{\mu\alpha} g^{\nu\beta} + g^{\mu\alpha} \delta g^{\nu\beta}) \\ &= 2F_{\mu\alpha} F_\nu{}^\alpha \delta g^{\mu\nu}. \end{aligned} \quad (124)$$

Thus,

$$\begin{aligned} \delta \left[e \left(-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \right) \right]_{e,g} &= -\frac{1}{4\mu_0} \delta e F_{\alpha\beta} F^{\alpha\beta} \\ &\quad - \frac{e}{4\mu_0} \delta (F_{\mu\nu} F^{\mu\nu})_g \\ &= -\frac{1}{4\mu_0} \delta e F_{\alpha\beta} F^{\alpha\beta} - \frac{e}{2\mu_0} F_{\mu\alpha} F_\nu{}^\alpha \delta g^{\mu\nu}. \end{aligned} \quad (125)$$

Now the determinant and inverse-metric variations are written in terms of the tetrad variation:

$$\delta e = e E_a{}^\nu \delta e^a{}_\nu, \quad (126)$$

and

$$\delta g^{\mu\nu} = - (g^{\lambda\nu} E_a{}^\mu + g^{\lambda\mu} E_a{}^\nu) \delta e^a{}_\lambda. \quad (127)$$

Substituting these expressions into Eq. (125) gives

$$\begin{aligned} \delta \left[e \left(-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \right) \right]_{e,g} &= -\frac{e}{4\mu_0} E_a^\lambda F_{\alpha\beta} F^{\alpha\beta} \delta e^a{}_\lambda \\ &\quad + \frac{e}{2\mu_0} F_{\mu\alpha} F_\nu{}^\alpha (g^{\lambda\nu} E_a{}^\mu + g^{\lambda\mu} E_a{}^\nu) \delta e^a{}_\lambda. \end{aligned} \quad (128)$$

Here the symmetry

$$F_{\mu\alpha} F_\nu{}^\alpha = F_{\nu\alpha} F_\mu{}^\alpha \quad (129)$$

is used. Thus, the second term becomes

$$\begin{aligned} \frac{e}{2\mu_0} F_{\mu\alpha} F_\nu{}^\alpha (g^{\lambda\nu} E_a{}^\mu + g^{\lambda\mu} E_a{}^\nu) \delta e^a{}_\lambda \\ = \frac{e}{\mu_0} E_a{}^\rho F_{\rho\alpha} F^{\lambda\alpha} \delta e^a{}_\lambda \end{aligned} \quad (130)$$

Therefore, the total expression is

$$\delta \left[e \left(-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \right) \right]_{e,g} = e E_a{}^\rho \left[\frac{1}{\mu_0} F_{\rho\alpha} F^{\lambda\alpha} - \frac{1}{4\mu_0} \delta_\rho{}^\lambda F_{\alpha\beta} F^{\alpha\beta} \right] \delta e^a{}_\lambda. \quad (131)$$

When the mixed-index form of the Maxwell energy-momentum tensor is defined as

$$\Theta^{\text{EM}}{}_\rho{}^\lambda = \frac{1}{\mu_0} F_{\rho\alpha} F^{\lambda\alpha} - \frac{1}{4\mu_0} \delta_\rho{}^\lambda F_{\alpha\beta} F^{\alpha\beta} \quad (132)$$

the kinetic contribution arising from determinant and metric dependence is obtained as

$$\delta S_{\text{kin}}|_{e,g} = \frac{1}{c} \int d^4x e E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu \delta e^a{}_\nu \quad (133)$$

4.3.2 Total Variation of the Kinetic Term

In the previous two subsections, the kinetic term was varied with respect to the tetrad, and the calculation for the field equation was performed. In this case, it was seen that two basic contributions arise from the variation of the kinetic term, and in this section these two contributions will be combined.

Combining Eq. 121 and Eq. 133, the part arising from the variation of the kinetic term is obtained as

$$\delta S_{\text{EM}} = \frac{1}{c} \int d^4x e \left[\frac{1}{\kappa_{\text{EM}}} \delta^0{}_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} \right) + E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu \right] \delta e^a{}_\nu.$$

However, the Maxwell equation does not consist only of the kinetic term; in order to obtain the full field equation, the source term must also be examined.

4.4 Variation of the Source Term

In the classical Maxwell Lagrangian, in addition to the kinetic term, there is also an interaction term with the source. This term is written as

$$S_{\text{int}} = -\frac{1}{c} \int d^4x e J^\mu A_\mu \quad (134)$$

Here J^μ is regarded as a source current externally inserted into the electromagnetic field. Therefore, in this variation J^μ is not a dynamical field and is not varied independently.

The general variation of the interaction action can first be written as

$$\delta S_{\text{int}} = -\frac{1}{c} \int d^4x [\delta e J^\mu A_\mu + e \delta J^\mu A_\mu + e J^\mu \delta A_\mu] \quad (135)$$

However, since the source current is treated here as an externally given fixed source,

$$\delta J^\mu = 0 \quad (136)$$

is taken. Therefore, the intermediate term

$$e \delta J^\mu A_\mu = 0 \quad (137)$$

vanishes, and the variation reduces to

$$\delta S_{\text{int}} = -\frac{1}{c} \int d^4x [\delta e J^\mu A_\mu + e J^\mu \delta A_\mu] \quad (138)$$

The reason for taking $\delta J^\mu = 0$ is that J^μ is treated in this theory not as a dynamical variable derived from the field equations, but as an externally given fixed source. In other words, in this variation J^μ is accepted not as a function of the electromagnetic field or of the tetrad, but as a predetermined external current distribution. If the source current were defined as a dynamical quantity derived from matter fields, that is, if it depended on fields such as $J^\mu = J^\mu(\psi, \bar{\psi})$ or $J^\mu = J^\mu(\phi, \partial_\mu \phi)$, then the term δJ^μ would also have to be included in the variation.

4.4.1 Direct A_μ Variation of the Source Term

First, the direct A_μ variation arising from the source term is considered. In standard Maxwell theory, A_μ is the fundamental dynamical variable, and e and J^μ are held fixed in this variation. In this case, only the term $e J^\mu \delta A_\mu$ is considered in Eq. (138):

$$\delta S_{\text{int}}|_A = -\frac{1}{c} \int d^4x e J^\mu \delta A_\mu. \quad (139)$$

However, in this work the electromagnetic potential A_μ is treated as a structure embedded in the temporal tetrad component. Therefore, the variation with

respect to A_μ can be expressed directly in terms of the tetrad variation. Using Eq. (105),

$$\delta A_\mu = \frac{1}{\kappa_{\text{EM}}} \delta^0_a \delta e^a_\mu \quad (140)$$

is written. Substituting this expression into the direct A_μ variation of the source term gives

$$\begin{aligned} e J^\mu \delta A_\mu &= e J^\mu \left(\frac{1}{\kappa_{\text{EM}}} \delta^0_a \delta e^a_\mu \right) \\ &= e \frac{1}{\kappa_{\text{EM}}} \delta^0_a J^\mu \delta e^a_\mu \\ &= e \frac{1}{\kappa_{\text{EM}}} \delta^0_a J^\nu \delta e^a_\nu. \end{aligned} \quad (141)$$

In the last line, the dummy index μ has been renamed as ν . This renaming is performed in order later to collect all contributions under the common variation δe^a_ν . Thus, the source contribution arising from the direct A_μ variation is obtained as

$$\delta S_{\text{int}}|_A = -\frac{1}{c} \int d^4x e \frac{1}{\kappa_{\text{EM}}} \delta^0_a J^\nu \delta e^a_\nu \quad (142)$$

4.4.2 Source Contribution from Determinant Dependence

In standard Maxwell theory, A_μ and the tetrad are regarded as independent variables. Therefore, when the variation with respect to A_μ is performed, the tetrad determinant e is held fixed, and no determinant variation arises from the source term. In this case, the variation of the source interaction with respect to A_μ consists only of the term $e J^\mu \delta A_\mu$.

In this work, however, since A_μ is embedded in the temporal tetrad component, the situation is different. A change in A_μ corresponds to a change in the tetrad components. When the tetrad changes, the determinant e also changes. Therefore, in the source term, not only the direct A_μ variation but also the δe contribution arising from the tetrad dependence of the determinant must be taken into account.

The part arising from the determinant variation in Eq. (138) is

$$\delta S_{\text{int}}|_e = -\frac{1}{c} \int d^4x \delta e J^\mu A_\mu \quad (143)$$

Since the tetrad variation of the determinant is

$$\delta e = e E_a{}^\nu \delta e^a_\nu \quad (144)$$

this expression is substituted into the determinant contribution of the source term. Thus,

$$\begin{aligned} \delta e J^\mu A_\mu &= (e E_a{}^\nu \delta e^a_\nu) J^\mu A_\mu \\ &= e E_a{}^\nu (J^\mu A_\mu) \delta e^a_\nu \end{aligned} \quad (145)$$

is obtained. Here $J^\mu A_\mu$ is a scalar contraction and is left outside as a coefficient with respect to the tetrad variation. Therefore, the source contribution arising from determinant dependence is written as

$$\delta S_{\text{int}}|_e = -\frac{1}{c} \int d^4x e E_a{}^\nu (J^\mu A_\mu) \delta e^a{}_\nu \quad (146)$$

4.4.3 Total Variation of the Source Term

The total variation of the source term is written as the sum of two separate contributions. The first of these is the contribution arising from the tetrad dependence of the determinant. The second is the direct A_μ variation that arises because A_μ is embedded in the temporal tetrad component. Therefore,

$$\delta S_{\text{int}} = \delta S_{\text{int}}|_e + \delta S_{\text{int}}|_A \quad (147)$$

is written.

Using Eq. (146) and Eq. (142), one obtains

$$\begin{aligned} \delta S_{\text{int}} = & -\frac{1}{c} \int d^4x e E_a{}^\nu (J^\mu A_\mu) \delta e^a{}_\nu \\ & - \frac{1}{c} \int d^4x e \frac{1}{\kappa_{\text{EM}}} \delta^0{}_a J^\nu \delta e^a{}_\nu \end{aligned} \quad (148)$$

Since both terms contain the common variation $\delta e^a{}_\nu$, the expressions can be collected under the same integral:

$$\delta S_{\text{int}} = \frac{1}{c} \int d^4x e \left[-E_a{}^\nu (J^\mu A_\mu) - \frac{1}{\kappa_{\text{EM}}} \delta^0{}_a J^\nu \right] \delta e^a{}_\nu. \quad (149)$$

Thus, the total contribution of the source term with respect to the tetrad variation is obtained explicitly.

4.5 Total Variation

The total variation is the sum of three parts:

$$\delta S_{\text{EM}} = \delta S_{\text{kin}}|_{\delta A} + \delta S_{\text{kin}}|_{e,g} + \delta S_{\text{int}}. \quad (150)$$

Writing Eq. (121), Eq. (133), and Eq. (149) together gives

$$\delta S_{\text{EM}} = \frac{1}{c} \int d^4x e \left[\frac{1}{\kappa_{\text{EM}}} \delta^0{}_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} \right) + E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu - E_a{}^\nu (J^\mu A_\mu) - \frac{1}{\kappa_{\text{EM}}} \delta^0{}_a J^\nu \right] \delta e^a{}_\nu.$$

Collecting similar terms gives

$$\delta S_{\text{EM}} = \frac{1}{c} \int d^4x e \left[\frac{1}{\kappa_{\text{EM}}} \delta^0{}_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} - J^\nu \right) + E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu - E_a{}^\nu (J^\mu A_\mu) \right] \delta e^a{}_\nu.$$

For the stationarity of the action and the derivation of the field equations,

$$\delta S_{\text{EM}} = 0 \quad (151)$$

must hold. Since $\delta e^a{}_\nu$ is arbitrary, the expression inside the square brackets is set equal to zero:

$$\frac{1}{\kappa_{\text{EM}}} \delta^0{}_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} - J^\nu \right) + E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu - E_a{}^\nu (J^\mu A_\mu) = 0. \quad (152)$$

Multiplying both sides by κ_{EM} , one obtains

$$\delta^0{}_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} - J^\nu \right) + \kappa_{\text{EM}} [E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu - E_a{}^\nu (J^\mu A_\mu)] = 0 \quad (153)$$

Here, the field equations are determined according to the value taken by $\delta^0{}_a$. This expression is the Kronecker delta and is defined as

$$\delta^0{}_a = \begin{cases} 1, & a = 0, \\ 0, & a = i \end{cases} \quad (154)$$

Here $a = 0$ denotes the temporal tetrad component, while $a = i$ denotes the spatial tetrad components. Therefore, the field equations must be examined separately for the temporal tetrad projection and the spatial tetrad projection. Below, the cases $\delta^0{}_i$ and $\delta^0{}_0$ will be considered respectively.

4.6 Spatial Projections and the Geometric Backreaction Condition

In the previous part, the temporal internal Lorentz projection, that is, the case $a = 0$, was examined, and it was shown that this projection gives Maxwell equations in the appropriate weak-field limit. In this section, the spatial internal Lorentz projection, that is,

$$a = i \quad (155)$$

will be considered.

This projection is important for the physical interpretation of the theory. In the Maxwell-adapted tetrad structure, the electromagnetic potential A_μ is treated as a structure embedded in the temporal tetrad component. By contrast, the spatial tetrad components do not directly represent the electromagnetic potential, but the geometric sector. Therefore, the $a = i$ projection does not directly produce Maxwell dynamics. Instead, it encodes the backreaction of the energy-momentum content of the electromagnetic field and the source-potential interaction on the spatial tetrad geometry.

The total tetrad field equation is given by Eq. (153). The Kronecker delta is Eq. 126, and therefore for the spatial projection

$$\delta^0{}_i = 0 \quad (156)$$

holds.

In this case, the Maxwell operator and the direct source term associated with the temporal projection do not contribute in the spatial projection:

$$\delta^0_i \left[\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} - J^\nu \right] = 0. \quad (157)$$

Thus, under the $a = i$ projection, the field equation remains as

$$\kappa_{\text{EM}} [E_i^\rho \Theta_{\rho}^{\text{EM} \nu} - E_i^\nu (J^\mu A_\mu)] = 0 \quad (158)$$

Assuming $\kappa_{\text{EM}} \neq 0$, this expression can be written as

$$E_i^\rho \Theta_{\rho}^{\text{EM} \nu} - E_i^\nu (J^\mu A_\mu) = 0 \quad (159)$$

Eq. (159) is not directly a Maxwell equation. This is because the Maxwell operator

$$\nabla_\mu F^{\mu\nu} \quad (160)$$

does not appear in this equation. Instead, the equation expresses the balance between the electromagnetic energy-momentum tensor and the spatial tetrad projections of the source-potential interaction. Therefore, Eq. (159) should be interpreted as the backreaction condition of the electromagnetic field on the spatial tetrad geometry.

To see this interpretation more clearly, Eq. (159) can be rewritten as

$$E_i^\rho \Theta_{\rho}^{\text{EM} \nu} = E_i^\nu (J^\mu A_\mu). \quad (161)$$

The left-hand side gives the projection of the electromagnetic energy-momentum tensor along the spatial tetrad direction. The right-hand side represents the contribution formed by the source-potential interaction scalar along the same spatial tetrad direction. Therefore, the equation can be read as the balance condition of the effective backreaction transferred by the electromagnetic sector to the spatial geometry.

However, although this equation may at first appear to be an independent constraint equation, it should not be interpreted by itself. The reason is that the $a = i$ projection corresponds to the spatial tetrad components rather than to the temporal electromagnetic potential component. Spatial tetrad components are naturally related to the geometric or gravitational sector of the theory. Therefore, it is not appropriate to interpret Eq. (159) as an independent physical constraint in the general case by considering only the electromagnetic action.

In the full theory, the total action does not consist only of the electromagnetic action. In general, one should consider a total action of the form

$$S_{\text{tot}} = S_{\text{grav}} + S_{\text{EM}} + S_{\text{int}} \quad (162)$$

In this case, in the spatial tetrad projection, the field equation consists not only of electromagnetic contributions, but also of contributions from the gravitational or geometric action:

$$\left. \frac{\delta S_{\text{grav}}}{\delta e^i_\nu} \right|_{\text{spatial}} + \left. \frac{\delta S_{\text{EM}}}{\delta e^i_\nu} \right|_{\text{spatial}} + \left. \frac{\delta S_{\text{int}}}{\delta e^i_\nu} \right|_{\text{spatial}} = 0. \quad (163)$$

Therefore, Eq. (159) must be evaluated together with the spatial tetrad variations arising from the gravitational action in the full theory. In this context, the spatial projection becomes a geometric balance condition showing how the energy-momentum content of the electromagnetic field backreacts on the spatial geometry.

Consequently, while the temporal projection produces the dynamical Maxwell equations of the electromagnetic field, the spatial projection determines the way in which the electromagnetic sector couples to the geometric sector and the back-reaction effect on the spatial tetrad geometry. However, since the gravitational sector and the field equations derived from this sector are outside the scope of this work, it is correct to state only that it should not be interpreted as a constraint equation.

4.7 Temporal Projection

For the temporal internal Lorentz component, one takes $a = 0$. In this case,

$$\delta^0_0 = 1 \quad (164)$$

and therefore Eq. (153) takes the form

$$\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} - J^\nu + \kappa_{\text{EM}} [E_0^\rho \Theta^{\text{EM}}_\rho{}^\nu - E_0{}^\nu (J^\mu A_\mu)] = 0 \quad (165)$$

When this equation is arranged in Maxwell-like form, one obtains

$$\nabla_\mu F^{\mu\nu} = \mu_0 J^\nu - \mu_0 \kappa_{\text{EM}} E_0^\rho \Theta^{\text{EM}}_\rho{}^\nu + \mu_0 \kappa_{\text{EM}} E_0{}^\nu (J^\mu A_\mu) \quad (166)$$

This expression shows that, in the temporal tetrad projection, the sourced Maxwell equation emerges together with geometric backreaction terms. The first term on the right-hand side is the classical source term. The other two terms are the geometric contributions arising from the tetrad determinant and metric dependence. In this part, only the field equation under this condition has been given; in the next section, these field equations will be examined in greater detail.

5 Appendix-5: On the Consistency of the Tetrad with the Field Equations

In a tetrad-based theory, the physical meaning of a proposed *ansatz* is not determined only by its producing a reasonable metric or torsion. The real test is whether this structure is compatible with the field equations arising from the action and whether it reproduces the known classical equations in the appropriate limits. Therefore, the fundamental question to be examined here is whether the Maxwell-adapted tetrad *ansatz* reproduces classical Maxwell electrostatic theory and limits such as the Poisson limit in the weak-field and

low-energy-density regime. In this section, the condition δ_0^0 will be examined, and various tests will be carried out in order to test the consistency of the obtained field equations with the chosen tetrad ansatz Eq. 13.

5.1 Derivation of the Classical Sourced Maxwell Equation

In the previous part, it was shown how the Poisson limit is obtained through the electrostatic $\nu = 0$ component. However, the same field equation can also be used to obtain the four-dimensional classical sourced Maxwell equation directly, without reducing it only to the electrostatic limit. The aim of this part is to show under which conditions the additional geometric terms arising from the Maxwell-adapted tetrad structure can be neglected, and to demonstrate explicitly that the classical sourced Maxwell equation is recovered in this limit.

As the starting point, the Maxwell-type field equation obtained from the temporal projection is taken:

$$\nabla_\mu F^{\mu\nu} = \mu_0 J^\nu + \mu_0 \kappa_{\text{EM}} E_0^\nu (J^\mu A_\mu) + \mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{\nu\rho} e^0_\rho. \quad (167)$$

In this equation, the left-hand side carries the standard Maxwell operator. The first term on the right-hand side is the classical source term. The second and third terms are the geometric backreaction contributions arising from the Maxwell-adapted tetrad ansatz.

It is possible to collect these two additional contributions under a single effective geometric source term. For this purpose, the definition

$$\mathcal{J}_{\text{geom}}^\nu = \kappa_{\text{EM}} E_0^\nu (J^\mu A_\mu) + \kappa_{\text{EM}} \Theta_{\text{EM}}^{\nu\rho} e^0_\rho \quad (168)$$

is made. With this definition, Eq. (167) is written as

$$\nabla_\mu F^{\mu\nu} = \mu_0 (J^\nu + \mathcal{J}_{\text{geom}}^\nu) \quad (169)$$

Here, $\mathcal{J}_{\text{geom}}^\nu$ is not a source arising from a real matter current, but an effective geometric backreaction current that appears due to the A_μ -dependence of the tetrad.

The condition required to obtain the classical Maxwell limit is that this effective geometric source remain small compared with the physical current. That is,

$$|\mathcal{J}_{\text{geom}}^\nu| \ll |J^\nu| \quad (170)$$

must hold. In this appendix, this will be tested numerically and analytically. For this purpose, Eq. 168 must be examined.

The point that should be emphasized here is that the geometric backreaction terms do not depend only on A_μ , but also appear together with the coefficient κ_{EM} . Numerically, κ_{EM} is a very small constant. The weak-backreaction regime is the regime in which the part of the tetrad depending on the electromagnetic potential is small. This limit can generally be expressed symbolically as

$$\left| \frac{\kappa_{\text{EM}} A}{1} \right| \ll 1 \quad (171)$$

In the electrostatic case, this condition takes the more concrete form

$$\left| \kappa_{\text{EM}} \frac{\Phi}{c} \right| \ll 1 \quad (172)$$

In addition, in the weak-field limit, the A_μ -dependence contained in the tetrad and inverse tetrad structure also remains a small correction.

$$E_0^\nu = \delta_0^\nu + O(\kappa_{\text{EM}}), \quad e^0_\rho = \delta^0_\rho + O(\kappa_{\text{EM}}). \quad (173)$$

Thus, the additional geometric contributions contain both the small coefficient κ_{EM} and the A_μ -dependence that is small in the weak-field regime. Therefore, these contributions are further suppressed relative to the classical Maxwell source term.

In this case, the backreaction current appearing in Eq. (168) is of order

$$\mathcal{J}_{\text{geom}}^\nu = O(\kappa_{\text{EM}}) \quad (174)$$

Therefore, Eq. (169) becomes

$$\nabla_\mu F^{\mu\nu} = \mu_0 J^\nu + O(\kappa_{\text{EM}}) \quad (175)$$

Here, the $O(\kappa_{\text{EM}})$ terms represent the geometric corrections arising from the source-potential contraction and from the projection of the electromagnetic energy-momentum tensor onto the temporal tetrad. For this reason, in the regime where the magnitude of $\kappa_{\text{EM}} A_\mu$ is small, the field equation reduces at leading order to the classical sourced Maxwell equation:

$$\nabla_\mu F^{\mu\nu} = \mu_0 J^\nu \quad (176)$$

To clarify this expression, a numerical calculation will be given in the next subsection.

5.1.1 An Example Numerical Calculation for the Maxwell Limit

In this section, a simple electrostatic order-of-magnitude estimate will be made in order to see the size of this suppression. The model used is based on the assumption that the charge of an electron is uniformly distributed over a sphere of radius equal to the Bohr radius a_0 . This model does not represent the real quantum-mechanical charge distribution; it is only a simple numerical test used for an order-of-magnitude comparison.

The numerical values used are

$$e_{\text{ch}} = 1.60218 \times 10^{-19} \text{ C}, \quad a_0 = 5.29177 \times 10^{-11} \text{ m} \quad (177)$$

The volume of the sphere is

$$\begin{aligned} V_{a_0} &= \frac{4\pi}{3} a_0^3 \\ &= \frac{4\pi}{3} (5.29177 \times 10^{-11} \text{ m})^3 \\ &\simeq 6.21 \times 10^{-31} \text{ m}^3 \end{aligned} \quad (178)$$

Thus, the approximate charge density is found to be

$$\begin{aligned}\rho &= \frac{e_{\text{ch}}}{V_{a_0}} \\ &= \frac{1.60218 \times 10^{-19} \text{ C}}{6.21 \times 10^{-31} \text{ m}^3} \\ &\simeq 2.58 \times 10^{11} \text{ C m}^{-3}\end{aligned}\tag{179}$$

At the Bohr radius, the magnitude of the electrostatic potential is

$$\begin{aligned}\phi(a_0) &= \frac{1}{4\pi\epsilon_0} \frac{e_{\text{ch}}}{a_0} \\ &\simeq 27.2 \text{ V}\end{aligned}\tag{180}$$

For the spatial gradient of the electrostatic potential, one obtains at the order of magnitude

$$\begin{aligned}|\nabla\phi| &\sim \frac{\phi(a_0)}{a_0} \\ &= \frac{27.2 \text{ V}}{5.29177 \times 10^{-11} \text{ m}} \\ &\simeq 5.14 \times 10^{11} \text{ V m}^{-1}\end{aligned}\tag{181}$$

In this section, the weak-backreaction limit will be used. This limit is expressed by the condition

$$\left| \kappa_{\text{EM}} \frac{\phi}{c} \right| \ll 1\tag{182}$$

At the Bohr scale, this dimensionless parameter is found to be

$$\begin{aligned}\left| \kappa_{\text{EM}} \frac{\phi}{c} \right| &\simeq \frac{(2.87 \times 10^{-19} \text{ m/(Vs)}) (27.2 \text{ V})}{2.998 \times 10^8 \text{ m s}^{-1}} \\ &\simeq 2.61 \times 10^{-26}\end{aligned}\tag{183}$$

Therefore, in this example, the approximations

$$E_0^0 \simeq 1, \quad e_0^0 \simeq 1\tag{184}$$

are satisfied extremely well. This result shows that geometric backreaction effects remain very small compared with the classical Maxwell terms in the weak-field and low-energy-density regime.

5.1.2 Classical Source Term

First, the magnitude of the classical source term is calculated. Since $J^0 = c\rho$ for the $\nu = 0$ component,

$$\begin{aligned}|\mu_0 J^0| &= \mu_0 c \rho \\ &= (1.2566 \times 10^{-6} \text{ N A}^{-2}) (2.998 \times 10^8 \text{ m s}^{-1}) (2.58 \times 10^{11} \text{ C m}^{-3}) \\ &\simeq 9.72 \times 10^{13} \text{ V s m}^{-3}.\end{aligned}\tag{185}$$

This quantity is the dominant classical Maxwell source term to be used for comparison in the field equation.

5.1.3 Source-Potential Backreaction Term

Now the first geometric backreaction contribution in Eq. (167) is considered. In the electrostatic case, the source-potential contraction is

$$J^\mu A_\mu = -\rho\phi \quad (186)$$

Here, the sign depends on the metric and potential convention used. Since only a magnitude comparison will be made in this section, the absolute value will be considered.

In the weak-backreaction limit, since $E_0^0 \simeq 1$, the magnitude of the source-potential backreaction term is

$$\begin{aligned} |\mu_0 \kappa_{\text{EM}} E_0^0 (J^\mu A_\mu)| &\simeq \mu_0 \kappa_{\text{EM}} \rho \phi \\ &= (1.2566 \times 10^{-6}) (2.87 \times 10^{-19}) (2.58 \times 10^{11}) \quad (27.2) \\ &\simeq 2.54 \times 10^{-12} \text{ V s m}^{-3}. \end{aligned} \quad (187)$$

The ratio of this term to the classical Maxwell source is

$$\begin{aligned} \frac{\mu_0 \kappa_{\text{EM}} \rho \phi}{\mu_0 c \rho} &= \frac{\kappa_{\text{EM}} \phi}{c} \\ &\simeq 2.61 \times 10^{-26} \end{aligned} \quad (188)$$

This ratio is directly the weak-backreaction parameter defined by Eq. (182). Thus, the source-potential backreaction term is suppressed by approximately 10^{-26} relative to the classical Maxwell source at the Bohr scale.

5.1.4 Electromagnetic Energy-Momentum Backreaction Term

The second geometric backreaction contribution comes from the electromagnetic energy-momentum tensor. In the electrostatic weak-backreaction limit, one takes

$$F_{0i} = \frac{1}{c} \partial_i \phi, \quad E_i = -\partial_i \phi \quad (189)$$

Therefore, the electromagnetic energy density is, at the order of magnitude,

$$u_{\text{EM}} = \frac{\epsilon_0}{2} |\nabla \phi|^2 = \frac{1}{2\mu_0 c^2} |\nabla \phi|^2 \quad (190)$$

In this convention, the component Θ_{EM}^{00} is of the order

$$\Theta_{\text{EM}}^{00} \simeq \frac{1}{2\mu_0 c^2} |\nabla \phi|^2 \quad (191)$$

The energy-momentum backreaction contribution in the field equation is given by

$$\left| \mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{0\rho} e^0{}_\rho \right| \quad (192)$$

In the weak-backreaction limit, since $e^0{}_0 \simeq 1$ and $e^0{}_i \simeq 0$,

$$\begin{aligned} \left| \mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{0\rho} e^0{}_\rho \right| &\simeq \mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{00} \\ &\simeq \frac{\kappa_{\text{EM}}}{2c^2} |\nabla\phi|^2. \end{aligned} \quad (193)$$

Numerically,

$$\begin{aligned} \frac{\kappa_{\text{EM}}}{2c^2} |\nabla\phi|^2 &= \frac{2.87 \times 10^{-19}}{2 (2.998 \times 10^8)^2} (5.14 \times 10^{11})^2 \\ &\simeq 4.23 \times 10^{-13} \text{ V s m}^{-3}. \end{aligned} \quad (194)$$

The dimensional check is performed as

$$\left[\frac{\kappa_{\text{EM}}}{c^2} |\nabla\phi|^2 \right] = \frac{\text{m}}{\text{V s}} \frac{\text{s}^2}{\text{m}^2} \frac{\text{V}^2}{\text{m}^2} = \text{V s m}^{-3} \quad (195)$$

Thus, this term has the same dimension as the classical source term.

The ratio of the energy-momentum backreaction term to the classical Maxwell source is found as

$$\begin{aligned} \frac{\frac{\kappa_{\text{EM}}}{2c^2} |\nabla\phi|^2}{\mu_0 c \rho} &= \frac{4.23 \times 10^{-13}}{9.72 \times 10^{13}} \\ &\simeq 4.35 \times 10^{-27} \end{aligned} \quad (196)$$

This ratio shows that the electromagnetic energy-momentum backreaction contribution is suppressed by approximately 10^{-27} relative to the classical Maxwell source at the Bohr scale.

5.1.5 Derivation of the Classical Maxwell Limit

The numerical test above shows that, at the Bohr-radius scale, both geometric backreaction contributions are extremely small compared with the classical Maxwell source. For the source-potential backreaction,

$$\frac{|\mu_0 \kappa_{\text{EM}} E_0{}^0 (J^\mu A_\mu)|}{|\mu_0 J^0|} \simeq 2.61 \times 10^{-26} \quad (197)$$

and for the electromagnetic energy-momentum backreaction,

$$\frac{|\mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{0\rho} e^0{}_\rho|}{|\mu_0 J^0|} \simeq 4.35 \times 10^{-27} \quad (198)$$

are obtained.

Therefore, in the weak-field and low-energy-density regime, the geometric backreaction terms in Eq. (167) can be neglected compared with the dominant source term. In this limit, the field equation reduces at leading order to the classical sourced Maxwell equation

$$\nabla_\mu F^{\mu\nu} \simeq \mu_0 J^\nu \quad (199)$$

This result shows that the Maxwell-adapted tetrad *ansatz* recovers classical Maxwell electrostatic theory in the appropriate weak-field limit. The geometric backreaction terms do not vanish in this limit; however, since they are suppressed by 10^{-26} - 10^{-27} relative to the classical source term, they do not modify the dominant Maxwell dynamics.

Nevertheless, the analysis performed here is intended only to test whether the classical Maxwell limit is recovered in the weak-field and low-energy-density regime. Therefore, the behavior of geometric backreaction terms of order κ_{EM} in strong electromagnetic-field regimes has been left outside the scope of this appendix.

In particular, in astrophysical systems with very high electromagnetic energy densities, the magnitude of these terms may be reevaluated. However, a detailed investigation of such strong-field regimes is not the main purpose of this work. The result shown in this appendix is only that the Maxwell-adapted tetrad *ansatz* successfully recovers the classical sourced Maxwell equation in the weak-field limit.

5.2 On the Poisson Limit

In the previous subsection, it was shown that the classical sourced Maxwell equation is obtained by suppressing the geometric backreaction terms. In this section, three basic classical limits will be examined. First, it will be shown how the electrostatic Poisson limit is obtained in the sourced case.

5.2.1 Sourced Weak-Field and Poisson Limit

As the starting point, the geometric Maxwell-type field equation obtained from the temporal projection will be used:

$$\nabla_\mu F^{\mu\nu} = \mu_0 J^\nu + \mu_0 \kappa_{\text{EM}} E_0^\nu (J^\mu A_\mu) + \mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{\nu\rho} e^0_\rho. \quad (200)$$

Here, E_0^ν denotes the inverse tetrad component, e^0_ρ the temporal tetrad component, and $\Theta_{\text{EM}}^{\nu\rho}$ the electromagnetic energy-momentum tensor. This writing is the form obtained by rewriting the mixed-index energy-momentum contribution in an equivalent temporal-tetrad form using the relation

$$E_0^\rho \Theta_{\rho}^{\text{EM} \nu} = -\Theta_{\text{EM}}^{\nu\rho} e^0_\rho \quad (201)$$

Therefore, the term

$$-\mu_0 \kappa_{\text{EM}} E_0^\rho \Theta_{\rho}^{\text{EM} \nu} \quad (202)$$

in the previous section has here been written as

$$+\mu_0\kappa_{\text{EM}}\Theta_{\text{EM}}^{\nu\rho}e^0{}_\rho \quad (203)$$

The potential components to be used in the electrostatic case will be taken as

$$A_\mu = \left(-\frac{\Phi}{c}, 0, 0, 0\right) \quad (204)$$

With this choice, when

$$J^\mu = (c\rho, 0, 0, 0) \quad (205)$$

one obtains

$$J^\mu A_\mu = -\rho\Phi \quad (206)$$

Here, ρ is the physical charge density in the SI unit system.

5.2.2 Calculation of the Left-Hand Side

In the static electrostatic configuration, the main component of the field equation is the $\nu = 0$ component. The covariant divergence of an antisymmetric tensor can be written as

$$\nabla_\mu F^{\mu 0} = \frac{1}{e} \partial_\mu (e F^{\mu 0}) \quad (207)$$

In the static case, since $F^{00} = 0$, only spatial derivatives contribute. For this ansatz, the tetrad determinant is

$$e = 1 - \kappa_{\text{EM}} \frac{\Phi}{c} \quad (208)$$

and hence

$$\nabla_\mu F^{\mu 0} = \frac{1}{1 - \kappa_{\text{EM}} \Phi/c} \partial_i \left[\left(1 - \kappa_{\text{EM}} \frac{\Phi}{c} \right) F^{i0} \right]. \quad (209)$$

Now F^{i0} is calculated. By definition,

$$F^{i0} = g^{ij} g^{00} F_{j0}. \quad (210)$$

For this ansatz,

$$g^{ij} = \delta^{ij}, \quad (211)$$

$$g^{00} = -\frac{1}{(1 - \kappa_{\text{EM}} \Phi/c)^2}, \quad (212)$$

and

$$F_{j0} = -\frac{1}{c} \partial_j \Phi \quad (213)$$

so that

$$F^{i0} = g^{ij} g^{00} F_{j0}. \quad (214)$$

Writing explicitly,

$$F^{i0} = \delta^{ij} \left[-\frac{1}{(1 - \kappa_{\text{EM}}\Phi/c)^2} \right] \left(-\frac{1}{c} \partial_j \Phi \right). \quad (215)$$

From this,

$$F^{i0} = \frac{1}{c(1 - \kappa_{\text{EM}}\Phi/c)^2} \delta^{ij} \partial_j \Phi \quad (216)$$

is obtained. Since the flat spatial metric is used, one may write $\delta^{ij} \partial_j \Phi = \partial_i \Phi$. Therefore,

$$F^{i0} = \frac{1}{c(1 - \kappa_{\text{EM}}\Phi/c)^2} \partial_i \Phi. \quad (217)$$

Thus,

$$\left(1 - \kappa_{\text{EM}} \frac{\Phi}{c} \right) F^{i0} = \left(1 - \kappa_{\text{EM}} \frac{\Phi}{c} \right) \frac{1}{c(1 - \kappa_{\text{EM}}\Phi/c)^2} \partial_i \Phi \quad (218)$$

From this,

$$\left(1 - \kappa_{\text{EM}} \frac{\Phi}{c} \right) F^{i0} = \frac{1}{c(1 - \kappa_{\text{EM}}\Phi/c)} \partial_i \Phi \quad (219)$$

is obtained. Substituting into Eq. (209) gives

$$\nabla_\mu F^{\mu 0} = \frac{1}{1 - \kappa_{\text{EM}}\Phi/c} \partial_i \left[\frac{1}{c(1 - \kappa_{\text{EM}}\Phi/c)} \partial_i \Phi \right]. \quad (220)$$

Since c is constant, it can be taken outside, yielding

$$\nabla_\mu F^{\mu 0} = \frac{1}{c(1 - \kappa_{\text{EM}}\Phi/c)} \partial_i \left[\frac{\partial_i \Phi}{1 - \kappa_{\text{EM}}\Phi/c} \right] \quad (221)$$

Now the derivative is calculated explicitly:

$$\partial_i \left[\frac{\partial_i \Phi}{1 - \kappa_{\text{EM}}\Phi/c} \right] = \frac{1}{1 - \kappa_{\text{EM}}\Phi/c} \partial_i \partial_i \Phi + \partial_i \Phi \partial_i \left[\frac{1}{1 - \kappa_{\text{EM}}\Phi/c} \right]. \quad (222)$$

Here,

$$\partial_i \left[\frac{1}{1 - \kappa_{\text{EM}}\Phi/c} \right] = \frac{\kappa_{\text{EM}}}{c} \frac{\partial_i \Phi}{(1 - \kappa_{\text{EM}}\Phi/c)^2} \quad (223)$$

Thus,

$$\partial_i \left[\frac{\partial_i \Phi}{1 - \kappa_{\text{EM}}\Phi/c} \right] = \frac{1}{1 - \kappa_{\text{EM}}\Phi/c} \nabla^2 \Phi + \frac{\kappa_{\text{EM}}}{c(1 - \kappa_{\text{EM}}\Phi/c)^2} \delta^{ij} \partial_i \Phi \partial_j \Phi. \quad (224)$$

Substituting this expression into Eq. (221), one obtains

$$\nabla_\mu F^{\mu 0} = \frac{1}{c(1 - \kappa_{\text{EM}}\Phi/c)^2} \nabla^2 \Phi + \frac{\kappa_{\text{EM}}}{c^2(1 - \kappa_{\text{EM}}\Phi/c)^3} \delta^{ij} \partial_i \Phi \partial_j \Phi. \quad (225)$$

5.2.3 Calculation of the Right-Hand Side

For a static source,

$$J^\mu = (c\rho, 0, 0, 0) \quad (226)$$

is taken. Here, ρ is the physical charge density. Together with Eq. (204),

$$J^\mu A_\mu = J^0 A_0 \quad (227)$$

is obtained. Explicitly,

$$J^\mu A_\mu = c\rho \left(-\frac{\Phi}{c} \right) \quad (228)$$

and therefore

$$J^\mu A_\mu = -\rho\Phi \quad (229)$$

The right-hand side of the $\nu = 0$ component of the field equation is

$$\mu_0 J^0 + \mu_0 \kappa_{\text{EM}} E_0{}^0 (J^\mu A_\mu) + \mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{0\rho} e^0{}_\rho \quad (230)$$

The first term is

$$\mu_0 J^0 = \mu_0 c\rho \quad (231)$$

In the second term, using

$$E_0{}^0 = \frac{1}{1 - \kappa_{\text{EM}} \Phi/c} \quad (232)$$

and Eq. (229), one obtains

$$\mu_0 \kappa_{\text{EM}} E_0{}^0 (J^\mu A_\mu) = \mu_0 \kappa_{\text{EM}} \frac{1}{1 - \kappa_{\text{EM}} \Phi/c} (-\rho\Phi). \quad (233)$$

From this,

$$\mu_0 \kappa_{\text{EM}} E_0{}^0 (J^\mu A_\mu) = -\frac{\mu_0 \kappa_{\text{EM}} \rho \Phi}{1 - \kappa_{\text{EM}} \Phi/c}. \quad (234)$$

In the third term, since

$$e^0{}_\rho = \left(1 - \kappa_{\text{EM}} \frac{\Phi}{c}, 0, 0, 0 \right) \quad (235)$$

one has

$$\mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{0\rho} e^0{}_\rho = \mu_0 \kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\Phi}{c} \right) \Theta_{\text{EM}}^{00}. \quad (236)$$

The electromagnetic energy-momentum tensor is

$$\Theta_{\text{EM}}^{\mu\nu} = \frac{1}{\mu_0} \left(F^{\mu\rho} F^\nu{}_\rho - \frac{1}{4} g^{\mu\nu} F_{\rho\sigma} F^{\rho\sigma} \right) \quad (237)$$

For this ansatz,

$$\Theta_{\text{EM}}^{00} = \frac{\delta^{ij} \partial_i \Phi \partial_j \Phi}{2\mu_0 c^2 (1 - \kappa_{\text{EM}} \Phi/c)^4} \quad (238)$$

is obtained. Thus, the energy-momentum backreaction term becomes

$$\mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{0\rho} e^0{}_\rho = \mu_0 \kappa_{\text{EM}} \left(1 - \kappa_{\text{EM}} \frac{\Phi}{c}\right) \frac{\delta^{ij} \partial_i \Phi \partial_j \Phi}{2\mu_0 c^2 (1 - \kappa_{\text{EM}} \Phi/c)^4}. \quad (239)$$

Here μ_0 cancels, and one obtains

$$\mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{0\rho} e^0{}_\rho = \frac{\kappa_{\text{EM}} \delta^{ij} \partial_i \Phi \partial_j \Phi}{2c^2 (1 - \kappa_{\text{EM}} \Phi/c)^3} \quad (240)$$

Thus, the right-hand side of the $\nu = 0$ component of the field equation is

$$\mu_0 J^0 + \mu_0 \kappa_{\text{EM}} E_0^0 (J^\mu A_\mu) + \mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{0\rho} e^0{}_\rho = \mu_0 c \rho - \frac{\mu_0 \kappa_{\text{EM}} \rho \Phi}{1 - \kappa_{\text{EM}} \Phi/c} + \frac{\kappa_{\text{EM}} \delta^{ij} \partial_i \Phi \partial_j \Phi}{2c^2 (1 - \kappa_{\text{EM}} \Phi/c)^3}. \quad (241)$$

Combining Eq. (225) and Eq. (241), the field equation becomes

$$\frac{1}{c(1 - \kappa_{\text{EM}} \Phi/c)^2} \nabla^2 \Phi + \frac{\kappa_{\text{EM}}}{c^2 (1 - \kappa_{\text{EM}} \Phi/c)^3} \delta^{ij} \partial_i \Phi \partial_j \Phi = \mu_0 c \rho - \frac{\mu_0 \kappa_{\text{EM}} \rho \Phi}{1 - \kappa_{\text{EM}} \Phi/c} + \frac{\kappa_{\text{EM}} \delta^{ij} \partial_i \Phi \partial_j \Phi}{2c^2 (1 - \kappa_{\text{EM}} \Phi/c)^3}. \quad (242)$$

The weak geometric backreaction limit is defined by

$$\left| \kappa_{\text{EM}} \frac{\Phi}{c} \right| \ll 1 \quad (243)$$

In this limit,

$$\left(1 - \kappa_{\text{EM}} \frac{\Phi}{c}\right)^{-1} \simeq 1 + \kappa_{\text{EM}} \frac{\Phi}{c}, \quad (244)$$

$$\left(1 - \kappa_{\text{EM}} \frac{\Phi}{c}\right)^{-2} \simeq 1 + 2\kappa_{\text{EM}} \frac{\Phi}{c}, \quad (245)$$

and

$$\left(1 - \kappa_{\text{EM}} \frac{\Phi}{c}\right)^{-3} \simeq 1 + 3\kappa_{\text{EM}} \frac{\Phi}{c} \quad (246)$$

hold. Keeping only up to first order in κ_{EM} , Eq. (242) is written as

$$\frac{1}{c} \left(1 + 2\kappa_{\text{EM}} \frac{\Phi}{c}\right) \nabla^2 \Phi + \frac{\kappa_{\text{EM}}}{c^2} \delta^{ij} \partial_i \Phi \partial_j \Phi = \mu_0 c \rho - \mu_0 \kappa_{\text{EM}} \rho \Phi + \frac{\kappa_{\text{EM}}}{2c^2} \delta^{ij} \partial_i \Phi \partial_j \Phi \quad (247)$$

Here, in the source-potential term on the right-hand side,

$$-\frac{\mu_0 \kappa_{\text{EM}} \rho \Phi}{1 - \kappa_{\text{EM}} \Phi/c} = -\mu_0 \kappa_{\text{EM}} \rho \Phi + O(\kappa_{\text{EM}}^2) \quad (248)$$

has been used. In addition, in the energy-momentum backreaction term,

$$\frac{\kappa_{\text{EM}}}{2c^2 (1 - \kappa_{\text{EM}} \Phi/c)^3} \delta^{ij} \partial_i \Phi \partial_j \Phi = \frac{\kappa_{\text{EM}}}{2c^2} \delta^{ij} \partial_i \Phi \partial_j \Phi + O(\kappa_{\text{EM}}^2) \quad (249)$$

has been taken.

Multiplying both sides by c gives

$$\left(1 + 2\kappa_{\text{EM}} \frac{\Phi}{c}\right) \nabla^2 \Phi + \frac{\kappa_{\text{EM}}}{c} \delta^{ij} \partial_i \Phi \partial_j \Phi = \mu_0 c^2 \rho - \mu_0 \kappa_{\text{EM}} c \rho \Phi + \frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi. \quad (250)$$

Expanding the parenthesis on the left-hand side gives

$$\nabla^2 \Phi + 2\kappa_{\text{EM}} \frac{\Phi}{c} \nabla^2 \Phi + \frac{\kappa_{\text{EM}}}{c} \delta^{ij} \partial_i \Phi \partial_j \Phi = \mu_0 c^2 \rho - \mu_0 \kappa_{\text{EM}} c \rho \Phi + \frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi. \quad (251)$$

Solving for $\nabla^2 \Phi$, one obtains

$$\nabla^2 \Phi = \mu_0 c^2 \rho - \mu_0 \kappa_{\text{EM}} c \rho \Phi + \frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi - \frac{\kappa_{\text{EM}}}{c} \delta^{ij} \partial_i \Phi \partial_j \Phi - 2\kappa_{\text{EM}} \frac{\Phi}{c} \nabla^2 \Phi. \quad (252)$$

Combining the gradient terms,

$$\frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi - \frac{\kappa_{\text{EM}}}{c} \delta^{ij} \partial_i \Phi \partial_j \Phi = -\frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi \quad (253)$$

is obtained. Therefore,

$$\nabla^2 \Phi = \mu_0 c^2 \rho - \mu_0 \kappa_{\text{EM}} c \rho \Phi - \frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi - 2\kappa_{\text{EM}} \frac{\Phi}{c} \nabla^2 \Phi. \quad (254)$$

To first order, the zeroth-order Poisson result is used for $\nabla^2 \Phi$ in the last term:

$$\nabla^2 \Phi = \mu_0 c^2 \rho + O(\kappa_{\text{EM}}). \quad (255)$$

Substituting this result into the last term gives

$$-2\kappa_{\text{EM}} \frac{\Phi}{c} \nabla^2 \Phi = -2\kappa_{\text{EM}} \frac{\Phi}{c} \mu_0 c^2 \rho + O(\kappa_{\text{EM}}^2) \quad (256)$$

From this,

$$-2\kappa_{\text{EM}} \frac{\Phi}{c} \mu_0 c^2 \rho = -2\mu_0 \kappa_{\text{EM}} c \rho \Phi \quad (257)$$

is obtained. Thus,

$$\nabla^2 \Phi = \mu_0 c^2 \rho - \mu_0 \kappa_{\text{EM}} c \rho \Phi - 2\mu_0 \kappa_{\text{EM}} c \rho \Phi - \frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi. \quad (258)$$

Combining the source-potential corrections,

$$-\mu_0 \kappa_{\text{EM}} c \rho \Phi - 2\mu_0 \kappa_{\text{EM}} c \rho \Phi = -3\mu_0 \kappa_{\text{EM}} c \rho \Phi \quad (259)$$

Therefore, the result is

$$\nabla^2 \Phi = \mu_0 c^2 \rho - 3\mu_0 \kappa_{\text{EM}} c \rho \Phi - \frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi. \quad (260)$$

In the SI unit system,

$$c^2 = \frac{1}{\epsilon_0 \mu_0} \quad (261)$$

and therefore

$$\mu_0 c^2 \rho = \mu_0 \frac{1}{\epsilon_0 \mu_0} \rho \quad (262)$$

Thus,

$$\mu_0 c^2 \rho = \frac{\rho}{\epsilon_0} \quad (263)$$

is obtained. Hence,

$$\nabla^2 \Phi = \frac{\rho}{\epsilon_0} - 3\mu_0 \kappa_{\text{EM}} c \rho \Phi - \frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi. \quad (264)$$

In the weak-backreaction and low-field-energy limit, under the conditions

$$|3\mu_0 \kappa_{\text{EM}} c \rho \Phi| \ll \left| \frac{\rho}{\epsilon_0} \right| \quad (265)$$

and

$$\left| \frac{\kappa_{\text{EM}}}{2c} \delta^{ij} \partial_i \Phi \partial_j \Phi \right| \ll \left| \frac{\rho}{\epsilon_0} \right| \quad (266)$$

the geometric corrections are neglected. This will also be evaluated numerically in the next heading. Thus, the standard electrostatic Poisson equation

$$\nabla^2 \Phi = \frac{\rho}{\epsilon_0} \quad (267)$$

is obtained. Here, a calculation can be presented in order to demonstrate the smallness of these terms.

5.2.4 A Numerical Calculation for the Poisson Limit

In this section, the result obtained for the Poisson limit will be tested with a numerical value. This criterion will be the same as the calculation performed previously for the Maxwell limit. It is based on the assumption that the charge of an electron is uniformly distributed over a sphere of radius equal to the Bohr radius a_0 . This model does not represent the real quantum-mechanical charge distribution; it is only a simple numerical test used for an order-of-magnitude comparison.

The weak-backreaction electrostatic limit of the tetrad-dependent field equation obtained in the previous section, under the sign convention of Eq. (200), is

$$\nabla^2 \Phi = \frac{\rho}{\epsilon_0} - 3\mu_0 \kappa_{\text{EM}} c \rho \Phi - \frac{\kappa_{\text{EM}}}{2c} |\nabla \Phi|^2 + O(\kappa_{\text{EM}}^2) \quad (268)$$

Here, the first term is the classical Poisson source. The second term is the geometric correction arising from the source-potential interaction, and the third term arises from electromagnetic energy-momentum backreaction.

This expression contains two approximations. The first is the expansion, up to first order, of the q^{-n} factors coming from $q = 1 - \kappa_{\text{EM}}\Phi/c$, since

$$\left| \kappa_{\text{EM}} \frac{\Phi}{c} \right| \ll 1 \quad (269)$$

The second is the neglect of $O(\kappa_{\text{EM}}^2)$ and higher-order geometric backreaction terms.

- Classical Poisson Source

The magnitude of the classical source term is

$$\frac{\rho}{\epsilon_0} = \frac{2.58 \times 10^{11} \text{ C m}^{-3}}{8.854 \times 10^{-12} \text{ C V}^{-1} \text{ m}^{-1}} \quad (270)$$

From this,

$$\frac{\rho}{\epsilon_0} \simeq 2.92 \times 10^{22} \text{ V m}^{-2}. \quad (271)$$

- Source-Potential Correction

The magnitude of the source-potential correction is

$$3\mu_0\kappa_{\text{EM}}c\rho\Phi = 3 (1.2566 \times 10^{-6}) (2.87 \times 10^{-19}) (2.998 \times 10^8) (2.58 \times 10^{11}) \quad (27.2) \quad (272)$$

From this,

$$3\mu_0\kappa_{\text{EM}}c\rho\Phi \simeq 2.28 \times 10^{-3} \text{ V m}^{-2}. \quad (273)$$

The ratio of this term to the classical Poisson source is

$$\frac{3\mu_0\kappa_{\text{EM}}c\rho\Phi}{\rho/\epsilon_0} = \frac{2.28 \times 10^{-3}}{2.92 \times 10^{22}} \quad (274)$$

From this,

$$\frac{3\mu_0\kappa_{\text{EM}}c\rho\Phi}{\rho/\epsilon_0} \simeq 7.82 \times 10^{-26}. \quad (275)$$

- Energy-Momentum Backreaction

The magnitude of the energy-momentum correction in Eq. (268) is

$$\frac{\kappa_{\text{EM}}}{2c} |\nabla\Phi|^2 = \frac{2.87 \times 10^{-19}}{2 (2.998 \times 10^8)} (5.14 \times 10^{11})^2 \quad (276)$$

From this,

$$\frac{\kappa_{\text{EM}}}{2c} |\nabla\Phi|^2 \simeq 1.27 \times 10^{-4} \text{ V m}^{-2}. \quad (277)$$

The dimensional check is

$$\left[\frac{\kappa_{\text{EM}}}{c} |\nabla\Phi|^2 \right] = \frac{\text{m}}{\text{V s}} \frac{\text{s}}{\text{m}} \frac{\text{V}^2}{\text{m}^2} \quad (278)$$

From this,

$$\left[\frac{\kappa_{\text{EM}}}{c} |\nabla \Phi|^2 \right] = \text{V m}^{-2} \quad (279)$$

is obtained. Therefore, the energy-momentum backreaction correction also has the same dimension as the source term of the Poisson equation.

The ratio of this term to the classical Poisson source is

$$\frac{\frac{\kappa_{\text{EM}}}{2c} |\nabla \Phi|^2}{\rho/\epsilon_0} = \frac{1.27 \times 10^{-4}}{2.92 \times 10^{22}} \quad (280)$$

From this,

$$\frac{\frac{\kappa_{\text{EM}}}{2c} |\nabla \Phi|^2}{\rho/\epsilon_0} \simeq 4.35 \times 10^{-27}. \quad (281)$$

Thus, at the Bohr-radius scale, the geometric corrections in the Poisson equation are suppressed by approximately 10^{-26} relative to the classical source term. The source-potential correction remains approximately of the order

$$7.82 \times 10^{-26} \quad (282)$$

while the energy-momentum backreaction remains approximately of the order

$$4.35 \times 10^{-27} \quad (283)$$

Under this limit, Eq. (268) reduces at leading order to the standard electrostatic Poisson equation

$$\nabla^2 \Phi = \frac{\rho}{\epsilon_0} \quad (284)$$

This calculation shows that tetrad-based geometric corrections are extremely small at laboratory and atomic scales, and that the classical Maxwell–Poisson limit is naturally recovered.

5.3 Laplace Limit and Electromagnetic-Field-Free Limit

In this section, first the uncharged region where no source current is present but an external field exists will be considered, and it will be shown that the Laplace limit emerges in this case. Finally, the completely empty limit in which neither source current nor an external electromagnetic field is present will be examined.

5.3.1 External Electrostatic Field in an Uncharged Medium and the Laplace Limit

Second, the case in which there is no source current but an external electrostatic field exists is considered. In this case,

$$J^\mu = 0, \quad A_\mu \neq 0 \quad (285)$$

The electrostatic potential can be taken as

$$A_\mu = \left(-\frac{\Phi(\mathbf{x})}{c}, 0, 0, 0 \right) \quad (286)$$

Since the source current is zero,

$$J^\mu A_\mu = 0 \quad (287)$$

is obtained.

In this case, Eq. 200 remains as

$$\nabla_\mu F^{\mu\nu} = -\mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{\nu\rho} e^0_\rho \quad (288)$$

That is, the classical source term and the source-potential backreaction term disappear. Only the geometric backreaction contribution arising from the electromagnetic energy-momentum tensor remains.

In the weak-field limit, this contribution is suppressed because it is proportional to κ_{EM} . Therefore, under the condition

$$|\mu_0 \kappa_{\text{EM}} \Theta_{\text{EM}}^{\nu\rho} e^0_\rho| \ll |\nabla_\mu F^{\mu\nu}|_{\text{char}} \quad (289)$$

the backreaction term can be neglected. Here, the expression on the right-hand side denotes the characteristic magnitude of the Maxwell operator for the relevant configuration.

In this case, the field equation takes the leading-order form

$$\nabla_\mu F^{\mu\nu} = 0 \quad (290)$$

This expression is the classical Maxwell equation in a region without sources.

In the electrostatic case, its $\nu = 0$ component reduces to the Laplace equation. Since there is no source, $\rho = 0$ is taken, and the source term in Eq. (284) disappears. Thus,

$$\nabla^2 \Phi = 0 \quad (291)$$

is obtained. This is the standard electrostatic Laplace equation in an uncharged region.

The physical meaning of this limit is the following: even if an external electrostatic field is present, since the geometric backreaction contribution arising from electromagnetic energy-momentum is suppressed by κ_{EM} in the weak-field regime, the classical source-free Maxwell equation is preserved.

5.3.2 Uncharged and External-Field-Free Limit

Finally, the limit in which neither source current nor an external electromagnetic field is present is considered. In this case,

$$J^\mu = 0, \quad A_\mu = 0 \quad (292)$$

Since the electromagnetic potential is zero, the field strength is also zero:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = 0. \quad (293)$$

Accordingly, the electromagnetic energy-momentum tensor also goes to zero:

$$\Theta_{\text{EM}}^{\mu\nu} = 0. \quad (294)$$

Moreover,

$$J^\mu A_\mu = 0 \quad (295)$$

When these results are substituted into Eq. (200), every term reduces to zero:

$$0 = 0. \quad (296)$$

Thus, when the electromagnetic sector is completely switched off, the electromagnetic backreaction terms arising from the Maxwell-adapted tetrad structure also disappear.

In this limit, the electromagnetic torsion embedded in the temporal tetrad also becomes zero. Since

$$T^0 = \kappa_{\text{EM}} F \quad (297)$$

for $F = 0$, one obtains

$$T^0 = 0 \quad (298)$$

This result shows that when the electromagnetic field is completely switched off, the temporal electromagnetic torsion also disappears. If the spatial tetrads are additionally chosen so as to represent a gravitational or cosmological background, other torsion components belonging to the geometric sector may exist. However, these are not the temporal electromagnetic torsion arising from the embedding of the electromagnetic potential into the temporal tetrad.

As a result, when the classical limits are considered together, it is seen that the Maxwell-adapted tetrad ansatz exhibits consistent classical behavior. In the sourced weak-field limit, the classical sourced Maxwell equation and the Poisson equation are recovered. In a source-free but external-field region, the source-free Maxwell equation and the electrostatic Laplace equation emerge. When both the source and the external field are switched off, the electromagnetic sector is completely zeroed out and the temporal electromagnetic torsion disappears.

6 Appendix-6: Writing the Maxwell Kinetic Term in Terms of the Hodge Dual and Temporal Torsion

In this appendix, how the Maxwell kinetic term is written in the language of differential forms and how it can be interpreted, under the Maxwell-adapted tetrad ansatz, as the quadratic norm of temporal torsion will be examined. The notation used in the previous appendices will be preserved exactly. The equality

Eq. 72 shows that the electromagnetic field strength is represented directly by the temporal torsion component. Therefore, whether the Maxwell kinetic term can also be written in terms of temporal torsion by using the same relation must be examined at the level of the action.

6.1 The Maxwell Kinetic Term with the Hodge Dual

The electromagnetic field strength is written as a two-form as

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu \quad (299)$$

The natural expression of the Maxwell kinetic term in the language of differential forms is constructed with the help of the Hodge dual. Since the Hodge dual is determined by the metric and the orientation, the expression $F \wedge \star F$ is the four-form that gives the metric norm of the electromagnetic field.

For two-forms, the identity

$$F \wedge \star F = \frac{1}{2} F_{\mu\nu} F^{\mu\nu} e^4 x \quad (300)$$

is used. Therefore, in SI units, the Maxwell kinetic action is written as

$$S_{\text{Maxwell}} = -\frac{1}{2\mu_0 c} \int F \wedge \star F \quad (301)$$

Substituting Eq. (300), one obtains

$$S_{\text{Maxwell}} = -\frac{1}{2\mu_0 c} \int \frac{1}{2} F_{\mu\nu} F^{\mu\nu} e^4 x = -\frac{1}{4\mu_0 c} \int d^4 x e F_{\mu\nu} F^{\mu\nu}.$$

This result shows that the Maxwell action in the form language is identical to the standard kinetic term in component notation. Here, one can go one step further and write the electromagnetic field tensor in terms of temporal torsion as in Eq. 72. The same relation also holds for the expression with raised indices:

$$F^{\mu\nu} = \frac{1}{\kappa_{\text{EM}}} T^{0\mu\nu}. \quad (302)$$

Substituting these two expressions into the Maxwell kinetic term gives

$$F_{\mu\nu} F^{\mu\nu} = \left(\frac{1}{\kappa_{\text{EM}}} T^0{}_{\mu\nu} \right) \left(\frac{1}{\kappa_{\text{EM}}} T^{0\mu\nu} \right) = \frac{1}{\kappa_{\text{EM}}^2} T^0{}_{\mu\nu} T^{0\mu\nu}$$

Thus, Eq. (302) becomes

$$S_{\text{Maxwell}} = -\frac{1}{4\mu_0 c \kappa_{\text{EM}}^2} \int d^4 x e T^0{}_{\mu\nu} T^{0\mu\nu} \quad (303)$$

The same result can also be seen directly in the form language;

$$F = \frac{1}{\kappa_{\text{EM}}} T^0 \quad (304)$$

is written. If this expression is substituted into Eq. (301),

$$S_{\text{Maxwell}} = -\frac{1}{2\mu_0 c} \int \left(\frac{1}{\kappa_{\text{EM}}} T^0 \right) \wedge \star \frac{1}{\kappa_{\text{EM}}} T^0 = -\frac{1}{2\mu_0 c \kappa_{\text{EM}}^2} \int T^0 \wedge \star T^0. \quad (305)$$

Therefore, under the Maxwell-adapted tetrad ansatz, the Maxwell kinetic action can be written as the Hodge norm of the temporal torsion two-form.

This result is not merely that the Maxwell kinetic term acquires a geometric expression in the language of differential forms. More strongly, since under the Maxwell-adapted tetrad ansatz the electromagnetic field strength is carried directly by temporal torsion, the kinetic term becomes the quadratic norm of temporal torsion.

6.2 Expression of the Lagrangian Terms in Terms of Geometric Torsions and Tetrads

In the previous parts, it was shown how the electromagnetic field is encoded within the Maxwell-adapted tetrad structure. In this structure, the electromagnetic field strength is directly related to the torsion component of the temporal tetrad. Therefore, it is a natural question whether the kinetic term and the interaction term with the source appearing in the standard Maxwell Lagrangian density can be rewritten in terms of torsion and tetrad variables under the chosen geometric ansatz.

In this section, this question will be answered in two steps. First, it will be shown how the Maxwell kinetic term is expressed in terms of the square of temporal torsion. Then, it will be explained how the interaction term with the source can be written in terms of the deviation of the temporal tetrad from the reference flow, instead of the electromagnetic potential.

6.2.1 Geometric Expression of the Kinetic Term

In the Maxwell-adapted tetrad structure, there is a direct relation between the electromagnetic field strength and temporal torsion. This is given by Eq. (72). This relation allows the electromagnetic kinetic term to be written as a torsional expression. When the indices are raised with the metric, one obtains

$$T^{0\mu\nu} = \kappa_{\text{EM}} F^{\mu\nu} \quad (306)$$

Thus, the square of temporal torsion is proportional to the standard Maxwell invariant:

$$T^0_{\mu\nu} T^{0\mu\nu} = \kappa_{\text{EM}}^2 F_{\mu\nu} F^{\mu\nu} \quad (307)$$

Therefore, the kinetic Lagrangian density can be written geometrically as

$$\mathcal{L}_{\text{kin}}^{(T)} = -\frac{1}{4\mu_0 \kappa_{\text{EM}}^2} T^0_{\mu\nu} T^{0\mu\nu} \quad (308)$$

When Eq. (306) is substituted into this expression,

$$\mathcal{L}_{\text{kin}}^{(T)} = -\frac{1}{4\mu_0\kappa_{\text{EM}}^2} (\kappa_{\text{EM}}F_{\mu\nu}) (\kappa_{\text{EM}}F^{\mu\nu}) = -\frac{1}{4\mu_0} F_{\mu\nu}F^{\mu\nu} \quad (309)$$

is obtained. Thus, it is seen that the kinetic term written with the square of temporal torsion exactly reproduces the standard Maxwell kinetic term in the Maxwell-adapted limit.

This result shows that, in this geometric structure, the electromagnetic field strength can be interpreted not as an independent two-form, but as the appropriately normalized form of temporal tetrad torsion. In other words, under this ansatz, the standard Maxwell kinetic term is obtained from the square of temporal torsion.

6.2.2 Geometric Expression of the Interaction Term

In standard sourced Maxwell theory, the interaction term with the source is written as the coupling of the electromagnetic potential with the current. At the level of the Lagrangian density, this term is

$$\mathcal{L}_{\text{int}} = J^\mu A_\mu \quad (310)$$

Here j^μ denotes the physical electric current, while A_μ denotes the electromagnetic potential.

In standard Maxwell theory, A_μ is treated as an independent dynamical field. In this work, however, A_μ is interpreted as a structure embedded inside the temporal tetrad component. When the temporal tetrad ansatz is given by Eq. (13), it is written as

$$e^0{}_\mu = u_\mu + \kappa_{\text{EM}} A_\mu \quad (311)$$

Here u_μ represents the reference temporal-flow one-form that remains when the electromagnetic potential is switched off.

Solving Eq. (311) for A_μ , one obtains

$$A_\mu = \frac{1}{\kappa_{\text{EM}}} (e^0{}_\mu - u_\mu) \quad (312)$$

This expression shows that the electromagnetic potential can be written as the deviation of the temporal tetrad from the reference flow.

Now, when Eq. (312) is substituted into the standard interaction term Eq. (310), one finds

$$\mathcal{L}_{\text{int}} = J^\mu A_\mu = J^\mu \frac{1}{\kappa_{\text{EM}}} (e^0{}_\mu - u_\mu) = \frac{1}{\kappa_{\text{EM}}} J^\mu (e^0{}_\mu - u_\mu) \quad (313)$$

At this stage, a geometrically rescaled current can be defined as

$$\mathcal{J}^\mu = \frac{1}{\kappa_{\text{EM}}} j^\mu \quad (314)$$

Using this definition, the interaction term becomes

$$\mathcal{L}_{\text{int}} = \mathcal{J}^\mu (e^0_\mu - u_\mu) \quad (315)$$

In this form, the interaction is no longer expressed directly through A_μ , but through the deviation of the temporal tetrad from the reference flow. Therefore, under the Maxwell-adapted tetrad ansatz, the interaction term acquires a geometric meaning. The source current couples to the electromagnetic deviation of the temporal tetrad.

This result does not mean that the physical content of the source term disappears. The source current still represents the externally given physical electric current. What changes is the interpretation of the variable to which this current couples. In standard Maxwell theory, the source couples directly to the potential A_μ , whereas here the same coupling has been rewritten as a coupling to the deviation of the temporal tetrad from the reference flow u_μ .

Thus, the geometric expression of the interaction term clearly shows how the Maxwell-adapted tetrad couples to sources. While the electromagnetic field strength is represented by temporal torsion in the kinetic term, the electromagnetic potential is represented by temporal tetrad deviation in the interaction term.

7 Appendix-7: δ^0_i Projection and Field Equations

In this appendix, it will be examined how the electromagnetic sector is placed not only through the temporal tetrad projection, but also within the unified gravitational–electromagnetic action. Since the main aim of this theory is not to construct a unified theory, but rather to express electromagnetic theory geometrically in an appropriate way, it will be left in the form S_{grav} . However, at this point, a representation will be made by using the normalization condition of the variable introduced through κ_{EM} . In addition, the Reissner–Nordström metric will be discussed, and the meaning of the δ^0_i component when gravitation enters will be checked. The notation used in the previous appendices will be preserved exactly. In particular, it will be used that the electromagnetic field is encoded within the Maxwell-adapted temporal tetrad.

7.1 Unified Action

The total action containing gravitational and electromagnetic contributions is written symbolically as

$$S_{\text{toplam}} = S_{\text{grav}} + S_{\text{EM}} \quad (316)$$

Since the gravitational coupling constant is

$$\kappa_G = \frac{8\pi G}{c^4} \quad (317)$$

the gravitational action is written as

$$S_{\text{grav}} = \frac{1}{c} \frac{1}{2\kappa_G} \int d^4x e L_{\text{grav}} \quad (318)$$

For the electromagnetic sector, the exterior region where there is no source term will be examined. Therefore,

$$J^\mu = 0 \quad (319)$$

is taken. When the Maxwell kinetic term is written in terms of temporal torsion using the relation Eq. (72), one obtains

$$-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} = -\frac{1}{4\mu_0 \kappa_{\text{EM}}^2} T^0{}_{\mu\nu} T^{0\mu\nu} \quad (320)$$

Thus, in the exterior region, the unified action is expressed as

$$S_{\text{toplam}} = \frac{1}{c} \int d^4x e \left[\frac{1}{2\kappa_G} L_{\text{grav}} - \frac{1}{4\mu_0 \kappa_{\text{EM}}^2} T^0{}_{\mu\nu} T^{0\mu\nu} \right] \quad (321)$$

It can be seen directly that this action gives the correct Maxwell limit. Because of Eq. (72),

$$T^0{}_{\mu\nu} = \kappa_{\text{EM}} F_{\mu\nu} \quad (322)$$

is written. Therefore,

$$T^{0\mu\nu} = \kappa_{\text{EM}} F^{\mu\nu} \quad (323)$$

holds. In this case,

$$-\frac{1}{4\mu_0 \kappa_{\text{EM}}^2} T^0{}_{\mu\nu} T^{0\mu\nu} = -\frac{1}{4\mu_0 \kappa_{\text{EM}}^2} (\kappa_{\text{EM}} F_{\mu\nu}) (\kappa_{\text{EM}} F^{\mu\nu}) \quad (324)$$

is obtained. From this,

$$-\frac{1}{4\mu_0 \kappa_{\text{EM}}^2} T^0{}_{\mu\nu} T^{0\mu\nu} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \quad (325)$$

is found. Therefore, the temporal torsional kinetic term appearing in the unified action reproduces the standard Maxwell kinetic term in the Maxwell-adapted limit.

7.2 Obtaining the Coefficient κ_{EM} from Einstein–Maxwell Normalization

In this section, it will be shown that the derivation of the constant κ_{EM} is ensured by the Einstein–Maxwell condition. The normalization coefficients in the unified action are not arbitrary. The gravitational sector is written with the prefactor

$$\frac{1}{2\kappa_G} = \frac{c^4}{16\pi G} \quad (326)$$

The electromagnetic temporal torsional kinetic term has the prefactor

$$-\frac{1}{4\mu_0\kappa_{\text{EM}}^2}T^0{}_{\mu\nu}T^{0\mu\nu} \quad (327)$$

If the electromagnetic torsional kinetic term is required to be written on the same geometric scale as the gravitational coefficient in the Einstein–Maxwell action, the condition

$$\frac{1}{4\mu_0\kappa_{\text{EM}}^2} = \frac{c^4}{16\pi G} \quad (328)$$

is obtained. From this equation,

$$16\pi G = 4\mu_0\kappa_{\text{EM}}^2 c^4 \quad (329)$$

is found. Dividing both sides by $4\mu_0 c^4$ gives

$$\kappa_{\text{EM}}^2 = \frac{4\pi G}{\mu_0 c^4} \quad (330)$$

In SI units,

$$c^2 = \frac{1}{\varepsilon_0 \mu_0} \quad (331)$$

and therefore

$$\frac{1}{\mu_0} = \varepsilon_0 c^2 \quad (332)$$

is written. Thus,

$$\frac{1}{\mu_0 c^4} = \frac{\varepsilon_0 c^2}{c^4} \quad (333)$$

From this,

$$\frac{1}{\mu_0 c^4} = \frac{\varepsilon_0}{c^2} \quad (334)$$

is obtained. Substituting this into Eq. (330) gives

$$\kappa_{\text{EM}}^2 = \frac{4\pi G \varepsilon_0}{c^2} \quad (335)$$

Therefore,

$$\kappa_{\text{EM}} = \frac{\sqrt{4\pi G \varepsilon_0}}{c} \quad (336)$$

is obtained.

This result shows that the coefficient κ_{EM} is not an arbitrary scale. It is the geometric normalization coefficient that ensures the torsional representation of the electromagnetic kinetic term is compatible with the Einstein–Maxwell normalization.

7.3 Field Equations with the Gravitational Sector

The spatial projection of the electromagnetic sector alone gives a consistency condition. However, physically, the spatial tetrad components are not expected to be determined only by the electromagnetic sector. Therefore, the total action is written symbolically as

$$S_{\text{toplam}} = S_{\text{grav}} + S_{\text{EM}} \quad (337)$$

Here S_{grav} will be left only as the symbolic action of the geometric sector without choosing an explicit gravitational model. In this work, a specific gravitational field equation will not be constructed.

The variation of the gravitational sector with respect to the tetrad can be shown symbolically as

$$\delta S_{\text{grav}} = \frac{1}{c} \int d^4x e \mathcal{E}_a{}^\nu \delta e^a{}_\nu \quad (338)$$

Here $\mathcal{E}_a{}^\nu$ is the symbolic geometric contribution arising from the tetrad variation of the chosen gravitational/geometric action. This symbol is not identified with any specific theory of gravitation.

The source-free electromagnetic variation, obtained from the results in the previous appendices, is

$$\delta S_{\text{EM}} = \frac{1}{c} \int d^4x e \left[\frac{1}{\kappa_{\text{EM}}} \delta^0{}_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} \right) + E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu \right] \delta e^a{}_\nu \quad (339)$$

Here, since there is no source, $J^\mu = 0$ has been taken.

Thus, the source-free total variation is

$$\delta S_{\text{toplam}} = \delta S_{\text{grav}} + \delta S_{\text{EM}} \quad (340)$$

Writing these two contributions together gives

$$\delta S_{\text{toplam}} = \frac{1}{c} \int d^4x e \left[\mathcal{E}_a{}^\nu + \frac{1}{\kappa_{\text{EM}}} \delta^0{}_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} \right) + E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu \right] \delta e^a{}_\nu. \quad (341)$$

Here, the variation must satisfy

$$\delta S_{\text{toplam}} = 0 \quad (342)$$

Since $\delta e^a{}_\nu$ is arbitrary, one obtains

$$\mathcal{E}_a{}^\nu + \frac{1}{\kappa_{\text{EM}}} \delta^0{}_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} \right) + E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu = 0 \quad (343)$$

Here the aim is to examine the gravitational part, and in this theory the gravitational part is encoded in the spatial component of the tetrad. For this reason, $a = i$ is taken for $\delta^0{}_a$. In this case,

$$\delta^0{}_i = 0 \quad (344)$$

and therefore the Maxwell-type derivative term again disappears:

$$\delta^0_i \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} \right) = 0. \quad (345)$$

In this projection, Eq. (343) becomes

$$\mathcal{E}_i{}^\nu + E_i{}^\rho \Theta_{\rho}^{\text{EM}}{}^\nu = 0 \quad (346)$$

Equivalently, one can write

$$\mathcal{E}_i{}^\nu = -E_i{}^\rho \Theta_{\rho}^{\text{EM}}{}^\nu \quad (347)$$

This equation should not be interpreted as the field equation of a particular gravitational theory. The purpose here is only to show the following: since A_μ is embedded in the temporal tetrad component, the Maxwell derivative operator appears directly only in the temporal projection; however, because of the metric and determinant dependence of the electromagnetic kinetic term, $\Theta_{\rho}^{\text{EM}}{}^\nu$ also appears in the spatial tetrad projections. Therefore, the fact that $\delta^0_i = 0$ does not mean that the electromagnetic sector is completely unrelated to the spatial tetrad geometry.

This result will be used within the scope of the work only as a symbolic consistency indicator. Since the gravitational sector is not chosen explicitly, the explicit form of $\mathcal{E}_i{}^\nu$ will not be written in this work. Nevertheless, Eq. (346) shows that the electromagnetic energy-momentum must be treated together with balancing geometric contributions in the spatial tetrad projections.

8 Appendix-8: Reissner–Nordström Geometry and Normalization Test

Reissner–Nordström geometry describes, in the Einstein–Maxwell system, the exterior spacetime of a static, spherically symmetric mass carrying electric charge. This solution is one of the fundamental examples showing that the electromagnetic field does not remain merely a test field, but produces backreaction on geometry through its energy-momentum tensor.

8.1 Reissner–Nordström Metric

In SI units, the Reissner–Nordström line element is written as

$$ds^2 = -f_{\text{RN}}(r)c^2 dt^2 + f_{\text{RN}}(r)^{-1} dr^2 + r^2 d\Omega^2 \quad (348)$$

Here

$$d\Omega^2 = d\theta^2 + \sin^2 \theta d\varphi^2 \quad (349)$$

and

$$f_{\text{RN}}(r) = 1 - \frac{2GM}{c^2 r} + \frac{GQ^2}{4\pi\epsilon_0 c^4 r^2} \quad (350)$$

hold.

The term

$$-\frac{2GM}{c^2 r} \quad (351)$$

is the Schwarzschild mass contribution. By contrast, the term

$$\frac{GQ^2}{4\pi\epsilon_0 c^4 r^2} \quad (352)$$

is the geometric backreaction contribution arising from the energy-momentum of the electromagnetic field. The fact that the charge enters the metric as Q^2/r^2 , rather than as Q/r , shows that the quantity sourcing geometry is not the potential, but the electromagnetic energy-momentum tensor.

8.2 Writing the Reissner–Nordström Electrostatic Field with Temporal Torsion

In the static and spherically symmetric electrostatic case, the Coulomb potential is taken as

$$\Phi(r) = \frac{Q}{4\pi\epsilon_0 r} \quad (353)$$

In agreement with the previous notation, the electromagnetic potential one-form is given by Eq. (8). Therefore, the components are

$$A_0 = -\frac{\Phi(r)}{c}, \quad A_r = 0 \quad (354)$$

Since the field tensor is defined by Eq. (68), one writes

$$F_{r0} = \partial_r A_0 - \partial_0 A_r \quad (355)$$

In the static case, since $\partial_0 A_r = 0$,

$$F_{r0} = \partial_r \left(-\frac{\Phi(r)}{c} \right) \quad (356)$$

Thus,

$$F_{r0} = -\frac{1}{c} \partial_r \Phi(r) \quad (357)$$

is obtained. For the Coulomb potential,

$$\partial_r \Phi(r) = \partial_r \left(\frac{Q}{4\pi\epsilon_0 r} \right) \quad (358)$$

and hence

$$\partial_r \Phi(r) = -\frac{Q}{4\pi\epsilon_0 r^2} \quad (359)$$

is found. Therefore,

$$F_{r0} = \frac{Q}{4\pi\epsilon_0 c r^2} \quad (360)$$

is obtained. Due to antisymmetry,

$$F_{0r} = -F_{r0} \quad (361)$$

and therefore

$$F_{0r} = -\frac{Q}{4\pi\epsilon_0 cr^2} \quad (362)$$

is written.

Since the physical electric field component is

$$E_r = -cF_{0r} \quad (363)$$

one obtains

$$E_r = -c \left(-\frac{Q}{4\pi\epsilon_0 cr^2} \right) \quad (364)$$

and therefore

$$E_r = \frac{Q}{4\pi\epsilon_0 r^2} \quad (365)$$

This is the standard Coulomb field.

Since the relation between temporal torsion and the electromagnetic field is

$$T^0_{\mu\nu} = \kappa_{\text{EM}} F_{\mu\nu} \quad (366)$$

for the Reissner–Nordström electrostatic field, one writes

$$T^0_{\text{EM},r0} = \kappa_{\text{EM}} F_{r0} \quad (367)$$

Using Eq. (360), one obtains

$$T^0_{\text{EM},r0} = \kappa_{\text{EM}} \frac{Q}{4\pi\epsilon_0 cr^2} \quad (368)$$

Similarly, since

$$T^0_{\text{EM},0r} = \kappa_{\text{EM}} F_{0r} \quad (369)$$

one obtains

$$T^0_{\text{EM},0r} = -\kappa_{\text{EM}} \frac{Q}{4\pi\epsilon_0 cr^2} \quad (370)$$

These equalities show that the Reissner–Nordström electrostatic field can be represented by the temporal torsion component in the Maxwell-adapted tetrad framework.

8.3 Energy-Momentum Origin of the Reissner–Nordström Charge Term

Since the electromagnetic energy-momentum tensor is

$$\Theta_{\text{EM}}^{\mu\nu} = \frac{1}{\mu_0} \left(F^{\mu\alpha} F^\nu_{\alpha} - \frac{1}{4} g^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right) \quad (371)$$

the energy density in the electrostatic Reissner–Nordström field scales with the square of the field strength. At the level of order of magnitude, one may write

$$\Theta_{\text{EM}} \sim \frac{1}{\mu_0} F^2 \quad (372)$$

In the electrostatic case, since $F \sim E_r/c$, one has

$$\Theta_{\text{EM}} \sim \frac{1}{\mu_0} \frac{E_r^2}{c^2} \quad (373)$$

In SI units,

$$\frac{1}{\mu_0 c^2} = \varepsilon_0 \quad (374)$$

and therefore

$$\Theta_{\text{EM}} \sim \varepsilon_0 E_r^2 \quad (375)$$

is obtained. Since the Coulomb field is

$$E_r = \frac{Q}{4\pi\varepsilon_0 r^2} \quad (376)$$

one obtains

$$\Theta_{\text{EM}} \sim \varepsilon_0 \left(\frac{Q}{4\pi\varepsilon_0 r^2} \right)^2 \quad (377)$$

and hence

$$\Theta_{\text{EM}} \sim \frac{Q^2}{16\pi^2\varepsilon_0 r^4} \quad (378)$$

In the language of temporal torsion, the same scale is written through the relation

$$F_{\mu\nu} = \frac{1}{\kappa_{\text{EM}}} T^0{}_{\mu\nu} \quad (379)$$

Thus, the expression

$$\Theta_{\text{EM}} \sim \frac{1}{\mu_0} F^2 \quad (380)$$

is read as

$$\Theta_{\text{EM}} \sim \frac{1}{\mu_0 \kappa_{\text{EM}}^2} (T_{\text{EM}}^0)^2 \quad (381)$$

Since the component T_{EM}^0 scales as Q/r^2 , the energy-momentum tensor is of order Q^2/r^4 .

In the Reissner–Nordström electrostatic field, the magnetic field vanishes and only the radial electric-field component is present:

$$B_1 = B_2 = B_3 = 0, \quad E_{\hat{r}} = \frac{Q}{4\pi\varepsilon_0 r^2}, \quad E_{\hat{\theta}} = E_{\hat{\varphi}} = 0. \quad (382)$$

In this case, the explicit 4×4 form of the electromagnetic energy-momentum tensor in the local orthonormal frame is

$$\Theta_{\text{EM}}^{\hat{a}\hat{b}} = \begin{pmatrix} \frac{\varepsilon_0}{2} \left(\frac{Q}{4\pi\varepsilon_0 r^2} \right)^2 & 0 & 0 & 0 \\ 0 & -\frac{\varepsilon_0}{2} \left(\frac{Q}{4\pi\varepsilon_0 r^2} \right)^2 & 0 & 0 \\ 0 & 0 & \frac{\varepsilon_0}{2} \left(\frac{Q}{4\pi\varepsilon_0 r^2} \right)^2 & 0 \\ 0 & 0 & 0 & \frac{\varepsilon_0}{2} \left(\frac{Q}{4\pi\varepsilon_0 r^2} \right)^2 \end{pmatrix} \quad (383)$$

Here, the negative sign of the radial pressure comes from the fact that the electric field creates tension in the radial direction. The tangential components have positive signs.

Therefore, the charge contribution in the Reissner–Nordström metric function arises from the quadratic energy-momentum tensor of the electromagnetic field strength represented by temporal torsion sourcing the gravitational field equations. In other words, the term

$$\frac{GQ^2}{4\pi\varepsilon_0 c^4 r^2} \quad (384)$$

in the metric function comes not from the component T_{EM}^0 itself, but from its quadratic energy-momentum content.

8.4 κ_{EM} Normalization from the Reissner–Nordström Metric

Now it is tested whether the coefficient κ_{EM} is compatible with the Reissner–Nordström charge term. The electromagnetic expression placed in the temporal component within the Maxwell-adapted tetrad and the second-order geometric contribution arising from charge are measured by

$$\left(\frac{\kappa_{\text{EM}} \Phi(r)}{c} \right)^2 \quad (385)$$

Substituting the Coulomb potential Eq. (353), one obtains

$$\left(\frac{\kappa_{\text{EM}} \Phi(r)}{c} \right)^2 = \left[\frac{\kappa_{\text{EM}}}{c} \frac{Q}{4\pi\varepsilon_0 r} \right]^2 \quad (386)$$

From this,

$$\left(\frac{\kappa_{\text{EM}} \Phi(r)}{c} \right)^2 = \frac{\kappa_{\text{EM}}^2 Q^2}{16\pi^2 \varepsilon_0^2 c^2 r^2}. \quad (387)$$

This expression is required to have the same normalization as the standard charge term in the Reissner–Nordström metric function:

$$\frac{\kappa_{\text{EM}}^2 Q^2}{16\pi^2 \varepsilon_0^2 c^2 r^2} = \frac{GQ^2}{4\pi\varepsilon_0 c^4 r^2}. \quad (388)$$

The factor Q^2/r^2 cancels on both sides:

$$\frac{\kappa_{\text{EM}}^2}{16\pi^2\varepsilon_0^2c^2} = \frac{G}{4\pi\varepsilon_0c^4}. \quad (389)$$

From this,

$$\kappa_{\text{EM}}^2 = 16\pi^2\varepsilon_0^2c^2 \frac{G}{4\pi\varepsilon_0c^4} \quad (390)$$

is obtained. After simplification,

$$\kappa_{\text{EM}}^2 = \frac{4\pi G\varepsilon_0}{c^2} \quad (391)$$

is found. Therefore,

$$\kappa_{\text{EM}} = \frac{\sqrt{4\pi G\varepsilon_0}}{c} \quad (392)$$

is obtained.

This result is identical to Eq. (336), obtained from the unified-action normalization. Therefore, the coefficient κ_{EM} is obtained in the same form both from the relative normalization of the Einstein–Maxwell action and from the geometric normalization of the Reissner–Nordström charge term.

The point to be noted in this comparison is the following. The naive temporal tetrad

$$e^0 = \left(1 - \kappa_{\text{EM}} \frac{\Phi}{c}\right) dx^0 \quad (393)$$

does not by itself produce the full Reissner–Nordström metric. This is because this expression gives

$$g_{00} = - \left(1 - \kappa_{\text{EM}} \frac{\Phi}{c}\right)^2 \quad (394)$$

When this square is written explicitly,

$$g_{00} = - \left[1 - 2\kappa_{\text{EM}} \frac{\Phi}{c} + \kappa_{\text{EM}}^2 \frac{\Phi^2}{c^2}\right] \quad (395)$$

is obtained. Since the Coulomb potential satisfies $\Phi \sim Q/r$, this expression contains a term of linear order Q/r . By contrast, the Reissner–Nordström charge contribution is of order Q^2/r^2 . Therefore, the operation performed here is not the claim that the full metric solution is derived directly from the naive temporal tetrad. What is shown is that the second-order geometric normalization of the electromagnetic contribution is consistent with the Reissner–Nordström charge term.

9 Appendix-9: On the Derivation of the Particle Equation of Motion

The equation of motion for a particle can be derived both with the help of the Lagrangian and the metric. In this work, it is obtained naturally from the

Lagrangian structure and also from the metric structure, since A_μ is embedded in the tetrad. This will be shown explicitly in this appendix.

9.1 Derivation of the Particle Equation of Motion from the Temporal Tetrad Action

In this appendix, it will be shown in detail how the particle equation of motion used in the main text is obtained through the temporal tetrad action. The aim is to examine explicitly whether the motion of a charged particle reduces to the standard Lorentz force law when the electromagnetic potential is encoded not as an independent U(1) connection, but in the deviation of the temporal tetrad component from the reference temporal form.

In classical electromagnetic theory, the interaction between a charged particle and the electromagnetic field is written directly through the electromagnetic potential one-form $A = A_\mu dx^\mu$.

9.1.1 Temporal Tetrad Decomposition and Particle Interaction

The Maxwell-adapted temporal tetrad one-form is of the form Eq. 13. In this work, however, the fundamental geometric object is the tetrad field. The electromagnetic potential is encoded not as an independent structure, but within the temporal tetrad component. Therefore, when particle motion is examined, the starting point should not be directly A_μ , but the temporal tetrad one-form e^0 .

That is,

$$S_{\text{int}}^{(e^0)} = \frac{q}{\kappa_{\text{EM}}} \int_{\gamma} e^0. \quad (396)$$

Here γ is the worldline of the particle. Using Eq. (13), one obtains

$$S_{\text{int}}^{(e^0)} = \frac{q}{\kappa_{\text{EM}}} \int_{\gamma} (u + \kappa_{\text{EM}} A) = q \int_{\gamma} A + \frac{q}{\kappa_{\text{EM}}} \int_{\gamma} u.$$

Thus, the interaction written through e^0 contains the standard electromagnetic minimal-coupling term and, in addition, a contribution arising from the reference temporal one-form.

The important point here is the following: the interaction action is not postulated directly through A_μ . It is first written through the fundamental geometric variable e^0 , and then, by using the decomposition $e^0 = u + \kappa_{\text{EM}} A$, it is shown how the standard electromagnetic interaction emerges at the level of the equation of motion.

9.1.2 Variation of the $\int_{\gamma} u$ Term

Here, the contribution of the term

$$\int_{\gamma} u \quad (397)$$

to the worldline variation will be examined. For a general one-form

$$B = B_\mu dx^\mu \quad (398)$$

the worldline integral is written as

$$I_B = \int_{\lambda_1}^{\lambda_2} B_\mu(x) \frac{dx^\mu}{d\lambda} d\lambda \quad (399)$$

Here λ is an arbitrary parameter chosen on the worldline. Defining briefly

$$\dot{x}^\mu \equiv \frac{dx^\mu}{d\lambda} \quad (400)$$

one has

$$I_B = \int_{\lambda_1}^{\lambda_2} B_\mu(x) \dot{x}^\mu d\lambda. \quad (401)$$

The variation of this expression is

$$\delta I_B = \int_{\lambda_1}^{\lambda_2} [\partial_\nu B_\mu \delta x^\nu \dot{x}^\mu + B_\mu \delta \dot{x}^\mu] d\lambda. \quad (402)$$

The second term is integrated by parts:

$$\int_{\lambda_1}^{\lambda_2} B_\mu \delta \dot{x}^\mu d\lambda = [B_\mu \delta x^\mu]_{\lambda_1}^{\lambda_2} - \int_{\lambda_1}^{\lambda_2} \frac{dB_\mu}{d\lambda} \delta x^\mu d\lambda = [B_\mu \delta x^\mu]_{\lambda_1}^{\lambda_2} - \int_{\lambda_1}^{\lambda_2} \partial_\nu B_\mu \dot{x}^\nu \delta x^\mu d\lambda.$$

Since the endpoints are held fixed,

$$\delta x^\mu(\lambda_1) = \delta x^\mu(\lambda_2) = 0 \quad (403)$$

and the boundary term vanishes. Thus,

$$\delta I_B = \int_{\lambda_1}^{\lambda_2} [\partial_\nu B_\mu \dot{x}^\mu \delta x^\nu - \partial_\nu B_\mu \dot{x}^\nu \delta x^\mu] d\lambda. \quad (404)$$

If the dummy indices in the second term are renamed as

$$\nu \leftrightarrow \mu \quad (405)$$

then

$$\delta I_B = \int_{\lambda_1}^{\lambda_2} [\partial_\mu B_\nu - \partial_\nu B_\mu] \dot{x}^\nu \delta x^\mu d\lambda. \quad (406)$$

Therefore,

$$\boxed{\delta \int_\gamma B = \int_\gamma (dB)_{\mu\nu} \dot{x}^\nu \delta x^\mu d\lambda} \quad (407)$$

is obtained. Here

$$(dB)_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \quad (408)$$

is the exterior derivative of the one-form B .

Applying this result to $B = u$, one finds

$$\delta \int_{\gamma} u = \int_{\gamma} (du)_{\mu\nu} \dot{x}^{\nu} \delta x^{\mu} d\lambda. \quad (409)$$

If the reference temporal form is closed,

$$du = 0, \quad (410)$$

then

$$\boxed{\delta \int_{\gamma} u = 0} \quad (411)$$

is obtained. Therefore, under the condition $du = 0$,

$$\delta \left(\frac{q}{\kappa_{\text{EM}}} \int_{\gamma} e^0 \right) = \delta \left(q \int_{\gamma} A \right) \quad (412)$$

holds. This equality is valid not at the level of the absolute value of the actions, but at the level of the worldline variation and hence the equation of motion. This is because the term $\int_{\gamma} u$ produces no force under fixed endpoints when $du = 0$.

9.1.3 Variation of the Particle Action

For a charged relativistic particle, the total action is

$$S_p = S_{\text{free}} + S_{\text{int}}^{(e^0)} \quad (413)$$

The free particle action is given by

$$S_{\text{free}} = -mc \int_{\gamma} ds = -mc^2 \int_{\gamma} d\tau \quad (414)$$

Here

$$ds = c d\tau \quad (415)$$

has been used. The interaction action is

$$S_{\text{int}}^{(e^0)} = \frac{q}{\kappa_{\text{EM}}} \int_{\gamma} e^0. \quad (416)$$

Under the condition $du = 0$, at the level of the equation of motion,

$$\delta S_{\text{int}}^{(e^0)} = \delta \left(q \int_{\gamma} A \right) \quad (417)$$

and therefore the interaction variation gives the same force term as the standard minimal-coupling variation in deriving the particle equation of motion.

Let the worldline be parametrized by the proper time τ :

$$U^\mu = \frac{dx^\mu}{d\tau}. \quad (418)$$

Thus, the electromagnetic interaction is

$$q \int_\gamma A = q \int_{\tau_1}^{\tau_2} A_\mu(x) U^\mu d\tau. \quad (419)$$

Using Eq. (407) directly for $B = A$, one obtains

$$\delta \left(q \int_\gamma A \right) = q \int_{\tau_1}^{\tau_2} F_{\mu\nu} U^\nu \delta x^\mu d\tau, \quad (420)$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (421)$$

The variation of the free particle action is standardly written as

$$\delta S_{\text{free}} = - \int_{\tau_1}^{\tau_2} m U^\nu \nabla_\nu U_\mu \delta x^\mu d\tau \quad (422)$$

Thus, the total variation is

$$\delta S_p = \delta S_{\text{free}} + \delta S_{\text{int}}^{(e^0)} = \int_{\tau_1}^{\tau_2} [-m U^\nu \nabla_\nu U_\mu + q F_{\mu\nu} U^\nu] \delta x^\mu d\tau.$$

From the condition $\delta S_p = 0$ and the arbitrariness of δx^μ , the equation of motion is obtained as

$$m U^\nu \nabla_\nu U_\mu = q F_{\mu\nu} U^\nu \quad (423)$$

This is the covariant Lorentz force law. If this expression is inserted into the Lorentz force equation, one finds

$$m U^\nu \nabla_\nu U_\mu = \frac{q}{\kappa_{\text{EM}}} T^0_{\mu\nu} U^\nu \quad (424)$$

This equation shows that the electromagnetic force acting on a charged particle can be interpreted as the effect of temporal torsion on the particle four-velocity.

9.1.4 From the Covariant Equation to the Three-Dimensional Lorentz Force

Now it will be shown how Eq. (424) reduces to the standard three-dimensional Lorentz force in the low-velocity and weak-field limit.

From the covariant Lorentz force equation, for the spatial components one writes

$$m \frac{dU^i}{d\tau} = q F^i_{\nu} U^\nu. \quad (425)$$

The right-hand side is explicitly

$$qF^i{}_\nu U^\nu = q(F^i{}_0 U^0 + F^i{}_j U^j) = q \left[\left(\frac{E_i}{c} \right) (c\gamma) + (\epsilon_{ijk} B^k) (\gamma v^j) \right] = q\gamma [E_i + \epsilon_{ijk} v^j B^k].$$

Since the vector product is defined by

$$(\mathbf{v} \times \mathbf{B})_i = \epsilon_{ijk} v^j B^k \quad (426)$$

one has

$$qF^i{}_\nu U^\nu = q\gamma [E_i + (\mathbf{v} \times \mathbf{B})_i]. \quad (427)$$

On the left-hand side,

$$U^i = \gamma v^i \quad (428)$$

and hence

$$\frac{dU^i}{d\tau} = \frac{d}{d\tau} (\gamma v^i) = \gamma \frac{d}{dt} (\gamma v^i). \quad (429)$$

Therefore, the fully relativistic three-dimensional equation can be written as

$$m\gamma \frac{d}{dt} (\gamma v^i) = q\gamma [E_i + (\mathbf{v} \times \mathbf{B})_i]. \quad (430)$$

Dividing both sides by γ , one obtains

$$\frac{d}{dt} (m\gamma v^i) = q [E_i + (\mathbf{v} \times \mathbf{B})_i]. \quad (431)$$

Since the momentum is $p^i = m\gamma v^i$,

$$\frac{d\mathbf{p}}{dt} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (432)$$

is found. In the low-velocity limit, since $\gamma \simeq 1$ and $\mathbf{p} \simeq m\mathbf{v}$,

$$m \frac{d^2 \mathbf{x}}{dt^2} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (433)$$

and therefore

$$\boxed{\frac{d^2 \mathbf{x}}{dt^2} = \frac{q}{m} (\mathbf{E} + \mathbf{v} \times \mathbf{B})} \quad (434)$$

is obtained.

This result shows that the acceleration response of the particle to the electromagnetic field is not universal; it is determined by the particle's charge-to-mass ratio q/m .

9.1.5 Writing in Terms of Temporal Torsion

The electromagnetic field can be written in terms of temporal torsion as in Eq. 72. Therefore, the electric and magnetic field components can also be expressed in terms of temporal torsion components. As shown earlier in Section 101, if the electric and magnetic counterparts of temporal torsion are defined as

$$(T_E)_i \equiv -cT^0_{0i} \quad (435)$$

and

$$(T_B)^k \equiv \frac{1}{2}\epsilon^{kij}T^0_{ij} \quad (436)$$

then

$$E_i = \frac{1}{\kappa_{\text{EM}}}(T_E)_i, \quad B^i = \frac{1}{\kappa_{\text{EM}}}(T_B)^i \quad (437)$$

are obtained. Thus, the low-velocity Lorentz force becomes

$$\frac{d^2\mathbf{x}}{dt^2} = \frac{q}{m\kappa_{\text{EM}}}(\mathbf{T}_E + \mathbf{v} \times \mathbf{T}_B) \quad (438)$$

This expression does not change the physical content of the standard Lorentz force. It only shows that the electromagnetic fields are geometrically rewritten in terms of temporal tetrad torsion. Thus, at the level of particle motion, the electromagnetic force can also be represented as a torsion effect originating from temporal tetrad geometry.

9.2 Decomposition of the Electromagnetic Contribution from the Metric Connection and the Lorentz Force

In this appendix, the electromagnetic contribution produced by the Maxwell-adapted temporal tetrad on the metric will be examined, and it will be shown how this contribution corresponds to the standard Lorentz force under the appropriate coupling decomposition. The aim here is to reveal explicitly which electromagnetic field combination the Levi-Civita connection produces when the electromagnetic potential is represented within the temporal tetrad.

First, the reference part of the temporal tetrad and the electromagnetic part will be separated. This separation is necessary to show that the mass and charge couplings couple to different parts of the same tetrad. The metric components will be derived from the Maxwell-adapted tetrad and linearized in the weak-field limit. Then the electromagnetic contributions of the Levi-Civita connection will be calculated from this metric. Afterwards, the spatial geodesic combination in the low-velocity limit will be written, it will be shown that this combination gives the electromagnetic fields, and at the final stage the Lorentz force will be obtained by applying the physical charge coupling.

In this work, the electromagnetic interaction is not interpreted as pure metric geodesic motion. While the mass sector determines the inertial response of the particle, the charge sector couples to the part of the temporal tetrad containing the electromagnetic potential. Therefore, the physical Lorentz force is

obtained as a result of selecting, with the charge coupling, the electromagnetic field combination appearing in the metric connection.

9.2.1 Temporal Tetrad and Coupling Decomposition

The Maxwell-adapted temporal tetrad is given by Eq. (13). This expression can be written as

$$e^0 = dx^0 + \kappa_{\text{EM}} A_\mu dx^\mu \quad (439)$$

Here, when $x^0 = ct$ is taken,

$$dx^0 = c dt \quad (440)$$

Thus, the temporal tetrad can be read as two parts:

$$e^0 = \underbrace{dx^0}_{m \text{ sector}} + \underbrace{\kappa_{\text{EM}} A_\mu dx^\mu}_{q \text{ sector}} \quad (441)$$

This separation is important for the physical interpretation. The mass m determines the inertial response in the particle action through proper time or the reference temporal flow. Therefore, the mass sector is associated with

$$m \longleftrightarrow dx^0 = c dt. \quad (442)$$

By contrast, the charge q couples to the electromagnetic potential:

$$q \longleftrightarrow A = A_\mu dx^\mu. \quad (443)$$

Therefore, e^0 should not be taken as a single universal geodesic coupling. Within the same temporal tetrad, mass and charge select different physical parts. While mass determines inertial motion through the dx^0 part, charge determines the electromagnetic interaction through the $A_\mu dx^\mu$ part.

If this separation is not made, the pure metric geodesic equation gives only the geometric effect of the connection; however, the particle charge q has not yet been taken into account. Therefore, below, first the electromagnetic part of the metric connection will be calculated, and then this contribution will be weighted by the charge coupling.

9.2.2 Mass Sector and Reference Inertial Motion

First, let us consider only the reference temporal structure seen by the mass sector. Since this structure couples with mass, the electromagnetic potential is switched off:

$$A_\mu = 0. \quad (444)$$

The temporal tetrad becomes

$$e^0 = dx^0 = c dt. \quad (445)$$

If the spatial tetrads are chosen in the flat background form as

$$e^i = dx^i \quad (446)$$

the metric becomes

$$ds^2 = -(dx^0)^2 + \delta_{ij} dx^i dx^j. \quad (447)$$

Using $x^0 = ct$, one obtains

$$ds^2 = -c^2 dt^2 + \delta_{ij} dx^i dx^j. \quad (448)$$

In this case, the metric components are constant:

$$\partial_\lambda g_{\mu\nu} = 0. \quad (449)$$

Therefore, from the Levi-Civita connection

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2} g^{\mu\rho} (\partial_\alpha g_{\rho\beta} + \partial_\beta g_{\rho\alpha} - \partial_\rho g_{\alpha\beta}) \quad (450)$$

one finds

$$\Gamma^\mu_{\alpha\beta} = 0. \quad (451)$$

Thus, from the geodesic equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0 \quad (452)$$

one obtains

$$\frac{d^2 x^\mu}{d\tau^2} = 0 \quad (453)$$

This result shows that the mass sector alone does not produce electromagnetic acceleration. Electromagnetic acceleration emerges when the connection contribution coming from the A_μ -dependent part of the temporal tetrad and the charge coupling are considered together.

9.2.3 Writing the Metric Components in Terms of the Electromagnetic Potential

Now let the electromagnetic part of the temporal tetrad be included. The tetrad metric is defined by

$$g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu. \quad (454)$$

When the spatial tetrads are chosen in the flat background form,

$$e^i = dx^i, \quad e^i_\mu = \delta^i_\mu \quad (455)$$

is obtained.

The temporal tetrad components are

$$e^0_\mu = \delta^0_\mu + \kappa_{\text{EM}} A_\mu \quad (456)$$

If the electromagnetic potential is written as

$$A_\mu = \left(-\frac{\phi}{c}, A_i \right) \quad (457)$$

then

$$e^0_0 = 1 - \kappa_{\text{EM}} \frac{\phi}{c}, \quad e^0_i = \kappa_{\text{EM}} A_i \quad (458)$$

are obtained.

Since the signature is $(-+++)$, the metric components are given by

$$g_{\mu\nu} = -e^0_\mu e^0_\nu + \delta_{ij} e^i_\mu e^j_\nu \quad (459)$$

Now the components are calculated one by one.

First, for the g_{00} component,

$$\begin{aligned} g_{00} &= -e^0_0 e^0_0 + \delta_{ij} e^i_0 e^j_0 \\ &= -\left(e^0_0\right)^2 \end{aligned} \quad (460)$$

is written. Here, for the flat spatial tetrads,

$$e^i_0 = 0 \quad (461)$$

has been used. Substituting Eq. (458), one obtains

$$g_{00} = -\left(1 - \kappa_{\text{EM}} \frac{\phi}{c}\right)^2 \quad (462)$$

Second, the g_{0i} component is

$$\begin{aligned} g_{0i} &= -e^0_0 e^0_i + \delta_{kl} e^k_0 e^l_i \\ &= -e^0_0 e^0_i \end{aligned} \quad (463)$$

Here also,

$$e^k_0 = 0 \quad (464)$$

has been used. Thus,

$$g_{0i} = -\left(1 - \kappa_{\text{EM}} \frac{\phi}{c}\right) \kappa_{\text{EM}} A_i \quad (465)$$

is found.

Finally, the g_{ij} component is calculated:

$$\begin{aligned} g_{ij} &= -e^0_i e^0_j + \delta_{kl} e^k_i e^l_j \\ &= -\kappa_{\text{EM}}^2 A_i A_j + \delta_{kl} \delta^k_i \delta^l_j \\ &= \delta_{ij} - \kappa_{\text{EM}}^2 A_i A_j. \end{aligned} \quad (466)$$

Therefore, the full metric components are obtained as

$$g_{00} = -\left(1 - \kappa_{\text{EM}} \frac{\phi}{c}\right)^2, \quad (467)$$

$$g_{0i} = -\left(1 - \kappa_{\text{EM}} \frac{\phi}{c}\right) \kappa_{\text{EM}} A_i, \quad (468)$$

$$g_{ij} = \delta_{ij} - \kappa_{\text{EM}}^2 A_i A_j. \quad (469)$$

9.2.4 Linear Weak-Field Limit and Inverse Metric

In the weak geometric backreaction limit,

$$\left| \kappa_{\text{EM}} \frac{\phi}{c} \right| \ll 1, \quad |\kappa_{\text{EM}} A_i| \ll 1 \quad (470)$$

is assumed. Therefore, only terms of order $\mathcal{O}(\kappa_{\text{EM}})$ will be kept.

First, g_{00} is expanded:

$$\begin{aligned} g_{00} &= - \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right)^2 \\ &= - \left[1 - 2\kappa_{\text{EM}} \frac{\phi}{c} + \kappa_{\text{EM}}^2 \frac{\phi^2}{c^2} \right] \\ &= -1 + 2\kappa_{\text{EM}} \frac{\phi}{c} + \mathcal{O}(\kappa_{\text{EM}}^2). \end{aligned} \quad (471)$$

For the g_{0i} component,

$$\begin{aligned} g_{0i} &= - \left(1 - \kappa_{\text{EM}} \frac{\phi}{c} \right) \kappa_{\text{EM}} A_i \\ &= -\kappa_{\text{EM}} A_i + \kappa_{\text{EM}}^2 \frac{\phi}{c} A_i \\ &= -\kappa_{\text{EM}} A_i + \mathcal{O}(\kappa_{\text{EM}}^2) \end{aligned} \quad (472)$$

is obtained.

For the g_{ij} component, since

$$g_{ij} = \delta_{ij} - \kappa_{\text{EM}}^2 A_i A_j \quad (473)$$

at linear order one has

$$g_{ij} = \delta_{ij} + \mathcal{O}(\kappa_{\text{EM}}^2) \quad (474)$$

Thus, the linear metric to be used in the connection calculation is written as

$$g_{00} = -1 + 2\kappa_{\text{EM}} \frac{\phi}{c}, \quad g_{0i} = -\kappa_{\text{EM}} A_i, \quad g_{ij} = \delta_{ij} \quad (475)$$

To find the inverse metric, define

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}. \quad (476)$$

Here,

$$h_{00} = 2\kappa_{\text{EM}} \frac{\phi}{c}, \quad h_{0i} = -\kappa_{\text{EM}} A_i, \quad h_{ij} = 0 \quad (477)$$

holds. At linear order, the inverse metric is given by

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + \mathcal{O}(h^2). \quad (478)$$

When indices are raised and lowered with the background metric

$$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1), \quad (479)$$

one finds

$$\begin{aligned} h^{00} &= \eta^{0\alpha} \eta^{0\beta} h_{\alpha\beta} \\ &= \eta^{00} \eta^{00} h_{00} \\ &= h_{00} \\ &= 2\kappa_{\text{EM}} \frac{\phi}{c}, \end{aligned} \quad (480)$$

$$\begin{aligned} h^{0i} &= \eta^{00} \eta^{ij} h_{0j} \\ &= -\delta^{ij} h_{0j} \\ &= \kappa_{\text{EM}} A^i, \end{aligned} \quad (481)$$

and

$$h^{ij} = 0 \quad (482)$$

Here, since spatial indices are raised with the flat metric,

$$A^i = \delta^{ij} A_j \quad (483)$$

has been used.

Thus, the inverse metric components are obtained as

$$\begin{aligned} g^{00} &= \eta^{00} - h^{00} + \mathcal{O}(\kappa_{\text{EM}}^2) \\ &= -1 - 2\kappa_{\text{EM}} \frac{\phi}{c} + \mathcal{O}(\kappa_{\text{EM}}^2), \end{aligned} \quad (484)$$

$$\begin{aligned} g^{0i} &= \eta^{0i} - h^{0i} + \mathcal{O}(\kappa_{\text{EM}}^2) \\ &= -\kappa_{\text{EM}} A^i + \mathcal{O}(\kappa_{\text{EM}}^2), \end{aligned} \quad (485)$$

and

$$g^{ij} = \delta^{ij} + \mathcal{O}(\kappa_{\text{EM}}^2) \quad (486)$$

The connection is already of order $\mathcal{O}(\kappa_{\text{EM}})$ because it comes from derivatives of the metric. Therefore, inverse metric components such as g^{0i} , which are of order $\mathcal{O}(\kappa_{\text{EM}})$, give $\mathcal{O}(\kappa_{\text{EM}}^2)$ contributions when multiplied by metric derivatives. These contributions are neglected in the linear calculation.

9.2.5 Electromagnetic Contribution in the Levi-Civita Connection

The Levi-Civita connection is given by

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2} g^{\alpha\beta} (\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}) \quad (487)$$

In the low-velocity limit, the electromagnetic connection components needed for the spatial equation of motion are

$$\Gamma^i_{00}, \quad \Gamma^i_{0j}. \quad (488)$$

First, Γ^i_{00} is calculated:

$$\begin{aligned} \Gamma^i_{00} &= \frac{1}{2} g^{i\beta} (\partial_0 g_{\beta 0} + \partial_0 g_{\beta 0} - \partial_\beta g_{00}) \\ &= \frac{1}{2} g^{i\beta} (2\partial_0 g_{\beta 0} - \partial_\beta g_{00}). \end{aligned} \quad (489)$$

At linear order,

$$g^{ij} \simeq \delta^{ij}, \quad g^{i0} = \mathcal{O}(\kappa_{\text{EM}}) \quad (490)$$

and therefore only the $\beta = j$ contribution is kept:

$$\begin{aligned} \Gamma^i_{00} &= \frac{1}{2} \delta^{ij} (2\partial_0 g_{j0} - \partial_j g_{00}) \\ &= \delta^{ij} \partial_0 g_{j0} - \frac{1}{2} \delta^{ij} \partial_j g_{00} \\ &= \partial_0 g_{i0} - \frac{1}{2} \partial_i g_{00}. \end{aligned} \quad (491)$$

From the linear metric components,

$$g_{i0} = -\kappa_{\text{EM}} A_i \quad (492)$$

and hence

$$\partial_0 g_{i0} = -\kappa_{\text{EM}} \partial_0 A_i \quad (493)$$

is found. Moreover, since

$$g_{00} = -1 + 2\kappa_{\text{EM}} \frac{\phi}{c}, \quad (494)$$

one obtains

$$\partial_i g_{00} = 2\kappa_{\text{EM}} \frac{1}{c} \partial_i \phi. \quad (495)$$

Substituting these into Eq. (491) gives

$$\begin{aligned} \Gamma^i_{00} &= -\kappa_{\text{EM}} \partial_0 A_i - \frac{1}{2} \left(2\kappa_{\text{EM}} \frac{1}{c} \partial_i \phi \right) \\ &= -\kappa_{\text{EM}} \partial_0 A_i - \kappa_{\text{EM}} \frac{1}{c} \partial_i \phi \\ &= -\kappa_{\text{EM}} \left(\partial_0 A_i + \frac{1}{c} \partial_i \phi \right). \end{aligned} \quad (496)$$

On the other hand, since

$$A_0 = -\frac{\phi}{c} \quad (497)$$

one has

$$\begin{aligned}
F_{0i} &= \partial_0 A_i - \partial_i A_0 \\
&= \partial_0 A_i - \partial_i \left(-\frac{\phi}{c} \right) \\
&= \partial_0 A_i + \frac{1}{c} \partial_i \phi.
\end{aligned} \tag{498}$$

Thus,

$$\Gamma^i_{00} = -\kappa_{\text{EM}} F_{0i} \tag{499}$$

is obtained. In the convention used,

$$F_{0i} = -\frac{E_i}{c} \tag{500}$$

and therefore

$$\Gamma^i_{00} = \kappa_{\text{EM}} \frac{E_i}{c} \tag{501}$$

is found.

Now Γ^i_{0j} is calculated:

$$\Gamma^i_{0j} = \frac{1}{2} g^{i\beta} (\partial_0 g_{\beta j} + \partial_j g_{\beta 0} - \partial_\beta g_{0j}). \tag{502}$$

At linear order, only the spatial contribution $\beta = k$ is again kept:

$$\Gamma^i_{0j} = \frac{1}{2} \delta^{ik} (\partial_0 g_{kj} + \partial_j g_{k0} - \partial_k g_{0j}). \tag{503}$$

At this order,

$$g_{kj} = \delta_{kj} \tag{504}$$

and therefore

$$\partial_0 g_{kj} = 0 \tag{505}$$

is written. In addition,

$$g_{k0} = -\kappa_{\text{EM}} A_k, \quad g_{0j} = -\kappa_{\text{EM}} A_j \tag{506}$$

and therefore

$$\partial_j g_{k0} = -\kappa_{\text{EM}} \partial_j A_k, \quad \partial_k g_{0j} = -\kappa_{\text{EM}} \partial_k A_j \tag{507}$$

are obtained. Thus,

$$\begin{aligned}
\Gamma^i_{0j} &= \frac{1}{2} \delta^{ik} (-\kappa_{\text{EM}} \partial_j A_k + \kappa_{\text{EM}} \partial_k A_j) \\
&= \frac{\kappa_{\text{EM}}}{2} \delta^{ik} (\partial_k A_j - \partial_j A_k) \\
&= \frac{\kappa_{\text{EM}}}{2} (\partial_i A_j - \partial_j A_i).
\end{aligned} \tag{508}$$

Since the spatial component of the field tensor is

$$F_{ij} = \partial_i A_j - \partial_j A_i \tag{509}$$

one finds

$$\Gamma^i_{0j} = \frac{\kappa_{\text{EM}}}{2} F_{ij} \tag{510}$$

9.2.6 Magnetic Field Convention and Index Structure

In order to compare the result with the vectorial Lorentz force, the magnetic field convention must be stated explicitly. In this appendix,

$$B^i = \frac{1}{2}\epsilon^{ijk}F_{jk}, \quad F_{ij} = \epsilon_{ijk}B^k \quad (511)$$

will be taken. This convention, together with the choice $F_{0i} = -E_i/c$, gives the standard Lorentz-force sign.

In this case,

$$\begin{aligned} F_{ij}v^j &= \epsilon_{ijk}B^kv^j \\ &= -\epsilon_{ijk}v^kB^j \\ &= (\mathbf{v} \times \mathbf{B})_i \end{aligned} \quad (512)$$

can be written. Since spatial indices are lowered and raised with the flat background metric δ_{ij} ,

$$E_i = \delta_{ij}E^j, \quad B_i = \delta_{ij}B^j, \quad v_i = \delta_{ij}v^j \quad (513)$$

holds. Therefore, the position of the spatial indices does not change the sign.

9.2.7 Geodesic Combination in the Low-Velocity Limit

The spatial geodesic equation is

$$\frac{d^2x^i}{d\tau^2} + \Gamma^i_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (514)$$

In the low-velocity limit, when parametrization by t is used,

$$\frac{dx^0}{dt} = c, \quad \frac{dx^i}{dt} = v^i \quad (515)$$

are taken. In this case, the dominant electromagnetic connection combination in the spatial geodesic equation is

$$c^2\Gamma^i_{00} + 2c\Gamma^i_{0j}v^j \quad (516)$$

Therefore, the acceleration contribution arising from the electromagnetic connection in the pure geodesic equation is

$$\left. \frac{d^2x^i}{dt^2} \right|_{\text{geo,EM}} = -[c^2\Gamma^i_{00} + 2c\Gamma^i_{0j}v^j]_{\text{EM}} \quad (517)$$

Now Eq. (501) and Eq. (510) are used:

$$\begin{aligned} [c^2\Gamma^i_{00} + 2c\Gamma^i_{0j}v^j]_{\text{EM}} &= c^2 \left(\kappa_{\text{EM}} \frac{E_i}{c} \right) + 2c \left(\frac{\kappa_{\text{EM}}}{2} F_{ij} \right) v^j \\ &= \kappa_{\text{EM}} c E_i + \kappa_{\text{EM}} c F_{ij} v^j \\ &= \kappa_{\text{EM}} c (E_i + F_{ij} v^j). \end{aligned} \quad (518)$$

This expression shows that the part of the metric connection coming from the electromagnetic potential produces the combination of electric and magnetic fields. However, if it is inserted directly into the pure geodesic equation,

$$\left. \frac{d^2 x^i}{dt^2} \right|_{\text{geo,EM}} = -\kappa_{\text{EM}} c (E_i + F_{ij} v^j) \quad (519)$$

is obtained. This expression is not the physical Lorentz force; it only shows the electromagnetic part of the metric connection. For the physical force, the particle's charge coupling must also be applied.

9.2.8 Physical Coupling Decomposition and the Lorentz Force

The coupling decomposition is written as

$$e^0 = \underbrace{dx^0}_{m \text{ sector}} + \underbrace{\kappa_{\text{EM}} A_\mu dx^\mu}_{q \text{ sector}}. \quad (520)$$

The mass sector gives the inertial response of the particle:

$$m \frac{d^2 x^i}{dt^2}. \quad (521)$$

The charge sector couples to the connection combination arising from the electromagnetic part of the temporal tetrad. As shown by Eq. (518), this connection combination produces the electromagnetic fields with the factor $\kappa_{\text{EM}} c$. Therefore, the physical charge coupling is selected by

$$\frac{q}{\kappa_{\text{EM}} c}. \quad (522)$$

Accordingly, the physical equation of motion is written as

$$m \frac{d^2 x^i}{dt^2} = \frac{q}{\kappa_{\text{EM}} c} [c^2 \Gamma^i_{00} + 2c \Gamma^i_{0j} v^j]_{\text{EM}} \quad (523)$$

Here, the expression in square brackets denotes only the part of the metric connection arising from the electromagnetic potential.

Substituting Eq. (518) into this expression, one obtains

$$\begin{aligned} m \frac{d^2 x^i}{dt^2} &= \frac{q}{\kappa_{\text{EM}} c} [\kappa_{\text{EM}} c (E_i + F_{ij} v^j)] \\ &= q (E_i + F_{ij} v^j). \end{aligned} \quad (524)$$

Thus, in indexed form,

$$m \frac{d^2 x^i}{dt^2} = q (E_i + F_{ij} v^j) \quad (525)$$

is obtained.

Using Eq. (512), one has

$$F_{ij}v^j = (\mathbf{v} \times \mathbf{B})_i. \quad (526)$$

Therefore, the equation of motion is written in vector form as

$$m \frac{d^2 \mathbf{x}}{dt^2} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (527)$$

Consequently, the mass sector alone does not produce electromagnetic acceleration; it only determines the inertial motion of the particle. The electromagnetic field appears inside the Levi-Civita connection combination coming from the A_μ -dependent part of the temporal tetrad. When this connection contribution is coupled appropriately with the charge of the particle, the classical Lorentz force is obtained. Thus, in the low-velocity and weak-field limit, the equation of motion found with the Euler–Lagrange formalism is also reproduced consistently through the electromagnetic contribution in the metric connection.

10 Appendix-10: Derivation of the Aharonov–Bohm Phase in Terms of the Temporal Tetrad

In this appendix, it will be shown step by step how the Aharonov–Bohm phase can be written by using the Maxwell-adapted temporal tetrad decomposition. In standard quantum mechanics, the Aharonov–Bohm phase is defined by the closed-loop integral of the electromagnetic potential one-form. However, in this work, the electromagnetic potential A_μ is encoded not as an independent and external structure, but within the deviation of the temporal tetrad component from the reference temporal form. Therefore, it is necessary to examine whether the same phase can be rewritten in terms of the temporal tetrad one-form e^0 .

The basic idea here is the following: the classical Aharonov–Bohm phase is normally defined through

$$A = A_\mu dx^\mu \quad (528)$$

However, in the Maxwell-adapted tetrad ansatz,

$$e^0 = u + \kappa_{\text{EM}} A \quad (529)$$

so the electromagnetic potential is contained within the temporal tetrad. Therefore, under suitable conditions, the Aharonov–Bohm phase can also be expressed by using the closed-loop integral of e^0 , instead of the closed-loop integral of A .

10.1 Standard Expression of the Aharonov–Bohm Phase

The phase contribution acquired by a charged quantum particle under an electromagnetic potential is given by

$$\Delta\theta = \frac{1}{\hbar} S_{\text{int}} \quad (530)$$

In the Aharonov–Bohm effect, the physically measured quantity is the phase difference along the closed loop formed by two different paths. Therefore, the standard Aharonov–Bohm phase is written as

$$\Delta\theta_{\text{AB}} = \frac{q}{\hbar} \oint_{\gamma} A \quad (531)$$

10.2 Temporal Tetrad Decomposition

The Maxwell-adapted temporal tetrad one-form is decomposed as in Eq. 13. If this equality is integrated along a closed loop, one obtains

$$\oint_{\gamma} e^0 = \oint_{\gamma} (u + \kappa_{\text{EM}} A) = \oint_{\gamma} u + \kappa_{\text{EM}} \oint_{\gamma} A.$$

If this expression is inserted into the standard Aharonov–Bohm phase Eq. (531), one obtains

$$\Delta\theta_{\text{AB}} = \frac{q}{\hbar\kappa_{\text{EM}}} \oint_{\gamma} e^0 = \frac{q}{\hbar\kappa_{\text{EM}}} \left(\oint_{\gamma} u + \kappa_{\text{EM}} \oint_{\gamma} A \right)$$

In this case, u is a closed one-form. Along a closed loop, the reference temporal contribution is

$$\oint_{\gamma} u = \oint_{\gamma} dx^0 \quad (532)$$

If the closed loop corresponds to the same initial and final event, the coordinate difference is zero:

$$\oint_{\gamma} dx^0 = x_{\text{final}}^0 - x_{\text{initial}}^0 = 0. \quad (533)$$

Therefore,

$$\oint_{\gamma} u = 0 \quad (534)$$

is obtained. Under this condition, Eq. (532) simplifies:

$$\Delta\theta_{\text{AB}} = \frac{q}{\hbar\kappa_{\text{EM}}} \oint_{\gamma} e^0. \quad (535)$$

When $\oint_{\gamma} u = 0$, one has

$$\oint_{\gamma} A = \frac{1}{\kappa_{\text{EM}}} \oint_{\gamma} e^0 \quad (536)$$

Thus,

$$\Delta\theta_{\text{AB}} = \frac{q}{\hbar} \oint_{\gamma} A = \frac{q}{\hbar\kappa_{\text{EM}}} \oint_{\gamma} e^0 \quad (537)$$

is obtained.

This result shows that, in the Maxwell-adapted tetrad framework, the Aharonov–Bohm phase can be expressed by the closed-loop integral of the temporal tetrad. Here, it is not the entirety of e^0 , but the electromagnetic deviation of e^0 from the reference temporal form that carries the physical phase. Since $\oint_{\gamma} u = 0$ on the closed loop, this deviation can be read directly through $\oint_{\gamma} e^0$.

11 Appendix-11: Tetrad Diffeomorphism Gauge Test

In this section, it will be tested whether the Maxwell-adapted tetrad ansatz proposed in this work is only a coordinate choice, that is, a pure diffeomorphism gauge structure. In a tetrad-based geometry, the fundamental variable is not only the metric, but the tetrad field. Therefore, in order to understand whether a structure added to the tetrad carries physical content, it is not sufficient to look only at the metric components; the torsion obtained from the exterior derivative of the tetrad and the associated geometric invariants must also be examined.

The temporal tetrad ansatz used in this work is given by Eq. 13. Now suppose that the electromagnetic structure added to the tetrad is removed. In this case, the temporal tetrad reduces only to the background structure:

$$\bar{e}^0 = u. \quad (538)$$

Its torsion is

$$\bar{T}^0 = d\bar{e}^0 = du \quad (539)$$

In the same reference condition, since $du = 0$, one obtains

$$\bar{T}^0 = 0 \quad (540)$$

Thus, the electromagnetic field also becomes

$$\bar{F}_{\mu\nu} = \frac{1}{\kappa_{\text{EM}}} \bar{T}^0_{\mu\nu} = 0. \quad (541)$$

This result is important. This is because, when the A_μ structure in the ansatz is removed, not only the representation of the tetrad changes; the electromagnetic part of temporal torsion disappears completely.

- To see this result more precisely, let us examine the transformation

under a diffeomorphism. Under a diffeomorphism

$$x^\mu \mapsto x'^\mu(x) \quad (542)$$

the tetrad components transform as

$$e'^a_{\mu'} = \frac{\partial x^\nu}{\partial x'^{\mu'}} e^a_\nu \quad (543)$$

The torsion components transform tensorially as

$$T'^a_{\mu'\nu'} = \frac{\partial x^\rho}{\partial x'^{\mu'}} \frac{\partial x^\sigma}{\partial x'^{\nu'}} T^a_{\rho\sigma} \quad (544)$$

Therefore, whether torsion is zero or not is not dependent on the choice of coordinates. If in one frame

$$T^0_{\mu\nu} \neq 0 \quad (545)$$

then under a regular diffeomorphism this tensor only changes its components; but in general it does not become zero. Therefore, a configuration for which

$$T^0_{\mu\nu} = \kappa_{\text{EM}} F_{\mu\nu} \neq 0 \quad (546)$$

holds cannot be reduced by a diffeomorphism to a pure background tetrad for which

$$\bar{T}^0_{\mu\nu} = 0 \quad (547)$$

holds.

The same result can also be seen through the Lagrangian density. The Maxwell kinetic term is

$$\mathcal{L}_{\text{EM}} = -\frac{e}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \quad (548)$$

Using the temporal torsion relation, this expression is written as

$$\mathcal{L}_{\text{EM}} = -\frac{e}{4\mu_0 \kappa_{\text{EM}}^2} T^0_{\mu\nu} T^{0\mu\nu} \quad (549)$$

Thus, while for the tetrad with the ansatz one may have

$$\mathcal{L}_{\text{EM}} \neq 0 \quad (550)$$

when the electromagnetic structure is removed, since $\bar{T}^0_{\mu\nu} = 0$, one obtains

$$\bar{\mathcal{L}}_{\text{EM}} = 0 \quad (551)$$

This clearly shows that the transition

$$\mathcal{L}_{\text{EM}} \longrightarrow \bar{\mathcal{L}}_{\text{EM}} \quad (552)$$

is not merely a change of coordinate representation. This is because under a diffeomorphism the scalar action density remains formally covariant; however, here the value of the torsion invariant changes:

$$T^0_{\mu\nu} T^{0\mu\nu} \longrightarrow 0. \quad (553)$$

Therefore, removing the ansatz changes the physical and geometric content.

As a result, the proposed Maxwell-adapted tetrad ansatz is not merely a tetrad diffeomorphism gauge choice. This is because, when the electromagnetic structure in the ansatz is removed, torsion changes; accordingly, the electromagnetic tensor also changes, and the Maxwell Lagrangian density

$$\mathcal{L}_{\text{EM}} = -\frac{e}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \quad (554)$$

generally goes to a different value. Diffeomorphisms only re-express tensorial structures; they do not physically set a nonzero torsion two-form to zero. Therefore, the presence of A_μ inside the temporal tetrad is, in this formalism, not merely an artificial coordinate-induced term, but a physical component carrying measurable geometric and dynamical content through temporal torsion.

12 Appendix-12: Metric Diffeomorphism Gauge Test

In this section, it is tested whether the A_μ -dependent terms appearing in the metric derived from the tetrad arise only from a coordinate choice.

Initially, the temporal tetrad is taken as Eq. 13, and the spatial tetrads are taken as $e^i = a(t) dx^i$. Then ds^2 is written as

$$ds^2 = -(e^0)^2 + \delta_{ij} e^i e^j \quad (555)$$

Substituting the ansatz gives

$$ds^2 = - \left[(c - \kappa_{\text{EM}} \phi) dt + \kappa_{\text{EM}} A_i dx^i \right]^2 + a^2(t) \delta_{ij} dx^i dx^j \quad (556)$$

Expanding the square, one obtains

$$ds^2 = -(c - \kappa_{\text{EM}} \phi)^2 dt^2 - 2\kappa_{\text{EM}} (c - \kappa_{\text{EM}} \phi) A_i dt dx^i + (a^2 \delta_{ij} - \kappa_{\text{EM}}^2 A_i A_j) dx^i dx^j.$$

From this, the metric components are found as

$$g_{00} = -(c - \kappa_{\text{EM}} \phi)^2, g_{0i} = -\kappa_{\text{EM}} (c - \kappa_{\text{EM}} \phi) A_i, g_{ij} = a^2 \delta_{ij} - \kappa_{\text{EM}}^2 A_i A_j \quad (557)$$

Thus, the metric contains A_i -dependent cross components:

$$g_{0i} \neq 0. \quad (558)$$

Now let us consider a small coordinate transformation:

$$x^\mu \mapsto x'^\mu = x^\mu + \xi^\mu(x). \quad (559)$$

Under this transformation, the metric changes as

$$g'_{\mu\nu} = g_{\mu\nu} - \nabla_\mu \xi_\nu - \nabla_\nu \xi_\mu \quad (560)$$

In particular, the cross component becomes

$$g'_{0i} = g_{0i} - \nabla_0 \xi_i - \nabla_i \xi_0. \quad (561)$$

Even if one can locally set

$$g'_{0i} = 0 \quad (562)$$

by a suitable choice of ξ^μ , this alone does not show that the A_μ -dependent structure is unphysical.

For this reason, it is checked whether the physical quantities change. First, let us look at the electromagnetic field strength. In this theory,

$$F_{\mu\nu} = \frac{1}{\kappa_{\text{EM}}} T^0{}_{\mu\nu} \quad (563)$$

and therefore, if the A_μ -dependent structure is completely removed, in general one obtains

$$F_{\mu\nu} \longrightarrow 0. \quad (564)$$

This is not merely a change of coordinate representation, but a change of the field strength.

Second, the Maxwell Lagrangian density is examined:

$$\mathcal{L}_{\text{EM}} = -\frac{e}{4\mu_0} F_{\mu\nu} F^{\mu\nu}. \quad (565)$$

If the A_μ -dependent metric terms were merely coordinate artifacts, the physical value of this term should not change. However, when A_μ is completely removed,

$$F_{\mu\nu} F^{\mu\nu} \longrightarrow 0, \quad \mathcal{L}_{\text{EM}} \longrightarrow 0 \quad (566)$$

holds.

Third, particle motion is checked. The connection arising from the metric is given by

$$\Gamma^\rho{}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) \quad (567)$$

Since the metric is A_μ -dependent, the terms

$$\partial_\alpha g_{\mu\nu} \quad (568)$$

contain $\partial_\alpha A_\mu$ terms. These derivatives produce, in the appropriate limits, the structure

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (569)$$

Thus, the equation of motion contains, in the appropriate limit, the Lorentz force contribution in the form

$$m \frac{d^2 x^\mu}{d\tau^2} \simeq q F^\mu{}_\nu u^\nu. \quad (570)$$

If the A_μ -dependent metric terms are completely removed, then

$$\partial_\alpha A_\mu \longrightarrow 0 \quad (571)$$

and the electromagnetic contributions in the connection disappear. In this case, the same metric equation of motion cannot produce the Lorentz force term.

Therefore, the A_μ -dependent metric terms are not merely coordinate artifacts. Removing these terms changes the electromagnetic field strength, the Lagrangian density, and particle motion. For this reason, the A_μ -dependent structure appearing in the metric should be interpreted as the projection, at the metric level, of the electromagnetic geometric content carried by the temporal tetrad.

13 Appendix-13: Local Lorentz Covariance and the Physical Content of the Maxwell-Adapted Tetrad

In this section, it will be examined whether the Maxwell-adapted tetrad ansatz is merely an artificial result of a particular local Lorentz frame. The aim is to show that the electromagnetic torsion content does not disappear under a local Lorentz transformation, but is only distributed covariantly among the internal Lorentz components.

Under a general local Lorentz transformation, the torsion components are not expected to remain individually unchanged. That is, in general, it is not correct to write

$$T'^a = T^a \quad (572)$$

The correct transformation rule is

$$T'^a = \Lambda^a_b T^b \quad (573)$$

The structures that remain unchanged are scalars in which the internal Lorentz indices are contracted appropriately.

Therefore, when checking whether the Maxwell-adapted tetrad carries physical content, the object that should be examined is not whether the component T_{EM}^0 alone remains unchanged. What must actually be checked is whether this torsion transforms correctly and covariantly under a local Lorentz transformation and whether the kinetic scalar constructed from it remains unchanged.

13.1 Tetrad, Spin Connection, and Torsion

The tetrad one-forms are defined as

$$e^a = e^a_\mu dx^\mu \quad (574)$$

Under a local Lorentz transformation, the tetrad one-forms transform as

$$e'^a = \Lambda^a_b(x) e^b \quad (575)$$

Here $\Lambda^a_b(x)$ is a Lorentz matrix depending on the spacetime point. Since the internal Minkowski metric is

$$\eta_{ab} = \text{diag}(-1, 1, 1, 1) \quad (576)$$

the Lorentz transformation satisfies the condition

$$\eta_{ab} \Lambda^a_c \Lambda^b_d = \eta_{cd} \quad (577)$$

In teleparallel geometry, the torsion two-form is defined by

$$T^a = De^a \quad (578)$$

Explicitly, it is

$$T^a = de^a + \omega^a_b \wedge e^b \quad (579)$$

Here ω^a_b is the spin connection.

In the covariant teleparallel formalism, the local Lorentz transformation is not applied only to the tetrad field. The spin connection must also be transformed compatibly. The transformation of the spin connection is

$$\omega'^a_b = \Lambda^a_c \omega^c_d (\Lambda^{-1})^d_b - d\Lambda^a_c (\Lambda^{-1})^c_b \quad (580)$$

Now the transformed torsion two-form is calculated explicitly:

$$T'^a = de'^a + \omega'^a_b \wedge e'^b. \quad (581)$$

Substituting the tetrad transformation gives

$$de'^a = d(\Lambda^a_c e^c) \quad (582)$$

Using the product rule for the exterior derivative, one obtains

$$de'^a = d\Lambda^a_c \wedge e^c + \Lambda^a_c de^c \quad (583)$$

Now the spin-connection term is calculated:

$$\omega'^a_b \wedge e'^b = [\Lambda^a_m \omega^m_n (\Lambda^{-1})^n_b - d\Lambda^a_m (\Lambda^{-1})^m_b] \wedge \Lambda^b_d e^d. \quad (584)$$

Here the identities

$$(\Lambda^{-1})^n_b \Lambda^b_d = \delta^n_d \quad (585)$$

and

$$(\Lambda^{-1})^m_b \Lambda^b_d = \delta^m_d \quad (586)$$

are used. Thus,

$$\omega'^a_b \wedge e'^b = \Lambda^a_m \omega^m_d \wedge e^d - d\Lambda^a_d \wedge e^d \quad (587)$$

is obtained.

Writing Eq. (583) and Eq. (587) together gives

$$T'^a = d\Lambda^a_c \wedge e^c + \Lambda^a_c de^c + \Lambda^a_m \omega^m_d \wedge e^d - d\Lambda^a_d \wedge e^d. \quad (588)$$

The first and last terms cancel each other by the same dummy-index renaming:

$$d\Lambda^a_c \wedge e^c - d\Lambda^a_d \wedge e^d = 0. \quad (589)$$

Therefore,

$$T'^a = \Lambda^a_c de^c + \Lambda^a_m \omega^m_d \wedge e^d \quad (590)$$

remains. Writing the dummy indices in a common form gives

$$T'^a = \Lambda^a_b (de^b + \omega^b_c \wedge e^c) \quad (591)$$

Since the expression inside the parentheses is T^b , one obtains

$$T'^a = \Lambda^a_b T^b \quad (592)$$

This result shows that torsion transforms as an internal Lorentz vector under a local Lorentz transformation. If the spin connection is not transformed together with the tetrad, the $d\Lambda$ terms enter torsion as spurious inertial contributions. Therefore, in the local Lorentz covariance test, the tetrad and spin connection must be considered together.

13.2 Electromagnetic Torsion in the Maxwell-Adapted Frame

In the Maxwell-adapted frame, the electromagnetic content is read through temporal torsion:

$$T_{\text{EM}}^0 = \kappa_{\text{EM}} F \quad (593)$$

and, in terms of components,

$$T_{\text{EM}\mu\nu}^0 = \kappa_{\text{EM}} F_{\mu\nu} \quad (594)$$

In this adapted frame, the electromagnetic torsion appears only in the $a = 0$ internal Lorentz component. This situation is written as

$$T_{\text{EM}}^a = \delta^a_0 T_{\text{EM}}^0 \quad (595)$$

Using Eq. (593), one obtains

$$T_{\text{EM}}^a = \delta^a_0 \kappa_{\text{EM}} F \quad (596)$$

Now a local Lorentz transformation is applied. Since the transformation rule of torsion is

$$T'_{\text{EM}}{}^a = \Lambda^a_b T_{\text{EM}}^b \quad (597)$$

substituting Eq. (596) gives

$$T'_{\text{EM}}{}^a = \Lambda^a_b \delta^b_0 \kappa_{\text{EM}} F. \quad (598)$$

Since the Kronecker delta selects the internal index,

$$\Lambda^a_b \delta^b_0 = \Lambda^a_0 \quad (599)$$

is obtained. Therefore,

$$T'_{\text{EM}}{}^a = \Lambda^a_0 \kappa_{\text{EM}} F \quad (600)$$

In terms of components, this is written as

$$T'_{\text{EM}}{}^a{}_{\mu\nu} = \Lambda^a_0 \kappa_{\text{EM}} F_{\mu\nu} \quad (601)$$

This expression shows that the electromagnetic field does not disappear under a local Lorentz transformation. While in the Maxwell-adapted frame the electromagnetic torsion appears only in the temporal internal component, in a boosted general frame the same content is distributed among the internal Lorentz components with the coefficients Λ^a_0 .

Therefore, the Maxwell-adapted tetrad is not a Lorentz-gauge trick that artificially produces the electromagnetic field. It is only the adapted frame in which the electromagnetic torsion is read most simply.

13.3 Transformation of the $\delta^a{}_0$ Projection Under a Boost

In the Maxwell-adapted writing, the structure selecting the temporal internal Lorentz component is denoted by

$$\delta^a{}_0 \quad (602)$$

In the adapted frame, this symbol selects only the $a = 0$ component:

$$\delta^0{}_0 = 1, \quad \delta^i{}_0 = 0. \quad (603)$$

However, when a local Lorentz boost is applied, the components of this projection must also be rewritten in the boosted frame. The correct transformation is

$$\delta'^a{}_0 = \Lambda^a{}_b \delta^b{}_0 \quad (604)$$

Since the delta symbol selects the $b = 0$ component, one obtains

$$\delta'^a{}_0 = \Lambda^a{}_0 \quad (605)$$

Explicitly,

$$\delta'^0{}_0 = \Lambda^0{}_0, \quad (606)$$

and

$$\delta'^i{}_0 = \Lambda^i{}_0 \quad (607)$$

This result shows the following: the electromagnetic source or torsion that is located only in the temporal component in the Maxwell-adapted frame also spreads into the spatial internal Lorentz components in the boosted frame. However, this spreading does not mean that the physical content disappears. It is only the component representation of the same content in a different local Lorentz frame.

13.4 Local Lorentz Covariance of the Hodge Dual Structure

The Hodge dual operation acts on spacetime indices; it does not mix the internal Lorentz index. Therefore, the local Lorentz matrix can be taken outside the Hodge dual operation. As the starting point, one writes

$$T'^a{}_{\text{EM}} = \Lambda^a{}_0 T^0_{\text{EM}} \quad (608)$$

Its Hodge dual is

$$\star T'^a{}_{\text{EM}} = \star (\Lambda^a{}_0 T^0_{\text{EM}}) \quad (609)$$

Since $\Lambda^a{}_0$ is an internal Lorentz coefficient, it is taken outside the Hodge dual:

$$\star T'^a{}_{\text{EM}} = \Lambda^a{}_0 \star T^0_{\text{EM}} \quad (610)$$

This can also be shown explicitly with components. The Hodge-dual two-form components are

$$(\star T'_{\text{EM}}{}^a)_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu}{}^{\rho\sigma} T'_{\text{EM}}{}^a{}_{\rho\sigma} \quad (611)$$

Substituting Eq. (601), one obtains

$$(\star T'_{\text{EM}}{}^a)_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu}{}^{\rho\sigma} \Lambda^a{}_0 T_{\text{EM}\rho\sigma}^0 \quad (612)$$

Since $\Lambda^a{}_0$ does not carry spacetime indices, it is taken out:

$$(\star T'_{\text{EM}}{}^a)_{\mu\nu} = \Lambda^a{}_0 \frac{1}{2} \epsilon_{\mu\nu}{}^{\rho\sigma} T_{\text{EM}\rho\sigma}^0. \quad (613)$$

The expression in parentheses is the component of $\star T_{\text{EM}}^0$. Thus,

$$(\star T'_{\text{EM}}{}^a)_{\mu\nu} = \Lambda^a{}_0 (\star T_{\text{EM}}^0)_{\mu\nu} \quad (614)$$

is obtained.

Therefore, just as torsion transforms as an internal Lorentz vector, its Hodge dual also transforms with the same internal Lorentz coefficient.

13.5 Preservation of the Kinetic Scalar Under a Local Lorentz Transformation

The electromagnetic kinetic structure must be written in the torsional language by contracting the internal Lorentz indices. Since the metric signature is

$$\eta_{ab} = \text{diag}(-1, 1, 1, 1) \quad (615)$$

and since only the temporal internal component exists in the Maxwell-adapted frame, one has

$$\eta_{ab} T_{\text{EM}}^a \wedge \star T_{\text{EM}}^b = \eta_{00} T_{\text{EM}}^0 \wedge \star T_{\text{EM}}^0. \quad (616)$$

Since $\eta_{00} = -1$, one obtains

$$\eta_{ab} T_{\text{EM}}^a \wedge \star T_{\text{EM}}^b = -T_{\text{EM}}^0 \wedge \star T_{\text{EM}}^0. \quad (617)$$

This sign comes from the internal metric signature $(-+++)$.

Now it is shown that the same scalar remains unchanged in the boosted frame. In the transformed frame,

$$T'_{\text{EM}}{}^a = \Lambda^a{}_0 T_{\text{EM}}^0 \quad (618)$$

and

$$\star T'_{\text{EM}}{}^b = \Lambda^b{}_0 \star T_{\text{EM}}^0 \quad (619)$$

Therefore,

$$\eta_{ab} T'_{\text{EM}}{}^a \wedge \star T'_{\text{EM}}{}^b = \eta_{ab} (\Lambda^a{}_0 T_{\text{EM}}^0) \wedge (\Lambda^b{}_0 \star T_{\text{EM}}^0) \quad (620)$$

is written. Taking the internal Lorentz coefficients outside gives

$$\eta_{ab} T'_{\text{EM}}{}^a \wedge \star T'_{\text{EM}}{}^b = \eta_{ab} \Lambda^a{}_0 \Lambda^b{}_0 T_{\text{EM}}^0 \wedge \star T_{\text{EM}}^0. \quad (621)$$

Due to the preservation of the Lorentz metric,

$$\eta_{ab} \Lambda^a{}_0 \Lambda^b{}_0 = \eta_{00} \quad (622)$$

holds. Therefore,

$$\eta_{ab} T'_{\text{EM}}{}^a \wedge \star T'_{\text{EM}}{}^b = \eta_{00} T_{\text{EM}}^0 \wedge \star T_{\text{EM}}^0. \quad (623)$$

Since $\eta_{00} = -1$,

$$\eta_{ab} T'_{\text{EM}}{}^a \wedge \star T'_{\text{EM}}{}^b = -T_{\text{EM}}^0 \wedge \star T_{\text{EM}}^0. \quad (624)$$

This is the same as the result in the initial frame. Thus, the kinetic scalar whose internal Lorentz indices are contracted with η_{ab} remains unchanged under a local Lorentz transformation.

13.6 Electromagnetic Lagrangian Density

The electromagnetic kinetic Lagrangian density is standardly

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \quad (625)$$

Using the Maxwell-adapted torsion relation

$$T_{\text{EM}\mu\nu}^0 = \kappa_{\text{EM}} F_{\mu\nu} \quad (626)$$

one obtains, in the adapted frame,

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4\mu_0 \kappa_{\text{EM}}^2} T_{\text{EM}\mu\nu}^0 T_{\text{EM}}^{0\mu\nu}. \quad (627)$$

To write this in a general internal Lorentz covariant form, one uses

$$\eta_{ab} T_{\text{EM}\mu\nu}^a T_{\text{EM}}^{b\mu\nu} = \eta_{00} T_{\text{EM}\mu\nu}^0 T_{\text{EM}}^{0\mu\nu}. \quad (628)$$

Since $\eta_{00} = -1$, this becomes

$$\eta_{ab} T_{\text{EM}\mu\nu}^a T_{\text{EM}}^{b\mu\nu} = -T_{\text{EM}\mu\nu}^0 T_{\text{EM}}^{0\mu\nu}. \quad (629)$$

Therefore,

$$T_{\text{EM}\mu\nu}^0 T_{\text{EM}}^{0\mu\nu} = -\eta_{ab} T_{\text{EM}\mu\nu}^a T_{\text{EM}}^{b\mu\nu} \quad (630)$$

is obtained. Substituting this expression into the Lagrangian density gives

$$\mathcal{L}_{\text{EM}} = \frac{1}{4\mu_0 \kappa_{\text{EM}}^2} \eta_{ab} T_{\text{EM}\mu\nu}^a T_{\text{EM}}^{b\mu\nu} \quad (631)$$

This sign is required because of the signature $\eta_{ab} = \text{diag}(-1, 1, 1, 1)$.

In the transformed frame, the Lagrangian density is

$$\mathcal{L}'_{\text{EM}} = \frac{1}{4\mu_0\kappa_{\text{EM}}^2} \eta_{ab} T'_{\text{EM}}{}^a{}_{\mu\nu} T'_{\text{EM}}{}^{b\mu\nu}. \quad (632)$$

Since the transformation rule is

$$T'_{\text{EM}}{}^a{}_{\mu\nu} = \Lambda^a{}_0 T_{\text{EM}\mu\nu}^0, \quad (633)$$

one obtains

$$\mathcal{L}'_{\text{EM}} = \frac{1}{4\mu_0\kappa_{\text{EM}}^2} \eta_{ab} (\Lambda^a{}_0 T_{\text{EM}\mu\nu}^0) (\Lambda^b{}_0 T_{\text{EM}}^{0\mu\nu}). \quad (634)$$

The internal Lorentz coefficients are taken outside:

$$\mathcal{L}'_{\text{EM}} = \frac{1}{4\mu_0\kappa_{\text{EM}}^2} \eta_{ab} \Lambda^a{}_0 \Lambda^b{}_0 T_{\text{EM}\mu\nu}^0 T_{\text{EM}}^{0\mu\nu}. \quad (635)$$

Since the Lorentz metric is preserved,

$$\eta_{ab} \Lambda^a{}_0 \Lambda^b{}_0 = \eta_{00}, \quad (636)$$

and therefore

$$\mathcal{L}'_{\text{EM}} = \frac{1}{4\mu_0\kappa_{\text{EM}}^2} \eta_{00} T_{\text{EM}\mu\nu}^0 T_{\text{EM}}^{0\mu\nu}. \quad (637)$$

Since $\eta_{00} = -1$,

$$\mathcal{L}'_{\text{EM}} = -\frac{1}{4\mu_0\kappa_{\text{EM}}^2} T_{\text{EM}\mu\nu}^0 T_{\text{EM}}^{0\mu\nu}. \quad (638)$$

This expression is the initial Maxwell-adapted Lagrangian density. Therefore,

$$\mathcal{L}'_{\text{EM}} = \mathcal{L}_{\text{EM}} \quad (639)$$

is obtained.

Consequently, the electromagnetic Lagrangian density remains unchanged under a local Lorentz transformation. While in the Maxwell-adapted frame the electromagnetic content appears only in the temporal torsion component, in a general tetrad frame the same content is distributed among the internal Lorentz components. However, since the Lagrangian density depends on the internal Lorentz scalar constructed with η_{ab} , this distribution does not change the physical Lagrangian structure.

13.7 Covariance of the Field Equation Under a Local Lorentz Transformation

For an ansatz proposed in the tetrad formalism to be consistent, the field equations must also preserve their covariant structure under local Lorentz transformations. This is because, in this formalism, the field equations carry an internal Lorentz index in addition to spacetime indices. Therefore, observers passing to

different local Lorentz frames are expected to express the same physical equation through different internal components.

The Maxwell-type equation obtained in the previous sections with the source choice $-J^\mu A_\mu$ is written in torsional representation as

$$\frac{1}{\mu_0 \kappa_{\text{EM}}} D_\mu T_{\text{EM}}^{a\mu\nu} - J^\nu \delta^a_0 + \kappa_{\text{EM}} (E^a_\rho \Theta_{\text{EM}}^{\rho\nu} - E^{a\nu} J^\mu A_\mu) = 0 \quad (640)$$

Here a is an internal Lorentz index. Therefore, under a local Lorentz transformation, all structures carrying the index a in the equation must transform with the same matrix Λ^a_b .

Under a local Lorentz transformation, torsion transforms as

$$T'_{\text{EM}}{}^{a\mu\nu} = \Lambda^a_b T_{\text{EM}}^{b\mu\nu} \quad (641)$$

In the covariant teleparallel formalism, since the spin connection is also transformed appropriately, the covariant-derivative torsion term obeys the transformation rule

$$D'_\mu T'_{\text{EM}}{}^{a\mu\nu} = \Lambda^a_b D_\mu T_{\text{EM}}^{b\mu\nu} \quad (642)$$

Thus, the left-hand side of the field equation transforms as

$$\frac{1}{\mu_0 \kappa_{\text{EM}}} D'_\mu T'_{\text{EM}}{}^{a\mu\nu} = \Lambda^a_b \left(\frac{1}{\mu_0 \kappa_{\text{EM}}} D_\mu T_{\text{EM}}^{b\mu\nu} \right) \quad (643)$$

In the source term, J^ν is a spacetime current and carries no internal Lorentz index. Therefore,

$$J'^\nu = J^\nu \quad (644)$$

is taken. However, the temporal projection δ^a_0 in the Maxwell-adapted frame is not kept fixed under a boost. In the boosted frame,

$$\delta'^a_0 = \Lambda^a_b \delta^b_0 \quad (645)$$

must hold. Since the Kronecker delta selects the $b = 0$ component, one obtains

$$\delta'^a_0 = \Lambda^a_0 \quad (646)$$

Thus, the source term becomes

$$J'^\nu \delta'^a_0 = J^\nu \Lambda^a_b \delta^b_0. \quad (647)$$

This is written as

$$J'^\nu \delta'^a_0 = \Lambda^a_b (J^\nu \delta^b_0) \quad (648)$$

Therefore, the source projection also transforms as an internal Lorentz vector.

In the geometric backreaction terms, the structure carrying the internal Lorentz index is the inverse tetrad. Under a local Lorentz transformation,

$$E'^a_\rho = \Lambda^a_b E^b_\rho \quad (649)$$

and

$$E'^{a\nu} = \Lambda^a_b E^{b\nu} \quad (650)$$

hold. By contrast, $\Theta_{\text{EM}}^{\rho\nu}$, $J^\mu A_\mu$, and J^ν carry no internal Lorentz index. Therefore,

$$E'^a{}_\rho \Theta_{\text{EM}}^{\rho\nu} = \Lambda^a_b E^b{}_\rho \Theta_{\text{EM}}^{\rho\nu} \quad (651)$$

and

$$E'^{a\nu} J^\mu A_\mu = \Lambda^a_b E^{b\nu} J^\mu A_\mu \quad (652)$$

are obtained. When the two terms are written together,

$$E'^a{}_\rho \Theta_{\text{EM}}^{\rho\nu} - E'^{a\nu} J^\mu A_\mu = \Lambda^a_b (E^b{}_\rho \Theta_{\text{EM}}^{\rho\nu} - E^{b\nu} J^\mu A_\mu) \quad (653)$$

is obtained.

Now the transformed field equation is written explicitly:

$$\frac{1}{\mu_0 \kappa_{\text{EM}}} D'_\mu T'_{\text{EM}}{}^{a\mu\nu} - J'^\nu \delta'^a{}_0 + \kappa_{\text{EM}} (E'^a{}_\rho \Theta_{\text{EM}}^{\rho\nu} - E'^{a\nu} J^\mu A_\mu) = 0. \quad (654)$$

When the transformation rules above are substituted, the left-hand side becomes

$$\Lambda^a_b \left(\frac{1}{\mu_0 \kappa_{\text{EM}}} D_\mu T_{\text{EM}}^{b\mu\nu} \right) \quad (655)$$

The right-hand side is written as

$$-\Lambda^a_b (J^\nu \delta^b{}_0) + \Lambda^a_b [\kappa_{\text{EM}} (E^b{}_\rho \Theta_{\text{EM}}^{\rho\nu} - E^{b\nu} J^\mu A_\mu)]. \quad (656)$$

Taking out the common factor Λ^a_b , one obtains

$$\Lambda^a_b \left[\frac{1}{\mu_0 \kappa_{\text{EM}}} D_\mu T_{\text{EM}}^{b\mu\nu} - J^\nu \delta^b{}_0 + \kappa_{\text{EM}} (E^b{}_\rho \Theta_{\text{EM}}^{\rho\nu} - E^{b\nu} J^\mu A_\mu) \right] = 0. \quad (657)$$

This expression is the entire field equation Eq. (640) in the initial frame multiplied by Λ^a_b . Therefore, if the equation is satisfied in one local Lorentz frame, it is also satisfied in the boosted frame.

To show this briefly, one may define

$$\mathcal{E}^{a\nu} = \frac{1}{\mu_0 \kappa_{\text{EM}}} D_\mu T_{\text{EM}}^{a\mu\nu} - J^\nu \delta^a{}_0 + \kappa_{\text{EM}} (E^a{}_\rho \Theta_{\text{EM}}^{\rho\nu} - E^{a\nu} J^\mu A_\mu) \quad (658)$$

The field equation is

$$\mathcal{E}^{a\nu} = 0. \quad (659)$$

Under a local Lorentz transformation,

$$\mathcal{E}'^{a\nu} = \Lambda^a_b \mathcal{E}^{b\nu} \quad (660)$$

Thus, if

$$\mathcal{E}^{a\nu} = 0 \quad (661)$$

then

$$\mathcal{E}'^{a\nu} = 0 \quad (662)$$

is also automatically satisfied.

As a result, the Maxwell-adapted tetrad ansatz is not an artificial expression depending only on a particular local Lorentz boost choice. The electromagnetic torsion is read in the temporal component in the adapted frame; however, under a boost, this content is distributed among the internal Lorentz components with the coefficients Λ^a_0 . Since torsion, the Hodge dual structure, the Lagrangian density, and the field equation preserve the same local Lorentz covariance structure, the Maxwell-adapted tetrad should be interpreted as a consistent adapted frame carrying physical torsion content.

14 Appendix-14: Detailed Demonstration of the Exact Coframe Shift Gauge Test

In this appendix, the exact coframe shift gauge test summarized in the main text will be shown at the level of the calculation. The aim is to show explicitly how the electromagnetic $U(1)$ gauge transformation is represented on the Maxwell-adapted tetrad and how the tetrad, metric, torsion, electromagnetic field tensor, Lagrangian density, field equations, and particle equation of motion are preserved under this transformation.

14.1 Starting Point: Maxwell-Adapted Temporal Tetrad

Initially, the temporal tetrad is taken as

$$e^0 = u + \kappa_{\text{EM}} A \quad (663)$$

When the electromagnetic gauge transformation is

$$A \longrightarrow A' = A + d\lambda \quad (664)$$

the temporal tetrad changes as

$$e^0 \longrightarrow e'^0 = u + \kappa_{\text{EM}}(A + d\lambda) = e^0 + \kappa_{\text{EM}} d\lambda. \quad (665)$$

Therefore, the $U(1)$ gauge transformation produces an exact coframe shift at the tetrad level.

14.2 Decomposition of the Four-Dimensional Exact Shift

The exact one-form $d\lambda$ is written in the tetrad basis as

$$d\lambda = \lambda_0 e^0 + \lambda_i e^i. \quad (666)$$

Here $\lambda_0 = E_0(\lambda)$ and $\lambda_i = E_i(\lambda)$ are defined. In this case,

$$e'^0 = e^0 + \kappa_{\text{EM}} d\lambda = e^0 + \kappa_{\text{EM}} (\lambda_0 e^0 + \lambda_i e^i) \quad (667)$$

is obtained. Therefore,

$$e'^0 = (1 + \kappa_{\text{EM}} \lambda_0) e^0 + \kappa_{\text{EM}} \lambda_i e^i. \quad (668)$$

This result explicitly separates the temporal and spatial parts of the exact shift.

14.3 Examination of the Temporal and Spatial Parts Under the Exact Gauge Shift

Since A_μ is embedded in the temporal tetrad in this work, it may be expected that the $U(1)$ gauge transformation can also be explained by a geometric transformation or transformations. This issue is very important, because when electromagnetic theory is geometric, the $d\lambda$ translation is expected not to affect physical observables. Due to this ontological information, it must be examined through geometric tests whether the A_μ structure embedded in the temporal tetrad affects physical observables under a $d\lambda$ translation. If the $U(1)$ gauge structure can also be explained geometrically as a gauge structure through geometric tests, this can be interpreted as showing the consistency of the work. Therefore, since the $U(1)$ gauge $d\lambda$ translation can geometrically be an exact shift gauge, the tetrad local Lorentz boost gauge and tetrad diffeomorphism gauge tests will be examined. If the $U(1)$ gauge structure can be explained as a result of these two tests, this is important for the consistency of the work.

First, let us examine the local Lorentz boost gauge. Can it provide something in terms of explaining the $U(1)$ gauge structure?

14.4 Spatial Part and Local Lorentz Boost Completion

The spatial part is

$$\delta_\perp e^0 = \kappa_{\text{EM}} \lambda_i e^i \quad (669)$$

The infinitesimal local Lorentz transformation is given by

$$e'^a = (\delta^a_b + \omega^a_b) e^b, \quad \omega_{ab} = -\omega_{ba} \quad (670)$$

For a pure boost, one takes

$$\omega^0_i = \chi_i, \quad \omega^i_0 = \chi^i. \quad (671)$$

In this case,

$$e'^0 = e^0 + \chi_i e^i \quad (672)$$

is obtained. If $d\lambda$ has no temporal component, that is, if

$$\lambda_0 = E_0(\lambda) = 0 \quad (673)$$

then from Eq. (672) one obtains

$$d\lambda = \lambda_i e^i. \quad (674)$$

In this case, the spatial shift coming from the gauge transformation is of the form

$$e'^0 = e^0 + \kappa_{\text{EM}} \lambda_i e^i. \quad (675)$$

When compared with the spatial part of the gauge shift, the choice

$$\chi_i = \kappa_{\text{EM}} \lambda_i \quad (676)$$

is made. Thus, for the full frame transformation, all tetrads must also transform:

$$e'^i = e^i + \chi^i e^0. \quad (677)$$

$$e'^0 = e^0 + \kappa_{\text{EM}} \lambda_i e^i = e^0 + \chi_i e^i \quad (678)$$

is obtained. Therefore, the spatial exact shift is completed into a full local Lorentz frame transformation not by itself, but together with the transformation of the spatial tetrads. However, the $\lambda_0 e^0$ part has still not been explained. We know from the structure of the $U(1)$ gauge transformation that this is a gauge. If A_μ is embedded in the temporal tetrad, the $\lambda_0 e^0$ part is also expected to be explained by a geometric gauge. For this reason, the tetrad diffeomorphism gauge will be examined.

For this purpose,

$$A \longrightarrow A' = A + d\lambda, \quad u \longrightarrow u' = u - \kappa_{\text{EM}} d\lambda \quad (679)$$

is taken. In particular, the shift corresponding to the temporal component,

$$u \rightarrow u - \kappa_{\text{EM}} \lambda_0 e^0 \quad (680)$$

can be read as a redefinition of the time coordinate. At the infinitesimal level, this transformation can be expressed as the coordinate transformation

$$t' = t - \frac{\kappa_{\text{EM}}}{c} \lambda. \quad (681)$$

Under this compensator transformation, the temporal tetrad remains unchanged:

$$e'^0 = u' + \kappa_{\text{EM}} A' = (u - \kappa_{\text{EM}} d\lambda) + \kappa_{\text{EM}} (A + d\lambda) = u + \kappa_{\text{EM}} A = e^0.$$

That is,

$$e'^0 = e^0. \quad (682)$$

Therefore, the $\lambda_0 e^0$ contribution should be treated not as a local boost, but as the time-diffeomorphism compensator of the temporal tetrad.

The point to be noted here is that this operation is different from the dynamic variation used in deriving the field equations. In Section 4, while the Lagrangian was varied with respect to the tetrad, u was kept fixed and

$$\delta u = 0 \quad (683)$$

was taken. In this section, since a gauge transformation is being considered, it is examined whether physicality changes under such a translation. However, this is

not the situation in the Lagrangian. There, due to the result of the gauge tests, one has $\delta u = 0$. Therefore, the variation represents a physical dynamical change; in general, the electromagnetic potential and, accordingly, the temporal tetrad $e^0 = u + \kappa_{\text{EM}} A$ change. This change allows the field equations to be obtained.

Consequently, the $d\lambda$ translation performed here is a gauge translation. It has been shown by Eq. 692 that this transformation does not change the electromagnetic field. Therefore, shifting u in the gauge test does not mean that u changes physically and dynamically; it is only a redefinition of representation that compensates the pure exact gauge part in A . Thus, there is no contradiction between taking $\delta u = 0$ in the dynamical variation and writing $u \rightarrow u - \kappa_{\text{EM}} d\lambda$ in the gauge test.

These two situations are different from each other, and there is no inconsistency. Here, the physicality has also been examined, and the $U(1)$ gauge structure is a gauge representation. $U(1)$ is an extended coframe gauge representation in which the local Lorentz boost and temporal compensators exist together.

14.5 Preservation of the Metric

In the completed local Lorentz transformation,

$$e'^a = \Lambda^a_b e^b \quad (684)$$

and the Lorentz matrix satisfies

$$\eta_{ab} \Lambda^a_c \Lambda^b_d = \eta_{cd} \quad (685)$$

Therefore,

$$g' = \eta_{ab} e'^a \otimes e'^b = \eta_{ab} \Lambda^a_c \Lambda^b_d e^c \otimes e^d = \eta_{cd} e^c \otimes e^d = g.$$

Thus, when the spatial boost-like part is completed into a full Lorentz transformation, the metric is preserved:

$$g' = g. \quad (686)$$

Therefore, even though the tetrad representative changes, the physical metric structure remains the same.

14.6 Preservation of Torsion

Temporal torsion is defined by

$$T^0 = de^0. \quad (687)$$

Under the exact shift,

$$T'^0 = de'^0 = d(e^0 + \kappa_{\text{EM}} d\lambda) = de^0 + \kappa_{\text{EM}} d^2\lambda = de^0 = T^0.$$

Here $d^2\lambda = 0$ has been used. Thus,

$$T'^0 = T^0 \quad (688)$$

is obtained.

Under the local Lorentz completion, the torsion components transform covariantly:

$$T'^a = \Lambda^a_b T^b. \quad (689)$$

Therefore, torsion invariants whose internal Lorentz indices are contracted with η_{ab} are preserved:

$$\eta_{ab} T'^a \wedge \star T'^b = \eta_{ab} T^a \wedge \star T^b. \quad (690)$$

14.7 Preservation of the Electromagnetic Field Tensor

The electromagnetic field two-form is defined by

$$F = dA. \quad (691)$$

Under the gauge transformation,

$$F' = dA' = d(A + d\lambda) = dA + d^2\lambda = dA = F.$$

Therefore,

$$F' = F \quad (692)$$

is obtained. In terms of components,

$$F'_{\mu\nu} = F_{\mu\nu} \quad (693)$$

holds.

14.8 Preservation of the Maxwell Kinetic Lagrangian Term

The Maxwell kinetic term is

$$\mathcal{L}_{\text{kin}} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \quad (694)$$

Since $F'_{\mu\nu} = F_{\mu\nu}$ under the exact shift and $g'_{\mu\nu} = g_{\mu\nu}$ under the local Lorentz completion, one obtains

$$F'^{\mu\nu} = F^{\mu\nu}. \quad (695)$$

Therefore,

$$\mathcal{L}'_{\text{kin}} = -\frac{1}{4\mu_0} F'_{\mu\nu} F'^{\mu\nu} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} = \mathcal{L}_{\text{kin}}.$$

As a result,

$$\mathcal{L}'_{\text{kin}} = \mathcal{L}_{\text{kin}} \quad (696)$$

holds.

14.9 Gauge Behavior of the Source Term

The source term is written as

$$\mathcal{L}_{\text{int}} = -J^\mu A_\mu. \quad (697)$$

Under the gauge transformation,

$$\mathcal{L}'_{\text{int}} = -J^\mu (A_\mu + \partial_\mu \lambda) = -J^\mu A_\mu - J^\mu \partial_\mu \lambda = \mathcal{L}_{\text{int}} - J^\mu \partial_\mu \lambda.$$

The change in the action is

$$\Delta S_{\text{int}} = - \int d^4x e J^\mu \partial_\mu \lambda = - \int d^4x e \nabla_\mu (J^\mu \lambda) + \int d^4x e \lambda \nabla_\mu J^\mu.$$

Under current conservation,

$$\nabla_\mu J^\mu = 0 \quad (698)$$

and under the boundary conditions, the total derivative contribution vanishes. Thus,

$$\Delta S_{\text{int}} = 0 \quad (699)$$

is obtained. Therefore, the source term is gauge-compatible at the level of the action, not at the level of the Lagrangian density.

14.10 Preservation of the Field Equations Under the Exact Coframe Shift

In this work, the field equation does not consist only of the classical Maxwell core. Since A_μ is encoded within the temporal tetrad, geometric backreaction terms arising from determinant and metric dependences also appear during the tetrad variation. Therefore, the exact coframe shift gauge test must be performed not only through the behavior of the $\nabla_\mu F^{\mu\nu}$ term, but through the behavior of the full tetrad field equation.

The field equation obtained from the full tetrad variation is

$$\delta^0_a \left(\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} - J^\nu \right) + \kappa_{\text{EM}} [E_a^\rho \Theta^{\text{EM}}_\rho{}^\nu - E_a{}^\nu (J^\mu A_\mu)] = 0 \quad (700)$$

Here, the first term represents the Maxwell core, while the terms of order κ_{EM} in the second line represent the geometric backreaction contributions arising from the fact that A_μ is contained in the tetrad.

14.10.1 Preservation of the $\nabla_\mu F^{\mu\nu}$ Term

Under the exact coframe shift,

$$A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu \lambda \quad (701)$$

and therefore the electromagnetic field tensor is unchanged:

$$F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu. \quad (702)$$

Substituting the transformation of the potential gives

$$F'_{\mu\nu} = \partial_\mu (A_\nu + \partial_\nu \lambda) - \partial_\nu (A_\mu + \partial_\mu \lambda). \quad (703)$$

Explicitly,

$$F'_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + \partial_\mu \partial_\nu \lambda - \partial_\nu \partial_\mu \lambda. \quad (704)$$

Since partial derivatives commute,

$$\partial_\mu \partial_\nu \lambda - \partial_\nu \partial_\mu \lambda = 0 \quad (705)$$

is obtained. Therefore,

$$F'_{\mu\nu} = F_{\mu\nu} \quad (706)$$

holds.

Thus, together with the preservation of the metric structure within the same physical representation class,

$$F'^{\mu\nu} = F^{\mu\nu} \quad (707)$$

and

$$\nabla'_\mu F'^{\mu\nu} = \nabla_\mu F^{\mu\nu} \quad (708)$$

are obtained. Since the external source current also represents the same physical source distribution,

$$J'^\nu = J^\nu \quad (709)$$

is taken. Thus, the Maxwell core in Eq. (700),

$$\frac{1}{\mu_0} \nabla_\mu F^{\mu\nu} - J^\nu \quad (710)$$

is preserved under the exact coframe shift.

14.10.2 Preservation of the $E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu$ Term

The electromagnetic energy-momentum tensor is constructed only from $F_{\mu\nu}$ and the metric:

$$\Theta^{\text{EM}}{}_\rho{}^\nu = \frac{1}{\mu_0} F_{\rho\alpha} F^{\nu\alpha} - \frac{1}{4\mu_0} \delta_\rho{}^\nu F_{\alpha\beta} F^{\alpha\beta}. \quad (711)$$

Since it has been shown in the previous sections that both $F_{\mu\nu}$ and the metric are preserved under the exact coframe shift, one has

$$\Theta'^{\text{EM}}{}_\rho{}^\nu = \Theta^{\text{EM}}{}_\rho{}^\nu. \quad (712)$$

The internal Lorentz index of the inverse tetrad is carried according to the same frame transformation. Therefore,

$$E'_a{}^\rho \Theta'^{\text{EM}}{}_\rho{}^\nu = E'_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu \quad (713)$$

is written. Thus, the term

$$E_a{}^\rho \Theta^{\text{EM}}{}_\rho{}^\nu \quad (714)$$

in the field equation carries the same physical energy-momentum content under the exact coframe shift.

14.10.3 Preservation of the $E_a{}^\nu J^\mu A_\mu$ Term

By contrast, the source-potential contraction depends directly on A_μ . Therefore, it must be examined separately how this term behaves under the gauge transformation. Let the gauge transformation for the electromagnetic potential be

$$A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu \lambda. \quad (715)$$

In this case, the source-potential contraction transforms as

$$J^\mu A_\mu \longrightarrow J^\alpha A'_\alpha. \quad (716)$$

Writing explicitly,

$$J^\alpha A'_\alpha = J^\alpha (A_\alpha + \partial_\alpha \lambda) \quad (717)$$

is obtained. From this,

$$J^\alpha A'_\alpha = J^\mu A_\mu + J^\alpha \partial_\alpha \lambda \quad (718)$$

follows. Therefore,

$$J^\mu A_\mu \longrightarrow J^\mu A_\mu + J^\alpha \partial_\alpha \lambda \quad (719)$$

This result shows that the expression $J^\mu A_\mu$ depends pointwise on the gauge representative. Therefore, this term appearing in Eq. (700) should not be interpreted pointwise like a pure Maxwell equation. Since the source-potential term comes directly from the source-interaction action, its gauge behavior must also be evaluated at the level of the action.

Let the source interaction action be written, with the convention $-J^\mu A_\mu$ chosen in this work, as

$$S_{\text{int}} = -\frac{1}{c} \int d^4x e J^\mu A_\mu. \quad (720)$$

Under the gauge transformation,

$$S_{\text{int}} \longrightarrow S'_{\text{int}} = -\frac{1}{c} \int d^4x e J^\mu A'_\mu \quad (721)$$

is obtained. Substituting the transformation of the potential gives

$$S'_{\text{int}} = -\frac{1}{c} \int d^4x e J^\mu (A_\mu + \partial_\mu \lambda). \quad (722)$$

Explicitly,

$$S'_{\text{int}} = -\frac{1}{c} \int d^4x e J^\mu A_\mu - \frac{1}{c} \int d^4x e J^\mu \partial_\mu \lambda. \quad (723)$$

Thus, the change in the action is

$$\Delta S_{\text{int}} = S'_{\text{int}} - S_{\text{int}} = -\frac{1}{c} \int d^4x e J^\mu \partial_\mu \lambda \quad (724)$$

Now this expression is arranged in terms of the covariant derivative. For the scalar λ ,

$$\nabla_\mu \lambda = \partial_\mu \lambda \quad (725)$$

so one can write

$$J^\mu \partial_\mu \lambda = J^\mu \nabla_\mu \lambda. \quad (726)$$

Using the Leibniz rule,

$$\nabla_\mu (J^\mu \lambda) = (\nabla_\mu J^\mu) \lambda + J^\mu \nabla_\mu \lambda. \quad (727)$$

From this,

$$J^\mu \nabla_\mu \lambda = \nabla_\mu (J^\mu \lambda) - \lambda \nabla_\mu J^\mu \quad (728)$$

is obtained. Therefore,

$$J^\mu \partial_\mu \lambda = \nabla_\mu (J^\mu \lambda) - \lambda \nabla_\mu J^\mu. \quad (729)$$

Substituting Eq. (729) into Eq. (724) gives

$$\Delta S_{\text{int}} = -\frac{1}{c} \int d^4x e [\nabla_\mu (J^\mu \lambda) - \lambda \nabla_\mu J^\mu]. \quad (730)$$

Expanding,

$$\Delta S_{\text{int}} = -\frac{1}{c} \int d^4x e \nabla_\mu (J^\mu \lambda) + \frac{1}{c} \int d^4x e \lambda \nabla_\mu J^\mu. \quad (731)$$

The first term is a total derivative. For the tetrad determinant $e = \sqrt{-g}$, the identity

$$e \nabla_\mu V^\mu = \partial_\mu (e V^\mu) \quad (732)$$

is used. If

$$V^\mu = J^\mu \lambda \quad (733)$$

is chosen, then

$$\int d^4x e \nabla_\mu (J^\mu \lambda) = \int d^4x \partial_\mu (e J^\mu \lambda) \quad (734)$$

is obtained. By Gauss' theorem, this expression becomes a boundary term:

$$\int d^4x \partial_\mu (e J^\mu \lambda) = \int_{\partial \mathcal{M}} d\Sigma_\mu e J^\mu \lambda. \quad (735)$$

Thus, the change of the source interaction action under the gauge transformation is written as

$$\Delta S_{\text{int}} = -\frac{1}{c} \int_{\partial \mathcal{M}} d\Sigma_\mu e J^\mu \lambda + \frac{1}{c} \int d^4x e \lambda \nabla_\mu J^\mu \quad (736)$$

If current conservation

$$\nabla_\mu J^\mu = 0 \quad (737)$$

is satisfied, the volume term disappears. In addition, if the gauge parameter and the current satisfy suitable conditions at the boundary, for example

$$\lambda J^\mu d\Sigma_\mu|_{\partial\mathcal{M}} = 0 \quad (738)$$

then the boundary term also gives no contribution. In this case,

$$\Delta S_{\text{int}} = 0 \quad (739)$$

is obtained. Therefore, the source-potential interaction is gauge-compatible at the level of the action.

This result should be interpreted as follows: the term $J^\mu A_\mu$ depends pointwise on the gauge representative, because A_μ appears directly in this expression. However, since this term comes to the field equation from the source-interaction action, its physical gauge behavior is determined at the level of the action. Under current conservation and suitable boundary conditions, the change produced by the gauge transformation in the action is only a vanishing boundary term. Therefore, the physical content is preserved within the same gauge class.

In the Maxwell-adapted tetrad structure, since A_μ is also encoded within the temporal coframe, the gauge transformation can be read not only as a pointwise change of the potential, but also as a change of the coframe representative. Therefore, the gauge compatibility of the $J^\mu A_\mu$ term is achieved at the level of the action when current conservation, suitable boundary conditions, and the compensating coframe transformation are considered together. By contrast, the local field equation must be read in a fixed gauge representative, because the $J^\mu A_\mu$ term appears pointwise dependent on the chosen representative.

As a result, under the exact coframe shift, the full field equation of this work is preserved at three levels. The Maxwell core is unchanged because $F'_{\mu\nu} = F_{\mu\nu}$; the energy-momentum backreaction term remains the same because F and the physical metric are preserved; and the source-potential term, although it appears pointwise representative-dependent, is gauge-compatible at the level of the action under current conservation and boundary conditions. Thus, Eq. (700) is preserved on the physical content associated with the Maxwell-adapted exact coframe gauge class, not with a single representative A_μ .

14.11 Preservation of the Lorentz Force and the Equation of Motion

For a charged particle, the Lorentz force is given by

$$f^\mu = qF^\mu{}_\nu U^\nu. \quad (740)$$

Since $F'_{\mu\nu} = F_{\mu\nu}$ under the exact shift and $F'^\mu{}_\nu = F^\mu{}_\nu$ under metric preservation, one obtains

$$f'^\mu = qF'^\mu{}_\nu U^\nu. \quad (741)$$

From this,

$$f'^\mu = qF^\mu{}_\nu U^\nu \quad (742)$$

and therefore

$$f'^{\mu} = f^{\mu} \quad (743)$$

are obtained. As a result, the equation of motion of a charged particle is preserved under the exact coframe shift.

The above operations clearly show the result of the exact coframe shift gauge test. The temporal part keeps e^0 in the same representation class through the diffeomorphism compensator. The spatial part is represented as local-Lorentz-boost-like and is completed into a full frame transformation by the spatial tetrad transformation. Under these two structures,

$$e'^0 = e^0, \quad g' = g, \quad F' = F, \quad \mathcal{L}'_{\text{kin}} = \mathcal{L}_{\text{kin}}, \quad f'^{\mu} = f^{\mu} \quad (744)$$

are obtained. Thus, the electromagnetic $U(1)$ gauge transformation is understood within the Maxwell-adapted tetrad as an exact coframe gauge transformation that does not produce new physical content, but is represented geometrically by being separated into temporal and spatial parts.